

Goertzel 4CP Route: Historical Audit and Current Lean Status

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Abstract

This note records the current formal status of Ben Goertzel’s four-color proof route as represented locally by the older branch history, the present `4cp_proof_v23.pdf` manuscript, and the corresponding Lean development in `Mettapedia/GraphTheory/FourColor`. The main correction made here is both historical and logical: the currently formalized Lemma 4.3 obstruction matches an older per-run purification claim, not the exact v23 Lemma 4.3 / Step 2 surface. The next important update is positive: the exact v23 Step 2 single-component purification statement is now formalized and proved in Lean. Separately, the current Lean development still exposes genuine manuscript-level bugs at the spanning-forest leaf reduction and the naive well-founded peeling step, but it also formalizes a stronger repaired replacement based on boundary-root adjacency-distance data, canonical shortest-path parents, and annulus-construction lowerings. The remaining live blocker is the boundary-order / annulus-geometry interface needed to instantiate that stronger route. At the weak collar-geometry layer, Lean now also shows that canonical lowering to the annulus construction does not by itself provide the strict parent-witness orientation needed by that repaired route. A further closed-walk obstruction now shows that the current all-roots-on-the-outer- boundary adjacency-distance package is incompatible with the honest two-component boundary split extracted from the closed-walk source; the first ground-up repair is now also formalized as a two-sided annulus-root package that allows roots to meet either boundary component while still lowering to the existing interior-dual Theorem 4.9 pipeline. A newer correction also isolates the interior-vertex side condition in Theorem 4.9: boundary erasure preserves Kirchhoff equations only after the target vertex set is purified away from the selected boundary support. So the route remains blocked later on, but the evidence has to be stated precisely. The repaired previous-boundary lane also has a new calibration: honest closed-walk source data plus an arbitrary same-embedding weak collar geometry does not force the missing current-boundary witness placement. The diamond-with-triangle benchmark now sharpens the root-distance side of this audit: it has an honest source and repaired previous-boundary / witness-cover geometry, but its forcing interior edge blocks the current interior-dual root-distance packages on the same embedding. The current crux evidence is strong enough to reject several weak interfaces, but not strong enough to certify a $> 90\%$ block of every Goertzel-route continuation; the next responsible move is ground-up formalization of the stronger annulus and boundary-order hypotheses.

1 Current Lean status

As of this note, the `FourColor` directory builds cleanly and contains no direct `sorry` or `axiom` placeholders. This matters because the negative results discussed below are not informal suspicions hidden behind unfinished proofs; they are exposed as checked lemmas, examples, and regression files.

The formalization currently contains:

- the algebraic generator layer for the Goertzel route;
- an exact Lean proof of the v23 Step 2 single-component purification identity;
- theorem-surface formalization work around the v23 Theorem 4.9 annulus and dual-peeling route;
- explicit regression files showing where claimed reductions are too weak or false;
- a boundary-order note documenting the present embedding-side gap.

2 Historical lineage

It helps to separate two very different histories.

First, there is already a canonical accepted formal proof of the Four Color Theorem: Gonthier’s Coq development, now available in the `rocq-community/fourcolor` repository. That line is not based on the Goertzel/Kauffman boundary-algebra route.

Second, there is the specific experimental lineage of the Goertzel route itself. In the local workspace history, that lineage runs through earlier machine-checked and semi-formal efforts in Megalodon, Lean, and Isabelle. Those efforts were not merely parallel reimplementations; they were part of the process by which the route was stress-tested, revised, and restated.

The key historical point for the present note is this:

- an earlier boundary-algebra effort already ran into a contradiction around an older per-run Lemma 4.3 purification identity;
- the older Lean branch explicitly records this as a fatal flaw and points to a Megalodon-checked refutation of that claim;
- an archival Isabelle/HOL branch then explored a revised disk/Kempe-closure-spanning presentation of the route, emphasizing strong-dual and run-purification structure;
- Goertzel v23 should be understood against that background: it is already a repair attempt after an earlier machine-exposed failure mode;
- in particular, v23 changes the local theorem surface: its Lemma 4.3 is a run-invariance statement, and its later purification step is phrased using the full boundary intersection $D \cap \partial f$, not a single distinguished run R ;
- the current mettapedia Lean work now finds that v23 itself still has formal obstructions.

So the present bug report is not an isolated first stumble. It is the next stage of an ongoing prover-guided audit of the route.

In the wider prover landscape, systems such as Isabelle, Lean, Coq/Rocq, and Mizar all matter as comparison points for formal mathematical practice. But for the specific Goertzel-route lineage discussed here, the concrete artifacts currently in hand are the earlier Megalodon-checked contradiction, an earlier Lean branch, an archival Isabelle branch, and the present mettapedia Lean development.

3 Verified obstructions and audit status

3.1 Older per-run purification obstruction

The file `Mettapedia/GraphTheory/FourColor/GoertzelLemma43Obstruction.lean` formalizes the older per-run purification surface using one distinguished run R together with two complementary arcs A and A' .

The key checked theorem is:

`goertzel_lemma43_boundary_claim_false`

Its content is that the two indicator-chain contributions cancel on the run itself, but their sum survives on the complementary arcs. So the claim “the sum equals $\gamma \cdot 1_R$ ” fails in that abstraction once complementary-arc mass is genuinely present.

This is a real formal obstruction, but it should be attributed carefully. It matches the older per-run purification pattern already recorded in the archival branch history. It does *not* by itself refute the exact v23 Step 2 claim whose target support is $D \cap \partial f$.

3.2 Exact v23 Lemma 4.3 / Step 2 status

The exact v23 local story is subtler:

- v23 Lemma 4.3 is a face-boundary run-invariance statement;
- the later purification step defines $Y_{f,D} := X_f^{ab}(C) \oplus X_f^{ab}(C^D)$ and claims $Y_{f,D} = c \cdot 1_{D \cap \partial f}$;
- the current Lean tree now contains an exact formalization of this local step in `GoertzelLemma44.lean`.

More precisely, the new definition `v23PolarizedFaceGenerator` formalizes the exact Step 1 polarized generator that marks only the a -colored half of the boundary, and the theorem

`v23_single_component_purification`

proves the claimed identity

$$X_f^{ab}(C) \oplus X_f^{ab}(C^D) = c \cdot 1_{D \cap \partial f}$$

for one ‘(a,b)’ Kempe component D . A second theorem, `sum_v23_single_component_purifications_eq_boundaryOnlyGenerator`, shows that summing these exact v23 Step 2 identities over all boundary-hitting components recovers the existing simplified boundary-only generator already used downstream.

So the present formal evidence is now enough to say something sharper: v23 does *not* simply repeat the older Lemma 4.3 / purification error. The old per-run claim is formally refuted, while the revised v23 local purification step is formally vindicated in the present abstraction.

3.3 Step 2 spanning-forest leaf bug and its replacement

The file `Mettapedia/GraphTheory/FourColor/Theorem49ForestRegression.lean` contains a concrete regression showing that a leaf of a chosen spanning forest need not be a leaf in the ambient dual graph.

The example uses the path graph on four vertices as a spanning-tree candidate inside the four-cycle. Vertex 0 is a leaf of the path, but the ambient cycle still has an extra adjacency from 0 to 3. Therefore a reduction step of the form

“leaf in the chosen spanning forest” $\Rightarrow$ “all other incident dual edges are already accounted for”

is unsound without additional argument.

However, this is no longer the live Lean endpoint. The current formal route replaces the false leaf implication with the packaged theorem surface `InteriorDualBoundaryRootAdjDistancePeelData` in `Theorem49InteriorDualForest.lean`. Instead of asserting that a forest leaf controls all remaining incident ambient edges, that interface only asks for a weaker and honest local condition: every non-witness edge on a peeled face lies either on the selected boundary support or on an adjacent peeled child face at strictly larger root distance in the chosen spanning forest.

From there the code canonically recovers the stronger parent-based packages: boundary-root parent data, explicit root-distance parent data, height-ordered peeling, and well-founded parent peeling. On the annulus side, these repaired packages are already lowered all the way to the BFS annulus construction layer in `PlanarBoundaryAnnulusConstruction.lean`, including the outer-boundary- root specialization. So the regression is best read as a refutation of a manuscript heuristic, not as a proof that no spanning-forest-based repair exists.

3.4 Well-founded peeling bug and its replacement

The same regression file also proves that

acyclic forest + parent-edge adjacency

does not by itself imply a well-founded parent peeling relation.

Even on the two-vertex tree, one may define a parent map sending each endpoint to the other. Every parent edge lies in the forest, the forest is acyclic, and yet the induced parent relation has a 2-cycle and is not well founded.

So any correct peeling argument needs an extra descending invariant such as root-distance, explicit height, or another strictly decreasing measure.

Again, the current Lean route has already internalized this repair. The packages just mentioned lower from adjacency-distance data to explicit root-distance parent data, then to height-ordered parent data, and finally to well-founded parent data. So the surviving burden is not to rediscover that an invariant is needed; it is to extract the stronger invariant-bearing package from the annulus geometry used later in the manuscript.

3.5 Annulus witness orientation gap

There is now a second checked crux at the annulus interface, separate from the endpoint-carrier issues discussed later. The file `Theorem49PlanarBoundaryAnnulusWitnessRegression.lean` contains an explicit `counterCollarGeometry` satisfying the current `PlanarBoundaryAnnulusCollarGeometry` fields. This object canonically lowers through `planarBoundaryAnnulusConstruction_of_annulusCollarGeometry` to `counterAnnulusConstruction`.

The theorem `not_parentWitnessOrientation_counterAnnulusConstruction` proves that the lowered construction fails `PlanarBoundaryAnnulusConstruction.ParentWitnessOrientation`. The bad middle-collar face has witness edge `eqEdge`; the only lower-distance face does not contain that edge, while every face that does contain it has distance at least as large as the bad face. The packaged existential theorem `exists_planarBoundaryAnnulusConstruction_without_parentWitnessOrientation` records this as a construction-layer counterexample.

Thus the current weak collar-geometry axioms cannot be used as a black box to obtain the parent-oriented, well-founded annulus route. A continuation must prove the missing orientation or positive-distance condition from stronger geometry, or replace this part of the strategy.

That repair is now checked more sharply too. The new file `Theorem49PlanarBoundaryAnnulusGeometryRegression.lean` builds a tiny two-face model `counterPreviousBoundaryWitnessGeometry` satisfying the repaired surface `PlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry`, equivalently the current-boundary form `WitnessOnCurrentBoundary`. But its canonical lowering `planarBoundaryAnnulusConstruction_of_annulusCollarGeometry` still fails `ParentWitnessOrientation`: the outer collar face itself becomes a peeled construction face of distance 0, so it cannot have any strictly earlier parent in the construction metric. The same repaired geometry now also explicitly yields an attached boundary-rooted face-peel certificate on that same embedding, so this is not just a failure of one packaging choice: the canonical construction-parent shell can fail even after the honest certificate-side endpoint is already present. So even the repaired annulus geometry should not be funneled back through the canonical construction-parent shell. The honest ground-up annulus continuation is the direct route from weak collar geometry to attached certificates, and from repaired geometry to well-founded peeling, already formalized in `Theorem49PlanarBoundaryAnnulusGeometry.lean`.

That conclusion is now calibrated on an honest source model too. The new file `Theorem49PlanarBoundaryAnnulusHonestGeometryRegression.lean` puts a weak one-collar annulus geometry `diamondWithTriangleOneCollarGeometry` on the genuine closed-walk source model `diamondWithTriangleEmbedding`. On that same embedding the one-collar geometry already yields an attached boundary-rooted face-peel certificate via `diamondWithTriangleOneCollarGeometry_hasAttachedRootedFacePeelCertificate`, but the canonical lowering `diamondWithTriangleOneCollarConstruction` still fails `ParentWitnessOrientation` by `not_diamondWithTriangleOneCollarConstructionParentWitnessOrientation`. More sharply, the source-side summary theorem `closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_attachedRootedFacePeelCertificate_without_interiorDualBoundaryRootAdjDistancePeelData_diamondWithTriangle` shows that the honest facial closed-walk source, weak annulus geometry, and attached certificate can coexist on one embedding even while the current interior-dual boundary-root adjacency-distance package is absent there. So the ground-up annulus certificate lane is not only earlier than the canonical construction-parent shell; it is also formally earlier than the present interior-dual route on a genuine cyclic source model. The same file now also proves `closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_attachedRootedFacePeelCertificate_without_planarBoundaryAnnulusRootParentPeelData_diamondWithTriangle`, so even that honest source plus weak one-collar annulus geometry and attached certificate still does not force the current parent-oriented annulus-root target on the same cyclic source model. Thus the direct certificate lane is already strictly earlier than the live parent-peel burden too. The same file now also proves `diamondWithTriangle_explicitTait_currentSufficientSameEmbeddingGeometry_consolidatedRouteDiagnosis`, which packages the whole same-embedding calibration on that genuine cyclic source model: the full current positive geometry stack and explicit-Tait Theorem 4.9 synthesis coexist with a nonempty raw interior-edge endpoint carrier, an empty purified carrier, failure of the projected nonempty endpoint, failure of generic `InteriorDualBoundaryRootAdjDistancePeelData`, and failure of the live parent-peel package. The more specialized theorems `diamondWithTriangle_explicitTait_currentSufficientSameEmbeddingGeometry_`

`without_interiorDualBoundaryRootAdjDistancePeelData` and `diamondWithTriangle_explicitTait_currentSufficientSameEmbeddingGeometry_without_planarBoundaryAnnulusRootParentPeelData` remain available as focused projections of that same diagnosis. The same file now also proves `diamondWithTriangle_explicitTait_synthesis_without_planarBoundaryAnnulusRootParentPeelData`, so even the full Theorem 4.9 synthesis endpoint under the explicit Tait coloring coexists with failure of the current parent-peel package on that same embedding. Thus this live parent-peel burden is not necessary even for a genuine cyclic positive synthesis model. This is now lifted to a reusable converse-failure theorem too: `Theorem49ForcingInteriorEdgeObstruction.lean` proves `not_forall_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_of_exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_and_forcingInteriorEdgeWitness, not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_and_forcingInteriorEdgeWitness`, and `Theorem49ForcingInteriorEdgeObstructionRegression.lean` instantiates them on the diamond model via `exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_and_forcingInteriorEdgeWitness_diamondWithTriangleGraph` and `not_forall_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_of_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_diamondWithTriangle_via_forcingInteriorEdgeWitness` and `not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_diamondWithTriangle_via_forcingInteriorEdgeWitness`. So any synthesis model carrying a forcing interior edge now formally refutes the universal converse from full Theorem 4.9 synthesis back to either same-embedding annulus-root adjacency-distance data or same-embedding parent-peel data. The same file now also proves `diamondWithTriangle_explicitTait_currentSufficientSameEmbeddingGeometry_without_nonemptyProjectedSynthesis`, which seals that comparison on the same genuine cyclic source model: the full current same-embedding positive geometry stack, together with explicit-Tait Theorem 4.9 synthesis, already exists on `diamondWithTriangleEmbedding`, yet the newer projected nonempty endpoint still fails there because the selected-boundary-purified carrier is empty. The new wheel theorems `exists_theorem49BoundaryRootSynthesis_wheelWithInnerTriangleGraph_via_degenerateAttachedCertificate` and `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_IDBRAD_obstruction` show the complementary graph-vs-embedding separation: the wheel graph still admits the full Theorem 4.9 synthesis endpoint for the explicit Tait coloring on some embedding via the degenerate all-boundary certificate route, while the honest source-preserving wheel embedding with the same coloring and an explicit unblocked interior endpoint still fails generic same-embedding `InteriorDualBoundaryRootAdjDistancePeelData`. So the current wheel obstructions are source-preserving route failures, not graph-level impossibility of Theorem 4.9 synthesis for that colored graph. The new theorem `wheelWithInnerTriangleGraph_explicitTait_degenerateSynthesis_blocks_nonemptyProjectedSynthesis` sharpens the positive side of that comparison: the current graph-level witness is exactly the degenerate all-boundary embedding, so it reaches full synthesis only on an embedding whose selected-boundary-purified interior-edge endpoint carrier is empty and therefore cannot witness `Theorem49BoundaryRootNonemptyProjectedSynthesis`. Thus the currently known wheel graph-level positive theorem still sits entirely off the projected nonempty lane. The new theorem `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_requires_embedding_change` sharpens that diagnosis one step further: the currently known graph-level witness is provably not the honest source-preserving wheel embedding at all, because the degenerate witness has empty selected-boundary-purified car-

rier while the honest wheel embedding has a surviving purified carrier. So the present wheel positivity result necessarily changes embeddings before it can fire. The stronger theorem `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_without_selector_or_acyclicEndpoint_or_IDBRAD` shows that this same separation already covers every currently sufficient same-embedding selector/acyclic endpoint on the honest wheel embedding too: with the same explicit Tait coloring, honest source, nonempty purified carrier, and explicit unblocked interior endpoint, the fixed wheel embedding still has no boundary-free selector, no well-founded/height-ordered/collar-layer witness surface, no weak annulus- collar geometry, and no generic same-embedding `InteriorDualBoundaryRootAdjDistancePeelData`. The new theorem `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_without_sameEmbedding_projectedSourceRoutes` pushes that same wheel diagnosis all the way to the currently formalized projected endpoint routes: the graph still admits full Theorem 4.9 synthesis on some embedding, but on the honest source-preserving wheel embedding the same explicit Tait coloring still cannot instantiate either the IDBRAD-based projected endpoint route or the source-fixed canonical-parent projected endpoint route. So the benchmark now separates graph-level Theorem 4.9 positivity not only from the live same-embedding sufficient shells, but from both concrete same-embedding projected source routes already formalized in Lean. The new theorem `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_without_replacementPositiveProjectedGeometry_or_previousBoundaryWitness` extends that same graph-vs-fixed-embedding separation to the newer repaired positive route as well: the graph still admits full Theorem 4.9 synthesis on some embedding, but on the honest source-preserving wheel embedding the same explicit Tait coloring, honest source, and surviving purified carrier still fail both direct replacement-positive packages and the repaired previous-boundary witness geometry. So current graph-level positivity on this benchmark remains strictly upstream of every same-embedding positive route surface now formalized around the live annulus manuscript lane. Lean.

That route has now been pushed one step further on the honest source interface itself. In `PlanarBoundaryClosedWalkSourceRoute.lean` the theorems `zero_on_allEdges_of_exists_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_direct_of_annihilatesKempeClosureGeneratorFamily` and `zero_on_allEdges_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_annulusCollarGeometry_direct_of_annihilatesKempeClosureGeneratorFamily` show that once the same embedding carries weak annulus collar geometry, boundary vanishing on the selected support, and the Kempe-generator annihilation hypothesis, Step 2 edge-vanishing follows directly. So the remaining burden on the honest annulus lane is no longer to recover the old interior-dual package: it is to derive same-embedding weak collar geometry, or an equally strong direct substitute, from genuine cyclic boundary-order hypotheses. This source-facing collar lane now also reaches the full Theorem 4.9 endpoint: `PlanarBoundaryClosedWalkSourceRoute.lean` proves `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_direct` and `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_annulusCollarGeometry_direct`. So once honest closed-walk or successor-cycle source data is paired with same-embedding weak collar geometry, no additional theorem-4.9 algebra or certificate packaging remains.

The route picture is now even sharper at the pure geometry level. In `Theorem49PlanarBoundaryAnnulusDecomposition.lean` the definition `planarBoundaryAnnulusDecomposition_of_boundaryData` shows that under the current weak

boundary API, an annulus boundary split already yields a trivial one-collar annulus decomposition with all ambient faces in the unique collar. Then `PlanarBoundaryClosedWalkSource.lean` proves `admitsPlanarBoundaryAnnulusDecomposition_of_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource`. So the honest closed-walk source already reaches the pure annulus decomposition surface with no interior-dual or witness-edge hypotheses. The live burden is therefore not pure decomposition anymore; it is witness-edge ownership and the stronger same-embedding collar geometry needed to continue the annihilator branch.

That burden is now formalized more sharply as well. In `Theorem49PlanarBoundaryAnnulusWitness.lean` now proves the generic criterion `not_nonempty_planarBoundaryAnnulusWitnessAssignment_of_exists_two_distinct_interior_edges_on_faceBoundary_of_canonicalDecomposition`: for the canonical one-collar decomposition built from annulus boundary data, any ambient face carrying two distinct interior edges already blocks witness assignment. The concrete three-face model in `Theorem49PlanarBoundaryAnnulusWitnessRegression.lean` is now just an instance of that general obstruction. Thus the new source-to-decomposition lowering does not secretly solve Step 2: even after decomposition becomes trivial, witness-edge ownership remains a genuine additional geometric payload, and the obstruction is not tied to one hand-crafted counterexample.

That obstruction now reaches the honest source lane itself. In `Theorem49PlanarBoundaryAnnulusWitness.lean` the new theorem `not_nonempty_planarBoundaryAnnulusWitnessAssignment_of_exists_two_distinct_interior_edges_on_faceBoundary_of_closedWalkAnnulusBoundarySource` pushes the same argument through the canonical one-collar decomposition extracted from an honest `PlanarBoundaryClosedWalkAnnulusBoundarySource`. Then the new regression file `Theorem49PlanarBoundaryAnnulusHonestWitnessRegression.lean` builds a finite source model `sharedInteriorPairAnnulusEmbedding` with two outer faces sharing the consecutive interior edges `sip01` and `sip12`, plus a disjoint inner triangular boundary component. This model carries the full honest source package, but `not_nonempty_planarBoundaryAnnulusWitnessAssignment_sharedInteriorPairCanonicalDecomposition` shows that the canonical decomposition extracted from that very source still admits no witness assignment. So the tempting repair “derive boundary data from an honest source, then use the canonical one-collar decomposition” is now blocked on an honest same-embedding source model, not just on an abstract annulus-boundary witness.

The same obstruction now reaches beyond the canonical decomposition and into the weak geometry lane itself, as long as that lane still tries to stay one-collar. `Theorem49PlanarBoundaryAnnulusWitness.lean` now proves `not_nonempty_planarBoundaryAnnulusWitnessAssignment_of_numCollars_eq_one_and_exists_two_distinct_interior_edges_on_faceBoundary`, and `Theorem49PlanarBoundaryAnnulusGeometry.lean` lifts this to `not_exists_planarBoundaryAnnulusCollarGeometry_of_numCollars_eq_one_and_exists_two_distinct_interior_edges_on_faceBoundary`. Instantiating those results on the same honest source model gives `not_exists_oneCollar_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair` in `Theorem49PlanarBoundaryAnnulusHonestWitnessRegression.lean`. So the remaining repair “derive weak annulus collar geometry directly from the honest source” is now formally blocked on the entire one-collar branch for that model, not only on the canonical decomposition recovered from it. Any workaround now has to produce a genuinely multi-collar geometry or leave the current weak-collar theorem surface altogether.

That honest-source obstruction first reached the same-split multi-collar repairs via the current-boundary bridge `not_exists_planarBoundaryAnnulusCollarGeometry_of_not_exists_`

`planarBoundaryAnnulusCurrentBoundaryData_of_numCollars_eq_and_boundaryData`. The same witness now has a stronger obstruction that no longer fixes the ambient boundary split. In `Theorem49PlanarBoundaryAnnulusWitness.lean`, `false_of_mem_terminal_collarFaces_and_exists_two_distinct_interior_edges_on_faceBoundary` shows that a terminal collar face cannot carry two distinct interior edges: once one of them is the local witness, the other has no deeper frontier on which to sit and would have to be an ambient boundary edge. The honest regression file applies this to the two outer faces of `sharedInteriorPairAnnulusEmbedding`; hence any terminal collar is forced to be exactly the inner triangular face, formalized as `terminal_collarFaces_eq_singleton_sipFace2`.

That terminal-singleton fact blocks the remaining two- and three-collar weak geometries even when their boundary split is chosen freely. If a predecessor stage has only `sipFace2` in its remainder, the theorem `false_of_intermediateBoundary_before_singleton_sipFace2` forces an intermediate boundary edge to lie both in `interiorEdgeSupport` and in the selected boundary support of the inner triangle, contradicting the global ambient-boundary/interior disjointness. This yields `not_exists_twoCollar_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair` and `not_exists_threeCollar_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair` without any equality to `sharedInteriorPairBoundaryData`. Together with the one-collar blocker and the finite-face bound `PlanarBoundaryAnnulusCollarGeometry.numCollars_le_card_faces`, specialized as `numCollars_le_three_of_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair`, the regression now proves `not_nonempty_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair`. Thus the packaged honest-source model `closedWalkAnnulusBoundarySource_without_annulusCollarGeometry_sharedInteriorPair` and its graph-level form `exists_embedding_closedWalkAnnulusBoundarySource_without_annulusCollarGeometry_sharedInteriorPair` refute any universal same-embedding theorem from honest closed-walk source data to weak annulus collar geometry on this theorem surface, regardless of how the annulus boundary split is later chosen. The corresponding failed converse is `not_forall_exists_planarBoundaryAnnulusCollarGeometry_of_closedWalkAnnulusBoundarySource_sharedInteriorPair`. Consequently the still stronger source-alone route to the repaired previous-boundary witness surface is false as well: the same regression now proves `not_nonempty_planarBoundaryAnnulusPreviousBoundaryWitnessGeometry_sharedInteriorPair`, `closedWalkAnnulusBoundarySource_without_previousBoundaryWitness_sharedInteriorPair`, and `not_forall_exists_planarBoundaryAnnulusPreviousBoundaryWitnessGeometry_of_closedWalkAnnulusBoundarySource_sharedInteriorPair` by projecting any repaired previous-boundary witness geometry back to weak collar geometry. This refutes only the unqualified honest-source implication; the stricter same-embedding question with weak collar geometry already assumed remains the separate repaired-witness burden. The repaired-witness investigation module now removes the earlier placeholder by proving the exact fixed-collar formulation `honest_source_plus_collar_geometry_extending_previous_boundary_witness_iff_witnessOnCurrentBoundary`: for a given weak collar geometry, an extension of that very collar to the previous-boundary surface is equivalent to proving its `WitnessOnCurrentBoundary` property. The same investigation now proves that this property is not forced by merely adding an honest source. On the connected chained-diamonds source, Lean defines the valid weak two-collar geometry `chainedDiamondsBadTwoCollarGeometry`; its terminal triangular face chooses `cdt97`, while the previous intermediate boundary is the singleton `cdt45`. The theorems `not_witnessOnCurrentBoundary_chainedDiamondsBadTwoCollarGeometry`, `not_forall_honest_source_plus_collar_geometry_forces_previous_boundary_witness`, and `not_nonempty_`

previousBoundaryWitnessGeometry_extending_chainedDiamondsBadTwoCollarGeometry therefore refute the route “honest source + arbitrary weak collar $\Rightarrow$ repaired previous-boundary witness” on a connected honest source embedding. The same file now sharpens this calibration: `exists_atMostOne_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_without_witnessOnCurrentBoundary_chainedDiamonds`, `not_forall_atMostOne_source_plus_collar_geometry_forces_previous_boundary_witness`, and `not_forall_canonicalWitnessChoice_source_plus_collar_geometry_forces_previous_boundary_witness` show that even the local at-most-one source package, and even an already inhabited canonical witness choice on the source boundary split, do not make a separately supplied weak collar satisfy the repaired placement condition. The positive route must use the collar constructed from that choice or carry the placement condition explicitly.

The same model now blocks the weaker witness-cover repair surfaces that the positive synthesis layer currently accepts. The theorem `not_nonempty_planarBoundaryHeightOrderedFacePeelWitnessData_sharedInteriorPair` does not mention annulus boundary splits or collar remainders. It uses only the height-ordered witness-cover fields: covering `sip01` forces an outer face to choose `sip01`; the other shared interior edge `sip12` must then be chosen by a strictly later peeled face; applying the same argument to that face forces a still-later `sip01` witness. Since only `sipFace0` and `sipFace1` can choose either shared edge, this gives a strict height cycle. The finite collar-layer form is blocked by `not_nonempty_planarBoundaryCollarLayerFacePeelWitnessData_sharedInteriorPair`, via its canonical lowering to the height-ordered surface. The obstruction now also removes the height hypothesis entirely. The generic list lemma `not_exists_mem_of_twoWitnessPrefixCycle` proves that a finite peel order cannot contain a two-witness prefix cycle, even if the list repeats faces. Its support-level strengthening `not_mem_facePeelWitnessSupport_pair_of_twoWitnessPrefixCycle` now records that both mutually dependent witness edges are excluded from the realized peel support. The regression instantiates this as `not_nonempty_attachedBoundaryRootedFacePeelCertificate_sharedInteriorPair` for the raw attached-face `BoundaryRootedFacePeelCertificate` consumed by the boundary-root synthesis theorem. The well-founded-parent witness-cover surface is then blocked by `not_nonempty_planarBoundaryWellFoundedFacePeelWitnessData_sharedInteriorPair`, since it canonically lowers to that attached certificate. The source-packaged counterexamples `closedWalkAnnulusBoundarySource_without_heightOrderedFacePeelWitnessData_sharedInteriorPair` and `closedWalkAnnulusBoundarySource_without_collarLayerFacePeelWitnessData_sharedInteriorPair` now have the attached-certificate and well-founded variants `closedWalkAnnulusBoundarySource_without_attachedBoundaryRootedFacePeelCertificate_sharedInteriorPair` and `closedWalkAnnulusBoundarySource_without_wellFoundedFacePeelWitnessData_sharedInteriorPair`. This obstruction is now calibrated at the coloring layer as well: `sharedInteriorPairTaitEdgeColoring` gives a concrete Tait edge coloring of the same graph, proved by `sharedInteriorPairTaitEdgeColoring_isTait`. The package `closedWalkAnnulusBoundarySource_and_taitEdgeColoring_without_attachedBoundaryRootedFacePeelCertificate_sharedInteriorPair` therefore combines an honest closed-walk annulus source, an actual Tait coloring of its graph, and failure of the same-embedding raw attached certificate. The obstruction is not an artifact of a non-colorable source graph; the missing datum is genuinely the same-embedding geometric certificate. They show that the repaired positive endpoint “honest source plus certificate or witness-cover data” has a real extra geometric burden. Honest closed-walk annulus source data alone does not force even the raw same-embedding attached certificate, let alone the weaker finite witness-cover interfaces now sufficient

for Step 2 and Theorem 4.9 synthesis. The same regression now realizes the stronger successor-dart boundary source on this embedding as `sharedInteriorPairDartSuccessorCycleGeometry`; the selected-boundary arc proof is `sharedInteriorPairDartSuccessorCycleGeometry_selectedBoundaryArcOnFace`. Consequently `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_without_attachedBoundaryRootedFacePeelCertificate_sharedInteriorPair` and `not_forall_exists_attachedBoundaryRootedFacePeelCertificate_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_sharedInteriorPair` show that even replacing the closed-walk source by the route-facing successor-cycle source does not force the raw attached certificate on the same embedding. This has now been strengthened by `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_without_attachedBoundaryRootedFacePeelCertificate_sharedInteriorPair` and `not_forall_exists_attachedBoundaryRootedFacePeelCertificate_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_sharedInteriorPair`: even the successor-dart source shell plus selected-boundary arcs plus an actual Tait coloring of the graph does not force the same-embedding attached certificate. The stronger bundled witness `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_without_currentSufficientSameEmbeddingGeometry_sharedInteriorPair` packages the same source and coloring together with simultaneous failure of height-ordered witness data, collar-layer witness data, the raw attached certificate, well-founded witness data, and annulus collar geometry. The corresponding disjunctive failed converse `not_forall_exists_some_currentSufficientSameEmbeddingGeometry_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_sharedInteriorPair` shows that even one of those currently sufficient same-embedding geometric endpoints is not forced by the route-facing source plus Tait coloring. The same model now also proves that this is not merely another empty-carrier phenomenon: `vertex_one_mem_selectedBoundaryInteriorEdgeEndpointVertices_sharedInteriorPair` exhibits vertex 1 as an endpoint of an interior shared edge while avoiding all selected-boundary edges, and hence `selectedBoundaryInteriorEdgeEndpointVertices_nonempty_sharedInteriorPair` certifies a nonempty purified carrier. The strengthened package `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_without_currentSufficientSameEmbeddingGeometry_sharedInteriorPair` and failed converse `not_forall_exists_some_currentSufficientSameEmbeddingGeometry_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_sharedInteriorPair` therefore show that honest successor-cycle source data, Tait colorability, and nonempty selected-boundary interior carrier still do not force any of the currently sufficient same-embedding geometric endpoints. The same successor-source package now has parallel failed-converse endpoints for the individual upstream surfaces too: `not_forall_exists_planarBoundaryAnnulusCollarGeometry_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_sharedInteriorPair`, `not_forall_exists_planarBoundaryHeightOrderedFacePeelWitnessData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData`

and_selectedBoundaryArc_sharedInteriorPair, not_forall_exists_planarBoundaryCollarLayerFacePeelWitnessData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_sharedInteriorPair, and not_forall_exists_planarBoundaryWellFoundedFacePeelWitnessData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_sharedInteriorPair. Thus the stronger route-facing source shell fails all currently sufficient same-embedding geometric endpoints in this shared-interior-pair model, not only the raw attached certificate. In Lean both the honest closed-walk source failures and the successor-source failures are now factored through reusable same-embedding failed-converse lemmas, with generic source principles in `PlanarBoundaryClosedWalkSource.lean` and target-specific wrappers in `Theorem49PlanarBoundaryAnnulusGeometry.lean`; the shared-interior-pair theorems are instances obtained by supplying the explicit missing-target witnesses.

3.6 Closed-walk source versus all-outer roots

There is now a sharper non-endpoint obstruction at the intersection of the boundary-order source and the surviving root-distance annulus route. The new file `Theorem49ClosedWalkOuterRootObstruction.lean` proves `false_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_of_rootsOuterBoundary`. If the annulus boundary split is the honest split obtained from a `PlanarBoundaryClosedWalkAnnulusBoundarySource`, then `InteriorDualBoundaryRootAdjDistancePeelData` cannot have all of its chosen roots meeting the extracted outer boundary component.

The formal reason is short. Inner-boundary faces are disjoint from the peeled interior, hence the root-distance package makes them root faces. If every root face touches the extracted outer boundary, an inner-boundary face is also an outer-boundary face. But the closed-walk source supplies the facewise boundary-component separation that makes the extracted outer and inner boundary-face layers disjoint. The theorem `false_of_closedWalkAnnulusBoundarySource_and_outerBoundaryRootAdjDistancePeelData_sameBoundaryData` states the same contradiction for the current `PlanarBoundaryOuterBoundaryRootAdjDistancePeelData` package whenever it uses the same boundary split as the closed-walk source, and `not_exists_embedding_closedWalkAnnulusBoundarySource_and_outerBoundaryRootAdjDistancePeelData_sameBoundaryData` packages the graph-level nonexistence claim. The same file now also proves `false_of_closedWalkAnnulusBoundarySource_and_outerBoundaryRootParentPeelData_sameBoundaryData` and `not_exists_embedding_closedWalkAnnulusBoundarySource_and_outerBoundaryRootParentPeelData_sameBoundaryData`, so the same-split obstruction already kills the stronger `PlanarBoundaryOuterBoundaryRootParentPeelData` package too, not only the weaker adjacency-distance interface.

This substantially narrows the surviving route. The abstract `PlanarBoundaryOuterBoundaryRootAdjDistancePeelData` synthesis package is still a formally valid theorem surface, but it cannot be instantiated from the honest closed-walk two-component boundary source while keeping the current “all roots touch the outer boundary” hypothesis. A continuation has to replace that root condition with a two-sided/root-per-boundary formulation, or derive a different boundary split not governed by the closed-walk source’s component separation. The new parent-level obstruction shows that simply strengthening the same outer-root package downstream does not repair this lane. The same file now also proves `false_of_boundaryReachabilityData_and_interiorDualBoundaryRootParentPeelData_of_rootsOuterAmbientBoundary_and_boundarySelectedBoundaryArcOnFace`, so once a

boundary-order shell already carries facewise selected-boundary arcs, the stronger outer-root parent package is impossible there even before one asks for an honest closed-walk annulus source. The same file now also proves the same-boundary local nonexistence theorems `not_exists_planarBoundaryOuterBoundaryRootAdjDistancePeelData_sameBoundaryData_of_boundaryReachabilityData_and_boundarySelectedBoundaryArcOnFace` and `not_exists_planarBoundaryOuterBoundaryRootParentPeelData_sameBoundaryData_of_boundaryReachabilityData_and_boundarySelectedBoundaryArcOnFace`, so the live selected-arc boundary-order shell already rules out both outer-root packages as soon as they try to reuse that shell's own annulus boundary split.

The first replacement surface is now in Lean. The file `Theorem49PlanarBoundaryAnnulusRootInteriorDual.lean` defines `PlanarBoundaryAnnulusRootAdjDistancePeelData` and `PlanarBoundaryAnnulusRootParentPeelData`, where each root face may meet either the extracted outer or inner ambient boundary component. The constructors `planarBoundaryAnnulusRootAdjDistancePeelData_of_boundaryData_and_interiorDualBoundaryRootAdjDistancePeelData` and `planarBoundaryAnnulusRootAdjDistancePeelDataOfCoverSeparatedAnnulusBoundaryRootsCanonicalRooted` show that this is not a new obstruction layer: an annulus boundary split plus generic boundary-root interior-dual data already gives the two-sided package, and raw cover/separated-root data can use the full outer/inner boundary union as its root support. The closed-walk constructors `planarBoundaryAnnulusRootAdjDistancePeelData_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData` and `admitsPlanarBoundaryAnnulusRootAdjDistancePeelData_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData` make the positive companion explicit: the honest closed-walk boundary source can feed the two-sided root surface, provided the generic interior-dual data is available. The lowering theorems `admitsPlanarBoundaryInteriorDualBoundaryRootAdjDistancePeelData_of_admitsPlanarBoundaryAnnulusRootAdjDistancePeelData` and `admitsPlanarBoundaryCollarLayerFacePeelWitnessData_of_admitsPlanarBoundaryAnnulusRootAdjDistancePeelData` connect the new surface back to the existing collar-layer and Theorem 4.9 pipeline. The route file `PlanarBoundaryClosedWalkSourceRoute.lean` now also exposes this positive path all the way to the purified theorem-4.9 endpoint: `exists_theorem49AnnulusRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_direct` and its successor-cycle analogue `exists_theorem49AnnulusRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData_direct` show that, on one embedding, honest boundary-order source data plus generic interior-dual data already reaches the two-sided synthesis surface without restating the older shifted-construction side conditions. The same file now also exposes the more direct certificate-side corollaries `admitsPlanarBoundaryAttachedRootedFacePeelCertificate_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_direct`, `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_direct`, and their successor-cycle analogues, showing that once the generic boundary-root interior-dual package is present the honest source shells can land directly on the attached-certificate and boundary-root synthesis surfaces without routing back through annulus-construction parent orientation. The same successor-cycle shell now also reaches the non-vacuous two-sided annulus-root surface: `theorem49AnnulusRootNonemptySynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and`

`interiorDualBoundaryRootAdjDistancePeelData_of_peelFaces_endpoint_disjoint_selectedBoundarySupport` and `exists_theorem49AnnulusRootNonemptySynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData_with_peelFaces_endpoint_disjoint_selectedBoundarySupport_and_nonempty_interiorEdgeEndpointSupport` show that if the canonical annulus construction on that same embedding has peeled-face endpoint no-touch and the raw interior-edge endpoint carrier is nonempty, then the live boundary-order shell already reaches `Theorem49AnnulusRootNonemptySynthesis` before any source-fixed parent package is introduced. The same live successor-cycle shell now also reaches the generic source-fixed IDBRAD surface itself: `interiorDualBoundaryRootAdjDistancePeelData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover` is the same-embedding constructor, and `admitsPlanarBoundaryInteriorDualBoundaryRootAdjDistancePeelData_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover` is the graph-level route theorem. So once the successor-cycle boundary-order shell supplies the source boundary-face root cover/separation, boundary-free peeled faces, rooted shared-edge coverage, and strict root-distance child-membership fields, no additional source-to-IDBRAD repackaging step remains. The same rooted-cover package now also closes directly at the theorem-4.9 endpoint through `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover_direct`, so no further theorem-4.9-side repackaging remains below that concrete source-facing surface. The annulus-root file now also records the basic two-sided geometric forcing fact that was missing from the earlier route story: every ambient face meeting either distinguished annulus boundary component is automatically a non-peeled face and therefore a root face for the generic interior-dual adjacency-distance package. In particular, `exists_root_mem_outerBoundaryFaces_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData` and `exists_root_mem_innerBoundaryFaces_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData` show that an honest closed-walk annulus source plus the generic boundary-root interior-dual data forces at least one chosen root face on each boundary component of the extracted annulus split. The same file now also exposes the upstream parent-orientation surface itself: `parentWitnessOrientation_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_rootDistance` is the same-embedding theorem, and the graph-level wrappers `admitsPlanarBoundaryAnnulusConstructionParentWitnessOrientation_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_direct` and `admitsPlanarBoundaryAnnulusConstructionParentWitnessOrientation_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData_direct` show that honest closed-walk and successor-cycle boundary-order sources already reach `PlanarBoundaryAnnulusConstruction.ParentWitnessOrientation` through the root-distance construction, before invoking annihilator or synthesis theorems.

The converse route is now explicitly blocked on the diamond-with-triangle benchmark. The new regression theorem `diamondWithTriangle_source_previousBoundaryWitness_and_witnessCover_without_interiorDual_rootDistance` packages, on one embedding, an honest `PlanarBoundaryClosedWalkAnnulusBoundarySource`, repaired

PlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry, well-founded and height-ordered witness-cover data, and a ForcingInteriorEdgeWitness; the same theorem also proves absence of both InteriorDualBoundaryRootAdjDistancePeelData and the two-sided PlanarBoundaryAnnulusRootAdjDistancePeelData. The failed converses `not_forall_interiorDualBoundaryRootAdjDistancePeelData_of_previousBoundaryWitnessGeometry_diamondWithTriangle` and `not_forall_planarBoundaryAnnulusRootAdjDistancePeelData_of_closedWalkAnnulusBoundarySource_and_previousBoundaryWitnessGeometry_diamondWithTriangle` record the route implication directly: repaired previous-boundary witness geometry, even with the honest closed-walk source present, is not enough to derive the interior-dual root-distance route data. Thus root-distance data should be treated as a separate sufficient construction input, not as a hidden consequence of the current repaired witness geometry.

On the annulus-geometry side, the development now also exposes the hidden parent-carrying step inside the previous-boundary witness proof itself. `Theorem49PlanarBoundaryAnnulusDecomposition.lean` defines the canonical collar index `PlanarBoundaryAnnulusDecomposition.layerOf`, while `Theorem49PlanarBoundaryAnnulusWitness.lean` proves `PlanarBoundaryAnnulusWitnessAssignment.exists_parentFace_of_previousBoundaryWitness` and packages the chosen parent map as `PlanarBoundaryAnnulusWitnessAssignment.parentFaceOfPreviousBoundaryWitness`. So the current live geometric gap is no longer hidden inside the theorem-4.9 endpoint proof: the remaining task is to derive honest previous-boundary witness placement, or a stronger replacement such as annular no-touch, from real boundary-order geometry.

3.7 Boundary-order gap

The note `fourcolor-boundary-order-notes.tex` records a separate structural obstruction: the current `PlaneEmbeddingWithBoundary` API only exposes unordered face-boundary incidence, while the downstream selected-boundary and annulus arguments require ordered or cyclic face-boundary geometry.

This gap is not merely prose. The Lean development includes explicit regressions showing that unordered incidence alone is too weak to recover the ordered walk data needed downstream. The current workaround is to introduce stronger ordered and cyclic source interfaces explicitly, rather than pretending that the older API already determines them.

3.8 Interior-vertex discharge

The later quotient and boundary-erasure bridges in `Theorem49BoundaryProjection.lean` all carry the same side condition: the vertices where Kirchhoff equations are imposed must not be incident to the selected boundary support that is erased. The reviewer escalation around this point is valid. Without this condition, erasing selected-boundary coordinates can change a vertex Kirchhoff sum.

The Lean development now factors this issue into a small checked bridge in `Theorem49InteriorVertices.lean`. Given any finite candidate vertex set, `selectedBoundaryInteriorVertices` filters out the vertices incident to selected-boundary edges. The theorem `selectedBoundaryInteriorVertices_avoids_selectedBoundarySupport` discharges the side condition for this purified set. The exact obstruction is also recorded: `verticesAvoidingEdgeSupport_eq_self_iff` says that the purification is the identity precisely when the original candidate set already avoids the erased edge support, and `not_boundary_avoids_vertices_of_incident_boundary_edge` shows that a single candidate

vertex incident to a selected-boundary edge falsifies the side condition.

The route-facing payoff is deliberately narrow rather than another quotient API packet. First, `theorem49BoundaryZeroKirchhoffSubspace_selectedBoundaryInteriorVertices_eq_map_boundaryZeroProjection_kirchhoffSubmodule` discharges the side condition in the boundary-erasure image form of $W_0(H)$: for the purified vertex set, the concrete boundary-zero Kirchhoff target is exactly the selected-boundary-erased Kirchhoff image. Second, the theorem `theorem49BoundaryZeroKirchhoffSubspace_selectedBoundaryInteriorVertices_chainDot_projectedKempeClosureGeneratorSubspace_eq_zero_iff_of_planarBoundaryInteriorDualBoundaryRootAdjDistancePeelData` specializes the current $W_0(H)$ separation endpoint to the purified interior-vertex set, and the corresponding graph-level existential theorem exposes the same discharge from `AdmitsPlanarBoundaryInteriorDualBoundaryRootAdjDistancePeelData`.

The annulus route now has a same-statement bridge as well: `theorem49BoundaryZeroKirchhoffSubspace_selectedBoundaryInteriorVertices_image_and_separation_of_planarBoundaryOuterBoundaryRootAdjDistancePeelData` takes the outer-boundary-rooted Step 2 package and returns both the purified boundary-erasure image identity and the purified separation endpoint. The graph-level theorem `exists_theorem49BoundaryZeroKirchhoffSubspace_selectedBoundaryInteriorVertices_image_and_separation_of_admitsPlanarBoundaryOuterBoundaryRootAdjDistancePeelData` exposes this directly from the paper-facing annulus target.

The arbitrary candidate has now been replaced at one route endpoint by a concrete finite graph carrier. The definitions `edgeEndpointSupport` and `graphEdgeEndpointSupport` collect all endpoints of graph edges from a finite edge set, without assuming a global finite vertex type. The purified carrier `selectedBoundaryGraphInteriorVertices` then deletes endpoints of the selected boundary support. The theorem `theorem49BoundaryZeroKirchhoffSubspace_selectedBoundaryGraphInteriorVertices_image_and_separation_of_planarBoundaryOuterBoundaryRootAdjDistancePeelData` specializes the annulus bridge to this concrete carrier, and the graph-level theorem `exists_theorem49BoundaryZeroKirchhoffSubspace_selectedBoundaryGraphInteriorVertices_image_and_separation_of_admitsPlanarBoundaryOuterBoundaryRootAdjDistancePeelData` exposes it without a user-supplied `verticesOf` function. This is a real interior-vertex route integration, not a further quotient API extension.

The next refinement uses the annulus route's own face-incidence edge classification. The definition `interiorEdgeEndpointSupport` collects endpoints of `interiorEdgeSupport` edges, i.e. edges incident to two ambient faces, and `selectedBoundaryInteriorEdgeEndpointVertices` purifies this more annular carrier. The theorem `theorem49BoundaryZeroKirchhoffSubspace_selectedBoundaryInteriorEdgeEndpointVertices_image_and_separation_of_planarBoundaryOuterBoundaryRootAdjDistancePeelData` specializes the same boundary-erasure and separation endpoint to this carrier, with graph-level wrapper `exists_theorem49BoundaryZeroKirchhoffSubspace_selectedBoundaryInteriorEdgeEndpointVertices_image_and_separation_of_admitsPlanarBoundaryOuterBoundaryRootAdjDistancePeelData`. The same surviving carrier is now exposed in map-level and dimension forms. The theorem `ker_theorem49BoundaryZeroKirchhoffSubspaceProjectedGeneratorPairingMap_selectedBoundaryInteriorEdgeEndpointVertices_eq_bot_of_planarBoundaryOuterBoundaryRootAdjDistancePeelData` states that the concrete pairing map from this purified $W_0(H)$ target into the dual of the projected Definition 4.8 generator span has trivial kernel. Consequently `finrank_theorem49BoundaryZeroKirchhoffSubspace_`

`selectedBoundaryInteriorEdgeEndpointVertices_le_finrank_projectedKempeClosureGeneratorSubspace_of_planarBoundaryOuterBoundaryRootAdjDistancePeelData` gives the finite-dimensional bound $\dim W_0(H)_{\text{purified}} \leq \dim K_{\text{proj}}$, where K_{proj} is the projected Definition 4.8 generator span. Both statements also have graph-level existential wrappers from `AdmitsPlanarBoundaryOuterBoundaryRootAdjDistancePeelData`. The same purified carrier now reaches the operational projected-source form of Theorem 4.9. The theorem `theorem49BoundaryZeroKirchhoffSubspace_selectedBoundaryInteriorEdgeEndpointVertices_eq_kirchhoffSubmodule_inf_projectedKempeClosureGeneratorSubspace_of_planarBoundaryOuterBoundaryRootAdjDistancePeelData` identifies this $W_0(H)$ target with the Kirchhoff equations at the purified carrier intersected with K_{proj} , and `theorem49BoundaryZeroKirchhoffSubspace_selectedBoundaryInteriorEdgeEndpointVertices_le_projectedKempeClosureGeneratorSubspace_of_planarBoundaryOuterBoundaryRootAdjDistancePeelData` gives the direct inclusion into K_{proj} . Finally, `exists_projectedKempeClosureGeneratorFamily_finset_sum_of_mem_theorem49BoundaryZeroKirchhoffSubspace_selectedBoundaryInteriorEdgeEndpointVertices_of_planarBoundaryOuterBoundaryRootAdjDistancePeelData` and `exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_and_mem_kirchhoffSubmodule_of_mem_theorem49BoundaryZeroKirchhoffSubspace_selectedBoundaryInteriorEdgeEndpointVertices_of_planarBoundaryOuterBoundaryRootAdjDistancePeelData` turn membership in the purified $W_0(H)$ target into finite projected Definition 4.8 generator sums, respectively finite raw generator sums whose selected-boundary projection is the target chain and whose raw sum remains Kirchhoff at the purified carrier. The graph-level wrappers expose these finite-combination endpoints from `AdmitsPlanarBoundaryOuterBoundaryRootAdjDistancePeelData`. The raw-source quotient form is now specialized to the same carrier as well. `theorem49BoundaryZeroKirchhoffSubspace_selectedBoundaryInteriorEdgeEndpointVertices_eq_map_boundaryZeroProjection_kempeClosureGeneratorSubspace_inf_kirchhoffSubmodule_of_planarBoundaryOuterBoundaryRootAdjDistancePeelData` identifies the purified $W_0(H)$ target with the selected-boundary erasure of raw Definition 4.8 generator-span chains that are already Kirchhoff at the purified carrier. The preimage theorem `exists_kempeClosureGeneratorSubspace_inf_kirchhoffSubmodule_preimage_boundaryZeroProjection_of_mem_theorem49BoundaryZeroKirchhoffSubspace_selectedBoundaryInteriorEdgeEndpointVertices_of_planarBoundaryOuterBoundaryRootAdjDistancePeelData` packages this as an explicit source representative in “raw Kempe span” intersected with the purified Kirchhoff submodule. Finally, `exists_kempeClosureGeneratorFamily_finset_sum_boundaryKernel_error_decomposition_of_mem_kirchhoffSubmodule_selectedBoundaryInteriorEdgeEndpointVertices_of_planarBoundaryOuterBoundaryRootAdjDistancePeelData` records the quotient error term: every raw Kirchhoff chain at the purified carrier is a finite raw generator sum plus a Kirchhoff chain killed by selected-boundary erasure. The new synthesis module `Theorem49Synthesis.lean` packages these positive consequences without adding another carrier layer. Its core structure `Theorem49BoundaryRootSynthesis` is proved directly from generic `PlanarBoundaryInteriorDualBoundaryRootAdjDistancePeelData`; it collects the boundary-erased image identity, pointwise separation, kernel-zero and finite-dimensional forms, projected-generator spanning, and the corrected raw-source quotient and boundary-kernel decomposition for the selected-boundary interior-edge endpoint carrier. The outer-route theorem `exists_theorem49OuterBoundaryRootSynthesis_of_admitsPlanarBoundaryOuterBoundaryRootAdjDistancePeelData` is now just a wrapper through the outer package’s `interiorData`, and the same file

also exposes the identical synthesis endpoint on the two-sided surface via `Theorem49AnnulusRootSynthesis` and `exists_theorem49AnnulusRootSynthesis_of_`
`admitsPlanarBoundaryAnnulusRootAdjDistancePeelData`. So the purified Theorem 4.9
algebra no longer depends on the all-outer-root localization once generic boundary-
root interior-dual data exists. The same module now also exposes the primitive two-
sided annulus-split interface: `theorem49AnnulusRootSynthesis_of_boundaryData_and_`
`interiorDualBoundaryRootAdjDistancePeelData` and its graph-level existence wrappers
show that once one embedding carries `PlanarBoundaryAnnulusBoundaryData` together
with generic `InteriorDualBoundaryRootAdjDistancePeelData`, the theorem-4.9 synthe-
sis endpoint is already available before any stronger closed-walk source fields are added.
The next bridge now also exists directly from the concrete annulus construction sur-
face. The new file `Theorem49PlanarBoundaryAnnulusConstructionSpanning.lean` proves
`boundaryZeroAnnihilatorTrivialForEmbedding_of_planarBoundaryAnnulusConstruction_`
`of_parentWitness`: if a `PlanarBoundaryAnnulusConstruction` satisfies the explicit
`ParentWitnessOrientation` condition, then every selected-boundary-zero chain annihilating
the Definition 4.8 family is already zero on all edges. This is the exact non-geometric hypothesis
used by the algebraic spanning bridge in `Theorem49SpanningGap.lean`. Consequently the same file
now derives `projectedKempeClosureGeneratorSubspace_eq_planarBoundaryZeroSubmodule_`
`of_planarBoundaryAnnulusConstruction_of_parentWitness`, its explicit-family compan-
ion `span_projectedKempeClosureGeneratorFamily_eq_planarBoundaryZeroSubmodule_`
`of_planarBoundaryAnnulusConstruction_of_parentWitness`, the corrected target
identity `theorem49BoundaryZeroKirchhoffSubspace_eq_kirchhoffSubmodule_inf_`
`projectedKempeClosureGeneratorSubspace_of_planarBoundaryAnnulusConstruction_of_`
`parentWitness`, the corresponding target inclusion, pointwise separation on the purified endpoint
carrier, kernel-triviality of the pairing map, the resulting finrank bound, and a raw Kempe-span
preimage theorem. So the current annulus route no longer needs to recreate the older interior-dual
boundary-root package merely to reach the algebraic theorem-4.9 bridge: construction-side ge-
ometry plus the honest parent-witness orientation already suffices. `Theorem49SpanningGap.lean`
now exposes the same algebraic bridge directly from the weaker boundary-aware witness-
cover surfaces. The theorems `boundaryZeroAnnihilatorTrivialForEmbedding_of_`
`planarBoundaryHeightOrderedFacePeelWitnessData` and `boundaryZeroAnnihilatorTrivialForEmbedding_`
`of_planarBoundaryCollarLayerFacePeelWitnessData` construct the annihilator endpoint from
height-ordered or finite collar-layer witness-cover data. The same file now also has graph-level
height-ordered and collar-layer wrappers for both the ordinary and relative submodule-duality in-
terfaces, with chain-dot variants that explicitly construct the relevant duality data from the named
same-embedding subspaces and identification lemmas. `Theorem49KempeGeneratorSpan.lean` then
fixes the generator side canonically through `theorem49_kempeGeneratorSubspace_spanning_`
`of_planarBoundaryHeightOrderedFacePeelWitnessData` and its collar-layer analogue. It
now also exposes the graph-level wrappers `exists_theorem49_kempeGeneratorSubspace_`
`spanning_of_admitsPlanarBoundaryHeightOrderedFacePeelWitnessData`
and `exists_theorem49_kempeGeneratorSubspace_spanning_of_`
`admitsPlanarBoundaryCollarLayerFacePeelWitnessData`. These instantiate the con-
crete submodule-duality data with the actual Definition 4.8 Kempe-generator sub-
space, so the remaining caller-side algebraic inputs are just target containment
and selected-boundary-zero interpretation, not a separate orthogonality-to-annihilator
proof. Finally `Theorem49BoundaryProjection.lean` pushes both surfaces to the cor-
rected projected-source conclusion: `projectedKempeClosureGeneratorSubspace_eq_`
`planarBoundaryZeroSubmodule_of_planarBoundaryHeightOrderedFacePeelWitnessData`

and `projectedKempeClosureGeneratorSubspace_eq_planarBoundaryZeroSubmodule_of_planarBoundaryCollarLayerFacePeelWitnessData`, together with the corresponding boundary-zero Kirchhoff target equality and inclusion theorems. Thus the spanning gap no longer waits for the stronger interior-dual rooted surface once either boundary-aware witness-cover surface is available. The same file now gives operational finite-witness forms for these weaker surfaces: every selected-boundary-zero chain, and every concrete boundary-zero Kirchhoff target chain, has a finite representation by projected Definition 4.8 generators. It also gives raw-generator preimage and finite projection statements before the boundary erasure. The graph-level existence wrappers from `AdmitsPlanarBoundaryHeightOrderedFacePeelWitnessData` and `AdmitsPlanarBoundaryCollarLayerFacePeelWitnessData` now include the explicit-family target identity, projected and explicit-family inclusion forms, raw Definition 4.8 generator-span preimages, and finite raw-generator projection witnesses for every boundary-zero Kirchhoff target chain. The same projected-source layer now also proves the chain-dot separation form

$$W_0(H) \cap (\text{projectedKempeClosureGeneratorSubspace})^\perp = 0$$

from the boundary-zero annihilator endpoint, together with detector and pointwise orthogonality iff forms. These theorems have height-ordered, finite-collar, and graph-level admissibility wrappers, so the projected Definition 4.8 source supplies actual nondegenerate testing against the concrete boundary-zero Kirchhoff target without the additional quotient-side interior-vertex disjointness assumptions required by the raw-source quotient presentation. Downstream theorem 4.9 route files can therefore consume height-ordered or collar-layer witness data at the corrected projected-source interface without unpacking the same-embedding witness manually. `Theorem49PlanarBoundaryCollarPeeling.lean` now also exposes the weakest explicit local-remainder collar surface directly at the generator-family annihilator endpoint. The fixed-embedding theorem `zero_on_allEdges_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData_of_annihilatesKempeClosureGeneratorFamily` and the existential companion `zero_on_allEdges_of_exists_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData_of_annihilatesKempeClosureGeneratorFamily` consume Definition 4.8 generator-family orthogonality without first forgetting the local-remainder data to ordinary collar-layer data. Thus the root-free local-remainder route is now a theorem-facing Step 2 endpoint, not merely an intermediate representation of finite collars. `Theorem49SpanningGap.lean` and `Theorem49BoundaryProjection.lean` now push this endpoint through the linear-duality layer as well. The theorem `boundaryZeroAnnihilatorTrivialForEmbedding_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData` feeds the local-remainder package into the named annihilator-trivial bridge, and `exists_projectedKempeClosureGeneratorFamily_finset_sum_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData` then gives the concrete projected Definition 4.8 finite-combination form for every boundary-zero Kirchhoff target chain on the same embedding. The graph level companion `exists_projectedKempeClosureGeneratorFamily_finset_sum_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of_admitsPlanarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData` records the same endpoint without first converting the local-remainder witness to an ordinary collar witness. The same local-remainder branch now also reaches the raw manuscript-facing source language. The fixed-embedding theorem `exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of`

`of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData` chooses a finite sum from the unprojected Definition 4.8 generator family whose selected-boundary projection is the target chain, and `exists_kempeClosureGeneratorSubspace_preimage_boundaryZeroProjection_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData` records the corresponding raw generator-subspace preimage. Thus the local-remainder route is not merely a boundary-erased projected-source result: it now lands back on the raw generator family used by the proof route. The fixed-embedding strengthening `exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_and_mem_kirchhoffSubmodule_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData` adds the missing Kirchhoff-retention clause: when the chosen target vertices avoid the selected boundary support, the raw finite Definition 4.8 sum can be chosen inside the relevant Kirchhoff submodule, and its selected-boundary projection is still the prescribed boundary-zero target chain. The graph-level version records the same endpoint for an `AdmitsPlanarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData` witness without passing through the blocked interior-dual root-distance package. This local-remainder branch now also has the quotient/error endpoint formerly reserved for the interior-dual package. The theorem `exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_eq_boundaryZeroProjection_of_mem_kirchhoffSubmodule_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData` starts from an arbitrary raw Kirchhoff chain and chooses a finite raw Definition 4.8 generator-family sum, still Kirchhoff at the chosen target vertices, with the same selected-boundary-erased image. The subsequent boundary-supported and kernel-decomposition theorems record that the difference is a Kirchhoff correction supported on the selected boundary, equivalently an element of the boundary-erasure kernel inside the Kirchhoff submodule. The graph-level companions expose the same quotient/error package directly from `AdmitsPlanarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData`. The route-facing projected synthesis endpoint now exposes the same package without asking downstream source arguments to reach back into the raw projection layer: `Theorem49BoundaryRootNonemptyProjectedSynthesis.rawKirchhoffRepresentative_and_boundaryKernelDecomposition` combines the Kirchhoff-valid raw representative and the boundary-kernel error decomposition already carried by the nested `Theorem49BoundaryRootSynthesis`. Thus any closed-walk or collar source package that has reached the positive same-embedding projected endpoint can consume the quotient/error interpretation of the raw Definition 4.8 generators directly at that endpoint. This endpoint has now been named as the abbreviation `Theorem49BoundaryRawQuotientErrorPackage`. The viable replacement positive surfaces consume it directly: `theorem49BoundaryRawQuotientErrorPackage_of_heightOrderedPositiveProjectedGeometryOn` and `theorem49BoundaryRawQuotientErrorPackage_of_collarLayerPositiveProjectedGeometryOn` give fixed-embedding height-ordered and finite collar-layer versions, while the namespace methods `Theorem49HeightOrderedPositiveProjectedGeometry.rawQuotientErrorPackage` and `Theorem49CollarLayerPositiveProjectedGeometry.rawQuotientErrorPackage` give the graph-level package forms. Thus the route no longer merely produces a nonempty projected synthesis package from the replacement geometry; it also exposes the raw Definition 4.8 quotient/error reading at that same replacement surface. The core synthesis surface now also records the corresponding global submodule decomposition. The theorem `Theorem49BoundaryRootSynthesis.kirchhoffSubmodule_eq_rawSource_sup_boundaryKernel`

states that, once `Theorem49BoundaryRootSynthesis` is available, the purified Kirchhoff submodule is exactly the supremum of the Kirchhoff-valid raw Definition 4.8 Kempe-source submodule and the selected-boundary projection kernel inside that purified Kirchhoff submodule. This is the synthesis-level form of the quotient/error conclusion: every purified Kirchhoff chain splits into a raw source component plus a boundary-kernel error, and the reverse inclusion is immediate because both summands remain Kirchhoff-valid. The quotient/surjectivity form is now explicit as `Theorem49BoundaryRootSynthesis.rawQuery_mkQ_eq_top`. When the raw source is viewed as a submodule of the purified Kirchhoff submodule and the selected-boundary projection kernel is restricted to that Kirchhoff submodule, the quotient map sends the raw source onto the top submodule of the quotient. Equivalently, every purified Kirchhoff quotient class modulo selected-boundary kernel error has a raw Definition 4.8 Kempe-source representative. The statement is deliberately internal to the purified Kirchhoff submodule, avoiding the stronger and generally unjustified ambient quotient claim. The same quotient endpoint now has a finrank consequence: `Theorem49BoundaryRootSynthesis.finrank_quotient_le_rawSource` bounds the dimension of the purified Kirchhoff quotient by the dimension of the Kirchhoff-valid raw Definition 4.8 source submodule inside it. This is just the finite-dimensional shadow of the quotient surjectivity, but it exposes a rank-level invariant that downstream theorem-4.9 arguments can use without opening the quotient-map proof. The target-side version is now explicit as `Theorem49BoundaryRootSynthesis.finrank_target_le_rawSource`: the concrete boundary-zero Kirchhoff target itself has finrank bounded by the Kirchhoff-valid raw Definition 4.8 source submodule whose selected-boundary erasure is the target. This packages the dimension consequence of `target_eq_rawSourceImage` directly at the synthesis surface. The quotient-target bridge itself is now named at this surface: `Theorem49BoundaryRootSynthesis.boundaryProjectionQuotientEquivTarget` identifies the purified Kirchhoff submodule modulo the restricted selected-boundary projection kernel with the concrete boundary-zero Kirchhoff target. Its computation theorem `Theorem49BoundaryRootSynthesis.boundaryProjectionQuotientEquivTarget_apply_mk` sends the quotient class of a purified Kirchhoff chain to its selected-boundary-erased chain. Thus the corrected quotient/error layer is no longer merely an interpretation of representatives; it is a first-isomorphism interface for the actual theorem-4.9 target. The source-side first-isomorphism form is now present as well: `Theorem49BoundaryRootSynthesis.rawQueryQuotientEquivTarget` identifies the Kirchhoff-valid raw Definition 4.8 source submodule modulo the selected-boundary projection kernel on that source with the same concrete boundary-zero Kirchhoff target. Its computation theorem `Theorem49BoundaryRootSynthesis.rawQueryQuotientEquivTarget_apply_mk` sends the class of a raw Kirchhoff-valid source chain to its selected-boundary-erased chain. This gives the target a direct raw-source quotient model, separate from the quotient of the entire purified Kirchhoff submodule. The two quotient models are now connected directly: `Theorem49BoundaryRootSynthesis.rawQueryQuotientEquivKirchhoffQuotient` identifies the raw-source quotient with the purified Kirchhoff quotient through their common concrete theorem-4.9 target. Its computation theorem `Theorem49BoundaryRootSynthesis.rawQueryQuotientEquivKirchhoffQuotient_apply_mk` sends the class of a raw Kirchhoff-valid source representative to the class of the same chain regarded as an element of the purified Kirchhoff submodule. This is the synthesis-surface comparison map needed to pass between raw Definition 4.8 quotient classes and purified Kirchhoff quotient classes without reopening either first-isomorphism proof. The representative-level endpoint is now explicit: `Theorem49BoundaryRootSynthesis.exists_rawGeneratorSum_for_kirchhoffQuotientClass` proves that every class of the purified Kirchhoff quotient has a representative that is both Kirchhoff-valid and a finite linear combination of actual raw Definition 4.8 generator-family

terms. The theorem packages the selected- boundary projection equality as equality of quotient classes modulo the restricted selected-boundary kernel. Downstream arguments can therefore choose a finite raw generator-sum representative for a purified Kirchhoff quotient class without replaying the raw representative construction or the quotient kernel conversion. The separation endpoint has also been transported to the purified quotient: `Theorem49BoundaryRootSynthesis.kirchhoffQuotientClass_eq_zero_iff_projectedGenerator_orthogonal` states that the class of a purified Kirchhoff chain is zero modulo the restricted selected-boundary kernel exactly when its selected-boundary-erased representative pairs to zero with every projected Definition 4.8 generator. The detector form `Theorem49BoundaryRootSynthesis.exists_projectedGenerator_chainDot_ne_zero_of_ne_zero_kirchhoffQuotientClass` produces a projected generator with nonzero pairing for every nonzero purified Kirchhoff quotient class. Thus nondegeneracy is now available directly on the quotient/error surface, not only after passing to the concrete target. The combined downstream form is `Theorem49BoundaryRootSynthesis.exists_rawGeneratorSum_and_projectedGenerator_chainDot_ne_zero_of_ne_zero_kirchhoffQuotientClass`: each nonzero purified Kirchhoff quotient class admits a finite raw Definition 4.8 generator-sum representative, and that representative is detected by a projected Definition 4.8 generator after selected-boundary erasure. This packages representative choice, quotient nonzeroness transport, and projected-generator detection into one theorem at the synthesis surface. The same nondegeneracy is now packaged as a quotient-level linear map: `Theorem49BoundaryRootSynthesis.kirchhoffQuotientProjectedGeneratorPairingMap` sends a purified Kirchhoff quotient class to its chain-dot functional on the projected Definition 4.8 generator subspace after selected-boundary erasure. The computation theorem `Theorem49BoundaryRootSynthesis.kirchhoffQuotientProjectedGeneratorPairingMap_apply_mk` evaluates this map on quotient classes represented by purified Kirchhoff chains, and `Theorem49BoundaryRootSynthesis.ker_kirchhoffQuotientProjectedGeneratorPairingMap_eq_bot` proves that its kernel is trivial. This turns the quotient/error separation endpoint into a standard linear-map kernel statement. The same map-level surface now gives `Theorem49BoundaryRootSynthesis.kirchhoffQuotientProjectedGeneratorPairingMap_eq_zero_iff` and `Theorem49BoundaryRootSynthesis.exists_projectedGenerator_pairingMap_ne_zero_of_ne_zero_kirchhoffQuotientClass`: an arbitrary purified Kirchhoff quotient class is zero exactly when its projected-generator functional is zero, and every nonzero quotient class is detected by evaluating that functional at some projected generator. The finite-dimensional consequences are also available at this same surface: `Theorem49BoundaryRootSynthesis.finrank_kirchhoffQuotient_le_projectedGeneratorSubspace` derives a dimension bound for the purified Kirchhoff quotient directly from this trivial-kernel pairing map and the dual-rank identity for finite subspaces, while `Theorem49BoundaryRootSynthesis.finrank_range_kirchhoffQuotientProjectedGeneratorPairingMap` identifies the dimension of the pairing-map image with the dimension of the purified Kirchhoff quotient. The image is now an operational quotient model: `Theorem49BoundaryRootSynthesis.kirchhoffQuotientEquivPairingMapRange` is the linear equivalence from the purified Kirchhoff quotient onto the pairing-map image, and `Theorem49BoundaryRootSynthesis.kirchhoffQuotientEquivPairingMapRange_apply_mk` computes that image functional on representatives by selected-boundary erasure and the chain-dot pairing. The raw-source quotient now lands on the same image model directly: `Theorem49BoundaryRootSynthesis.rawSourceQuotientEquivPairingMapRange` identifies the Kirchhoff-valid raw Definition 4.8 source quotient with the pairing-map image, and `Theorem49BoundaryRootSynthesis.rawSourceQuotientEquivPairingMapRange_apply_mk` computes its representative functional by the same selected-boundary erasure and chain-dot pairing. The corresponding separation surface is also ex-

plicit: `Theorem49BoundaryRootSynthesis.rawSourceQuotientClass_eq_zero_iff_projectedGenerator_orthogonal` tests raw-source quotient classes by vanishing against all projected generators, while `Theorem49BoundaryRootSynthesis.exists_projectedGenerator_chainDot_ne_zero_of_ne_zero_rawSourceQuotientClass` turns every nonzero raw-source quotient class into a nonzero projected-generator pairing witness. The finite representative form `Theorem49BoundaryRootSynthesis.exists_rawGeneratorSum_and_projectedGenerator_chainDot_ne_zero_of_ne_zero_rawSourceQuotientClass` now gives the same conclusion with an explicit finite raw Definition 4.8 generator-family representative in the original raw-source quotient class. Thus a downstream nonzero raw-source quotient class can be replaced by a finite raw generator sum without losing the projected-generator detector. The concrete target-level version is now explicit as well: `Theorem49BoundaryRootSynthesis.exists_rawGeneratorSum_and_projectedGenerator_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace` starts from a nonzero element of the boundary-zero Kirchhoff target, chooses a finite raw Definition 4.8 generator-family sum whose selected-boundary erasure is exactly that target, and simultaneously returns a projected generator with nonzero chain-dot pairing against the target. The sharpened non-vacuity form `Theorem49BoundaryRootSynthesis.exists_nonempty_rawGeneratorSum_and_nonzero_projectedGenerator_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace` adds that the finite raw generator-family sum is genuinely nonempty and that the projected detector is a nonzero chain. Thus a nonzero theorem-4.9 target cannot be witnessed by empty raw data or by a zero projected detector. The further support-sensitive form `Theorem49BoundaryRootSynthesis.exists_nonzero_rawGeneratorSum_and_nonempty_coeffSupport_and_nonzero_projectedGenerator_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace` shows that the raw sum itself is nonzero and that the coefficient support is nonempty, not merely that the chosen ambient finite term set is nonempty. The term-level extraction `Theorem49BoundaryRootSynthesis.exists_rawGenerator_with_nonzero_coeff_and_nonzero_projectedGenerator_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace` then chooses an actual raw Definition 4.8 generator-family chain from that support with nonzero coefficient, while preserving the same nonzero raw sum, selected-boundary erasure, and projected detector. The strengthened term-level cleanup `Theorem49BoundaryRootSynthesis.exists_nonzero_rawGenerator_with_nonzero_coeff_and_nonzero_projectedGenerator_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace` further proves that the extracted support chain satisfies $y \neq 0$. Thus the witness is a nonzero raw Definition 4.8 generator-family chain with a nonzero coefficient, not merely a formal support element attached to the same raw representative. The boundary-facing refinement `Theorem49BoundaryRootSynthesis.exists_rawGenerator_with_nonzero_boundaryZeroProjection_and_nonzero_coeff_and_nonzero_projectedGenerator_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace` then chooses such a support chain with nonzero selected-boundary erasure $\pi_\partial(y) \neq 0$. Hence the nonzero theorem-4.9 target is detected through a raw generator-family term that actually survives the boundary-zero projection, rather than only through cancellation among projected-zero support terms. The finite-dimensional consequences are now attached to this raw-source image model too: `Theorem49BoundaryRootSynthesis.finrank_rawSourceQuotient_eq_pairingMapRange` identifies the raw-source quotient dimension exactly with the pairing-image dimension, and `Theorem49BoundaryRootSynthesis.finrank_rawSourceQuotient_le_projectedGeneratorSubspace` then bounds that quotient dimension by the projected Definition 4.8 generator subspace through the pairing-image/purified-quotient rank calculation. `PlanarBoundaryClosedWalkSourceProjection.lean` now lowers this separation interface to the

route-facing source shells as well. Canonical witness choice, the local at-most-one source criterion, and same-boundary one-collar geometry, for both honest closed-walk sources and successor-cycle boundary-order sources, now yield

$$W_0(H) \cap (\text{projectedKempeClosureGeneratorSubspace})^\perp = 0$$

through both height-ordered and finite collar-layer witness data. Thus the same corrected projected-source interface carries not only projected spanning but also the nondegenerate chain-dot separation endpoint needed by the downstream theorem-4.9 route. The same source-level lowering now also exposes the pointwise detector and orthogonality-characterization forms: every nonzero concrete boundary-zero Kirchhoff chain is detected by some projected Definition 4.8 generator, and vanishing against all projected generators is equivalent to the chain itself being zero. The same source packages now also expose the raw Definition 4.8 preimage and finite raw-generator sum forms: every concrete boundary-zero Kirchhoff chain is the selected-boundary projection of a raw generator-span element, and can be presented as the projection of a finite linear combination of actual raw Definition 4.8 generators. The constructive non-vacuous source lane is now packaged at this same interface by `Theorem49BoundaryRootNonemptyProjectedSynthesis`: on one fixed embedding it records both `Theorem49BoundaryRootNonemptySynthesis` and the corrected projected Definition 4.8 subspace and family-span equalities. The package is constructed from height-ordered or finite collar-layer witness data with a nonempty selected-boundary-purified carrier, and the closed-walk and successor-cycle source wrappers now expose it for canonical witness choice and same-boundary one-collar geometry. No wrapper is added for the obstructed at-most-one/nonempty-carrier branch. `Theorem49PositiveGeometricSource.lean` records this candidate construction burden as explicit geometric source packages: `Theorem49ClosedWalkCanonicalPositiveGeometry`, `Theorem49ClosedWalkOneCollarPositiveGeometry`, `Theorem49SuccessorCycleCanonicalPositiveGeometry`, and `Theorem49SuccessorCycleOneCollarPositiveGeometry`. Each package carries the source data, the canonical witness choice or same-boundary one-collar geometry, and the nonempty purified carrier on one embedding, and each lowers to `Theorem49BoundaryRootNonemptyProjectedSynthesis` for any supplied Tait coloring. The fixed-embedding predicates also have direct same-embedding lowering lemmas, so a candidate construction can target a named embedding without first passing through a graph-level existential wrapper. `Theorem49PositiveGeometricSourceRegression.lean` now calibrates this target on the wheel-with-inner-triangle benchmark. The model still has the useful pair of facts, `wheelWithInnerTriangleTaitEdgeColoring_isTait` and `selectedBoundaryInteriorEdgeEndpointVertices_nonempty_wheelWithInnerTriangle`, but the fixed embedding fails all four package predicates: `not_theorem49ClosedWalkCanonicalPositiveGeometryOn_wheelWithInnerTriangle`, `not_theorem49ClosedWalkOneCollarPositiveGeometryOn_wheelWithInnerTriangle`, `not_theorem49SuccessorCycleCanonicalPositiveGeometryOn_wheelWithInnerTriangle`, and `not_theorem49SuccessorCycleOneCollarPositiveGeometryOn_wheelWithInnerTriangle`. The summary theorem `wheelWithInnerTriangle_tait_nonempty_carrier_without_positiveGeometricSourcePackageOn` therefore isolates the missing positive ingredient as genuine source-tied canonical-choice or one-collar geometry, not coloring or carrier nonemptiness. The same fixed benchmark now also calibrates the source-IDBRAD bridge: `wheelWithInnerTriangle_closedWalkSource_tait_hasUnblockedEndpoint_without_interiorDualBoundaryRootAdjDistancePeelData` bundles an honest closed-walk source, the explicit Tait coloring, and a local unblocked endpoint, while proving that generic `InteriorDualBoundaryRootAdjDistancePeelData` is impossible on that embedding. `Theorem49PositiveGeometricSourceImpossibility.lean` then gives the structural correction:

the four current package surfaces are not merely uninhabited by that benchmark, but inconsistent with the current purified nonempty carrier. The theorem `facewiseAtMostOneInteriorEdge_of_planarBoundaryCanonicalWitnessChoice` shows that canonical witness choice forces facewise at-most-one interior edge; `selectedBoundaryInteriorEdgeEndpointVertices_eq_empty_of_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice` combines this with the closed-walk at-most-one obstruction to empty the carrier. Since same-boundary one-collar geometry induces the corresponding canonical witness choice, the summary theorem `not_nonempty_currentTheorem49PositiveGeometricSourcePackages` refutes all four graph-level package structures. The next constructive target must therefore replace this package surface rather than instantiate it. The same file now also states this in the live endpoint vocabulary: `not_hasUnblockedInteriorEndpoint_of_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice` and `not_hasUnblockedInteriorEndpoint_of_closedWalkAnnulusBoundarySource_and_oneCollarGeometry_with_sourceBoundaryData` say directly that the old canonical-choice and source-preserving one-collar branches have no local unblocked endpoint. Equivalently, `not_nonempty_planarBoundaryCanonicalWitnessChoice_of_closedWalkAnnulusBoundarySource_and_hasUnblockedInteriorEndpoint` rules out canonical witness choice from an honest source plus `HasUnblockedInteriorEndpoint`, and `not_exists_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_of_closedWalkAnnulusBoundarySource_and_hasUnblockedInteriorEndpoint` rules out source-preserving one-collar geometry on the same boundary split. `PlanarBoundaryClosedWalkSourceCanonicalWitness.lean` now records the exact source-side criterion. The theorem `nonempty_planarBoundaryCanonicalWitnessChoice_iff_facewiseAtMostOneInteriorEdge_of_closedWalkAnnulusBoundarySource` says that, for an honest closed-walk annulus source, the source boundary split carries a canonical witness choice exactly when each face has at most one interior boundary edge. Its existential graph-level companion `exists_closedWalkAnnulusBoundarySource_and_planarBoundaryCanonicalWitnessChoice_iff_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge` is the corresponding source-search form. Thus the canonical-witness source attack has been reduced to the facewise at-most-one obligation itself; it is not a hidden selector or annulus-construction compatibility condition. `PlanarBoundaryClosedWalkSourceRoute.lean` now also lowers this lane to the exact split-annulus witness-ownership surface itself: `admitsPlanarBoundaryAnnulusWitnessAssignment_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_direct,` `admitsPlanarBoundaryAnnulusWitnessAssignment_of_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge_direct,` `admitsPlanarBoundaryAnnulusWitnessAssignment_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_canonicalWitnessChoice_direct,` and `admitsPlanarBoundaryAnnulusWitnessAssignment_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_facewiseAtMostOneInteriorEdge_direct.` So on the canonical-witness lane, no annulus-decomposition or witness-assignment packaging burden remains below the facewise at-most-one condition itself. `Theorem49AtMostOneNonemptyCarrierImpossibility.lean` now instantiates that exact live successor-cycle facewise-at-most-one shell on the explicit diamond benchmark via `exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_facewiseAtMostOneInteriorEdge_diamondWithTriangleGraph,` `diamondWithTriangleGraph_admitsPlanarBoundaryAnnulusWitnessAssignment_via_successorCycle_facewiseAtMostOneInteriorEdge,` and `exists_`

`theorem49BoundaryRootSynthesis_diamondWithTriangleGraph_for_explicitTaitColoring_via_successorCycle_facewiseAtMostOneInteriorEdge`. So the live successor-cycle at-most-one branch is genuinely inhabited and already reaches witness assignment and `theorem-4.9 synthesis` on a concrete graph; the remaining burden on that branch is the nonempty-carrier upgrade, not basic route reachability. `Theorem49AtMostOneNonemptyCarrierImpossibility.lean` now also packages the stronger fallback/count/boundary-rest hypothesis surface on that benchmark via `exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_atMostOneInteriorEdgePerFace_diamondWithTriangleGraph`, and then instantiates the first corrected projected Definition 4.8 endpoints as `exists_projectedKempeClosureGeneratorSubspace_eq_planarBoundaryZeroSubmodule_diamondWithTriangleGraph_for_explicitTaitColoring_via_successorCycle_atMostOneInteriorEdgePerFace` and `exists_span_projectedKempeClosureGeneratorFamily_eq_planarBoundaryZeroSubmodule_diamondWithTriangleGraph_for_explicitTaitColoring_via_successorCycle_atMostOneInteriorEdgePerFace`. So this same live successor-cycle at-most-one shell already reaches the corrected projected spanning conclusions on an explicit graph as well; the remaining burden on that branch is the nonempty-carrier positive upgrade, not `theorem-4.9` or projected-spanning reachability. The same benchmark file now pushes one algebraic layer deeper still: `exists_theorem49BoundaryZeroKirchhoffSubspace_inf_chainDot_orthogonal_projectedKempeClosureGeneratorSubspace_eq_bot_diamondWithTriangleGraph_for_explicitTaitColoring_via_successorCycle_atMostOneInteriorEdgePerFace` and `exists_theorem49BoundaryZeroKirchhoffSubspace_chainDot_projectedKempeClosureGeneratorSubspace_eq_zero_iff_diamondWithTriangleGraph_for_explicitTaitColoring_via_successorCycle_atMostOneInteriorEdgePerFace` show that the explicit live shell also reaches the first chain-dot duality endpoints on the natural purified carrier vertex set. So the remaining burden on that constructive branch is not algebraic projection/duality reachability either; it is the positive nonempty-carrier upgrade itself. The same file now also reaches the raw finite corrected Definition 4.8 representation there: `exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_of_mem_theorem49BoundaryZeroKirchhoffSubspace_diamondWithTriangleGraph_for_explicitTaitColoring_via_successorCycle_atMostOneInteriorEdgePerFace` packages finite Kempe-closure generator sums realizing every class in the corresponding boundary-zero Kirchhoff subspace on that same explicit live shell. So the remaining burden on this constructive branch is not raw Definition 4.8 realization either; it is still the positive nonempty-carrier upgrade itself. The same benchmark file now also reaches the projected-generator detector there: `exists_projectedKempeClosureGeneratorSubspace_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace_diamondWithTriangleGraph_for_explicitTaitColoring_via_successorCycle_atMostOneInteriorEdgePerFace` shows that every nonzero class in that same boundary-zero Kirchhoff subspace is detected by some projected Kempe generator on the explicit live shell. So the remaining burden on this constructive branch is not nontrivial projected-generator detection either; it is still the positive nonempty-carrier upgrade itself. `Theorem49PositiveGeometricSourceReplacement.lean` records that replacement explicitly. The fixed-embedding predicates `Theorem49HeightOrderedPositiveProjectedGeometryOn` and `Theorem49CollarLayerPositiveProjectedGeometryOn` require only the corresponding boundary-aware witness-cover data plus nonempty purified carrier on the same embedding, and they lower directly to `Theorem49BoundaryRootNonemptyProjectedSynthesis`. The graph-level structures `Theorem49HeightOrderedPositiveProjectedGeometry` and `Theorem49CollarLayerPositiveProjectedGeometry` package the same endpoint, and Lean proves the fixed-embedding equivalence `heightOrderedPositiveProjectedGeometryOn_iff_`

`collarLayerPositiveProjectedGeometryOn` by the existing height-to-collar and collar-to-height conversions. The file also carries the structural obstruction into compatibility lemmas showing that nonempty purified carrier forbids adding the old closed-walk canonical-choice or source-preserving one-collar fields on the same embedding. Thus the next positive task is construction of these direct witness-data packages from boundary-order geometry, not another wrapper around the refuted surfaces. The same file now factors the construction burden into graph-level bridge theorems: `theorem49CollarLayerPositiveProjectedGeometry_of_annulusCollarGeometry` and `theorem49HeightOrderedPositiveProjectedGeometry_of_annulusCollarGeometry` package `annulus collar geometry` plus a surviving purified carrier directly as the replacement graph packages, and the corresponding existential theorems do the same from an arbitrary witness of those hypotheses. Repaired previous-boundary witness geometry has matching specializations through `theorem49CollarLayerPositiveProjectedGeometry_of_annulusPreviousBoundaryWitnessGeometry` and `theorem49HeightOrderedPositiveProjectedGeometry_of_annulusPreviousBoundaryWitnessGeometry`. The wheel-with-inner-triangle regression now reaches these direct predicates themselves: `not_theorem49HeightOrderedPositiveProjectedGeometryOn_wheelWithInnerTriangle` and `not_theorem49CollarLayerPositiveProjectedGeometryOn_wheelWithInnerTriangle` show that the strict three-spoke height cycle blocks the fixed-embedding replacement packages despite the honest source, Tait coloring, and purified carrier. The source-level failed universal `not_forall_some_replacementPositiveProjectedGeometry_or_previousBoundaryWitness_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle` also rules out deriving either direct replacement package or repaired previous-boundary witness geometry from those hypotheses alone. The acyclic parent-witness surface is now connected at the same level: `theorem49HeightOrderedPositiveProjectedGeometryOn_of_wellFoundedFacePeelWitnessData_and_hasUnblockedInteriorEndpoint` uses the canonical parent-height rank to turn `PlanarBoundaryWellFoundedFacePeelWitnessData` plus `HasUnblockedInteriorEndpoint` into the height-ordered replacement predicate, and `theorem49CollarLayerPositiveProjectedGeometryOn_of_wellFoundedFacePeelWitnessData_and_hasUnblockedInteriorEndpoint` gives the finite-collar form by grouping those heights. The projected endpoint is exposed directly as `theorem49BoundaryRootNonemptyProjectedSynthesis_of_wellFoundedFacePeelWitnessData_and_hasUnblockedInteriorEndpoint`, with graph-level nonempty and existential versions for the same hypotheses. The repaired previous-boundary lane now also consumes the named local endpoint witness directly: `theorem49BoundaryRootNonemptyProjectedSynthesis_of_annulusPreviousBoundaryWitnessGeometry_and_hasUnblockedInteriorEndpoint` reaches the projected endpoint from previous-boundary witness geometry plus `HasUnblockedInteriorEndpoint`, and the fixed predicates have matching finite-collar and height-ordered constructors. The closed-walk route file adds the source-facing traceability theorem `theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySource_and_annulusPreviousBoundaryWitnessGeometry_and_hasUnblockedInteriorEndpoint`. It also adds the acyclic source-facing theorem `theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySource_and_wellFoundedFacePeelWitnessData_and_hasUnblockedInteriorEndpoint`. The same acyclic input is now produced directly from the root-distance interior-dual route: `PlanarBoundaryClosedWalkSourceRoute`. `lean` defines `planarBoundaryWellFoundedFacePeelWitnessData_of_`

`closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData`,
then `planarBoundaryHeightOrderedFacePeelWitnessData_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData` as the source-facing
height-ordered witness-cover object itself, with the graph-level theo-
rem `admitsPlanarBoundaryHeightOrderedFacePeelWitnessData_of_exists_`
`closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_`
`direct`, and `Theorem49PositiveGeometricSourceReplacementRoute`.
`lean` proves `theorem49BoundaryRootNonemptyProjectedSynthesis_of_`
`closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_`
`and_hasUnblockedInteriorEndpoint`. The successor-cycle source fields have
the parallel endpoint `theorem49BoundaryRootNonemptyProjectedSynthesis_`
`of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_`
`selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData_and_`
`hasUnblockedInteriorEndpoint`. The same source-facing root-distance bridge now reaches
the raw quotient/error endpoint itself: `theorem49BoundaryRawQuotientErrorPackage_of_`
`closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_`
`and_hasUnblockedInteriorEndpoint` and `theorem49BoundaryRawQuotientErrorPackage_`
`of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_`
`selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData_and_`
`hasUnblockedInteriorEndpoint` return the raw Definition 4.8 representative plus selected-
boundary-kernel decomposition for every Kirchhoff chain once a Tait coloring is supplied. Thus
generic boundary-root interior-dual adjacency-distance data now supplies the acyclic witness-cover
side of the positive projected and raw quotient routes directly. If one starts below this layer,
from raw closed-walk or successor-cycle boundary-order source data alone, the minimal missing
lemma is a same-embedding construction of `InteriorDualBoundaryRootAdjDistancePeelData`,
or an equivalent cycle-breaking well-founded parent witness. The file
`Theorem49PlanarBoundaryAnnulusRootInteriorDual.lean` now makes this source-side target ex-
plicit through `interiorDualBoundaryRootAdjDistancePeelDataOfClosedWalkAnnulusBoundarySourceCoverSe`
an honest closed-walk source plus cover/separated roots, boundary-free peeled faces,
canonical rooted shared-edge coverage of all interior edges, child-face membership
with strict root-distance increase, and the two-incidence bound constructs the generic
interior-dual boundary-root package on the same embedding. Its graph-level compan-
ion `admitsPlanarBoundaryInteriorDualBoundaryRootAdjDistancePeelData_of_exists_`
`closedWalkAnnulusBoundarySource_and_coverSeparatedAnnulusBoundaryRootsCanonicalRootedSharedEdge`
states the same obligation as an admissibility theorem. Thus the source-side attack has been local-
ized to producing these rooted-peel fields from the boundary-order source, rather than to a selector-
internal compatibility claim or an unnamed construction shell. A sharper source-root constructor,
`interiorDualBoundaryRootAdjDistancePeelDataOfClosedWalkAnnulusBoundarySourceBoundaryFaceRootsC`
now fixes the root set to the outer-or-inner boundary faces extracted from the closed-
walk source, fixes the fallback edge to the source's honest facial closed-walk fallback,
and uses the embedding's built-in two-incidence bound. Thus the remaining IDBRAD
work is exactly cover/separation of the source boundary-face roots, boundary-free peeled
faces, interior-edge coverage by rooted shared edges, and strict root-distance child mem-
bership. The Lean development now also exposes the parent-oriented sufficiency route for
the same fixed roots: `PlanarBoundaryClosedWalkAnnulusBoundarySource.wellFounded_`
`parentRel_boundaryFaceRoots_parentTowardsRoot` proves that cover/separated source
boundary-face roots induce a well-founded canonical shortest-path parent orientation, while
`interiorDualBoundaryRootParentPeelDataOfClosedWalkAnnulusBoundarySourceBoundaryFaceRootsCanoni`

packages the same source roots and fallback edge into `InteriorDualBoundaryRootParentPeelData` once rooted shared-edge coverage is supplied. The canonical parent child-membership field is now derived inside Lean from that cover and the ambient two-incidence bound by `canonicalParentChildMembership_of_canonicalParentCover`; the earlier face-membership-cover constructor remains available for callers that already have the field. The graph-level endpoint `admitsPlanarBoundaryInteriorDualBoundaryRootParentPeelData_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover` records this raw-cover fixed-source acyclic-parent sufficiency theorem; the older graph-level `admitsPlanarBoundaryInteriorDualBoundaryRootParentPeelData_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover` remains as the compatibility theorem for callers that already supply the derived child-membership field. The actual successor-cycle boundary-order shell now has the matching lowering package `admitsPlanarBoundaryInteriorDualBoundaryRootParentPeelData_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover`, `admitsPlanarBoundaryInteriorDualBoundaryRootParentPeelData_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover`, and the direct graph-level endpoints `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_direct` and `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_direct`. All four lower through `PlanarBoundaryClosedWalkAnnulusBoundarySource.ofDartSuccessorCycleFields`, so once the live successor-cycle shell carries this concrete source-fixed canonical-parent cover package, no further projection step remains before full Theorem 4.9 synthesis. There is now also a proved base case for this sufficiency branch: `interiorDualBoundaryRootParentPeelDataOfClosedWalkAnnulusBoundarySourceBoundaryFaceRootsNoInteriorEdges` and `admitsPlanarBoundaryInteriorDualBoundaryRootParentPeelData_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsNoInteriorEdges` construct the same source-fixed parent package with empty peel set when the source boundary-face roots cover/separate and the interior-edge support is empty. Thus the nondegenerate parent-source task is specifically to produce or supply the pairwise-unique shared-edge selector input, cover/separated source roots, boundary-free peeled faces, and a rooted shared-edge cover of actual interior edges. It is no longer a separate canonical parent child-membership problem. Lean also derives the otherwise-required pairwise shared-edge uniqueness in this degenerate case via `pairwiseUniqueSharedInteriorEdges_of_interiorEdgeSupport_eq_empty`, so no extra uniqueness premise remains hidden in the base case. This raw parent-cover contract is now known to be degenerate at the honest closed-walk source layer. The walk source theorem `PlaneEmbeddingWithBoundary.FaceBoundaryClosedWalk.three_le_faceBoundary_card` shows that every honest nonempty simple facial closed walk has at least three boundary edges, while the height-terminal lemma for `PlanarBoundaryHeightOrderedFacePeelWitnessData` forces any nonempty raw-cover parent package to contain a terminal peeled face whose non-witness boundary edges are already selected-boundary edges. Since the raw contract also requires peeled faces to be disjoint from selected-boundary support, that terminal face would have boundary cardinality at most one. Lean packages the contradiction as `false_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover`

`of_interiorEdgeSupport_nonempty` and the route-facing degeneracy statement as `interiorEdgeSupport_eq_empty_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParent`. Thus the current honest closed-walk source-fixed raw-cover contract has only the no-interior-edge branch; any future nondegenerate sufficiency theorem must change the source-side contract or replace this terminal parent condition. This degeneracy is now also enough to close the downstream theorem-4.9 route on that shell itself. `Theorem49ResidualBoundaryPeeling.lean` proves `theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_via_residualBoundary` and `theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_via_residualBoundary`. So the source-fixed raw canonical-parent shared-edge-cover shell, and its successor-cycle boundary-order form, already reach the full Theorem 4.9 synthesis endpoint on the same embedding before any additional endpoint witness is added. The same cardinality obstruction is now stated at the IDBRAD layer itself: `InteriorDualBoundaryRootAdjDistancePeelData.false_of_mem_interiorEdgeSupport_of_faceBoundaryClosedWalkGeometry` combines honest facial closed walks with the terminal-leaf theorem for `InteriorDualBoundaryRootAdjDistancePeelData`. A covered interior edge would force a peeled face of boundary-cardinality at most one, contradicting the three-edge lower bound for every honest nonempty simple facial closed walk. The source-facing theorem `not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_of_interiorEdgeSupport_nonempty_of_faceBoundaryClosedWalkGeometry` therefore rules out any current boundary-free adjacent-distance/root-parent package on an honest closed-walk embedding with genuine interior support. This identifies the interface itself, not only the source-fixed raw-cover constructor, as the part that must be revised for a nondegenerate route. The nonempty branch now has a constructor-side sanity theorem. The generic lemma `exists_nonroot_peelFace_parent_rootedSharedInteriorEdge_eq_of_canonicalParentCover_of_mem_interiorEdgeSupport` proves that an actual interior edge covered by the canonical rooted shared-edge image is covered by a peeled non-root face with a real shortest-path parent, and that the rooted shared edge on that face is exactly the original interior edge. Its nonempty-support corollary `exists_nonroot_peelFace_parent_rootedSharedInteriorEdge_eq_of_canonicalParentCover_of_interiorEdgeSupport_nonempty` makes the nondegenerate forest branch explicit. The source-fixed specializations `exists_nonroot_peelFace_parent_rootedSharedInteriorEdge_eq_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_of_mem_interiorEdgeSupport` and `exists_nonroot_peelFace_parent_rootedSharedInteriorEdge_eq_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParent` of `of_interiorEdgeSupport_nonempty` instantiate this witness using the closed-walk source's boundary-face roots and fallback edge. Thus a nonempty canonical-parent cover cannot be witnessed by the fallback branch. The same constructor layer now also proves parent-side membership of canonical shared edges and `canonicalParentChildMembership_of_canonicalParentCover`: every non-witness edge on a peeled face is either selected boundary support or lies on a peeled child whose canonical parent is the original face. Hence the source-fixed parent constructor no longer takes parent child-membership as a raw source hypothesis; the remaining source-side burden is the pairwise-unique shared-edge selector input plus the root cover/separation, boundary-free peeled-face, and rooted shared-edge cover hypotheses from boundary-order geometry. The base case is now exercised on a concrete source rather than only exposed as an abstract constructor. The new positive regression `Theorem49PlanarBoundaryAnnulusRootInteriorDualPositiveRegression.lean` defines the disjoint two-triangle source `twoTriangleClosedWalkAnnulusBoundarySource`, proves

`twoTriangleAnnulusBoundaryFaceRoots_eq_univ`, proves root cover and root separation because the interior dual is $\perp$, and proves `twoTriangleAnnulus_interiorEdgeSupport_eq_empty`. It then constructs `twoTriangleClosedWalkSourceInteriorDualBoundaryRootParentPeelData` and the graph-level `admitsPlanarBoundaryInteriorDualBoundaryRootParentPeelData_twoTriangleAnnulusGraph`. Thus the empty-support sufficiency branch has both a generic theorem and a concrete same-source instantiation. Later the exact-v23 positive regression adds an explicit Tait coloring on the same graph and proves `theorem49BoundaryRootSynthesis_twoTriangleAnnulusEmbedding_for_explicitTaitColoring_via_noInteriorParentRoute`, so this no-interior parent branch is not merely inhabited: on that benchmark it already reaches full `Theorem49BoundaryRootSynthesis`. More sharply, that same benchmark now also proves `exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_without_planarBoundaryAnnulusConstructionPositiveFrontierData_twoTriangleAnnulusGraph` and the failed universal `not_forall_nonempty_planarBoundaryAnnulusConstructionPositiveFrontierData_of_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_twoTriangleAnnulusGraph`, so even the final theorem-4.9 endpoint under a real Tait coloring does not force the positive-frontier annulus-construction shell there. More sharply, `SomePositiveStageConstructionShell`, `NoPositiveStageConstructionShells`, `not_somePositiveStageConstructionShell_of_noPositiveStageConstructionShells`, and `noPositiveStageConstructionShells_of_interiorEdgeSupport_eq_empty` in `PlanarBoundaryAnnulusConstructionCore.lean`, as well as `twoTriangle_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_noPositiveStageConstructionShells`, and `exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_and_noPositiveStageConstructionShells_twoTriangleAnnulusGraph` show that on this actual successor-cycle benchmark full theorem-4.9 synthesis already coexists with simultaneous failure of all four currently formalized positive-stage construction shells, not only the last positive-frontier one. The generic lifting now lives in `Theorem49Synthesis.lean` as `not_forall_nonempty_of_exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_without`, `not_forall_somePositiveStageConstructionShell_of_exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_and_noPositiveStageConstructionShells`, `not_forall_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData_of_exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_without_planarBoundaryAnnulusConstructionBoundarySupportFaceData`, `not_forall_nonempty_planarBoundaryAnnulusConstructionFacePartitionData_of_exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_without_planarBoundaryAnnulusConstructionFacePartitionData`, `not_forall_nonempty_planarBoundaryAnnulusConstructionFaceLayerData_of_exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_without_planarBoundaryAnnulusConstructionFaceLayerData`, and `not_forall_nonempty_planarBoundaryAnnulusConstructionPositiveFrontierData_of_exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_without_planarBoundaryAnnulusConstructionPositiveFrontierData`, so any same-embedding synthesis witness of that shape already refutes a universal derivation of the corresponding positive-stage construction package. The exact two-triangle benchmark now instantiates that pattern not only for positive-frontier data but also for boundary-support face data, face-partition data, face-layer data, and the aggregated `SomePositiveStageConstructionShell` surface. It now also pushes that same failed-converse diagnosis back up to the exact-v23 source shells them-

selves: twoTriangle_closedWalkSource_tait_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_noPositiveStageConstructionShells, exists_embedding_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_without_somePositiveStageConstructionShell_twoTriangleAnnulusGraph, and not_forall_somePositiveStageConstructionShell_of_closedWalkAnnulusBoundarySource_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_twoTriangleAnnulusGraph do this on the honest closed-walk shell, while exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_without_somePositiveStageConstructionShell_twoTriangleAnnulusGraph and not_forall_somePositiveStageConstructionShell_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_twoTriangleAnnulusGraph do the same on the live successor-cycle selected-arc shell. The generic exact-v23 route-side converses now live in Theorem49ResidualBoundaryRoute. lean as not_forall_somePositiveStageConstructionShell_of_exists_embedding_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_noPositiveStageConstructionShells and not_forall_somePositiveStageConstructionShell_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_noPositiveStageConstructionShells, and the two-triangle source-shell failures simply instantiate those shell-specific helpers from the stronger witness package NoPositiveStageConstructionShells. This positive source now also reaches the selector-valued construction surface: boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData extracts a BoundaryFreeIncidentFaceSelector from any parent payload while preserving the actual parent cover: boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData_faceOf_mem_peelFaces says each selected face lies in the parent package's peelFaces, and rootedSharedInteriorEdge_boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData_faceOf_eq says its rooted shared edge is the queried interior edge. At the finite construction layer, boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData_peelFaces_subset shows that the selector image introduces no new peeled faces beyond the parent package. The source-fixed empty-support specialization boundaryFreeIncidentFaceSelectorOfClosedWalkAnnulusBoundarySourceBoundaryFaceRootsNoInteriorEdges packages the same closed-walk hypotheses, and twoTriangleClosedWalkSourceBoundaryFreeIncidentFaceSelector instantiates that selector on the two-triangle source. Thus the base case produces the actual boundary-free selector object consumed by the replacement-construction lane, not only an inhabited parent package. The parent-to-selector bridge now also supplies the selector strict-descent condition itself. The extracted selector is injective on interior edges by boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData_injective; its inverted selected witness is the canonical rooted shared edge by boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData_selectedWitnessEdge_eq; and

`boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData_`
`strictDescent` turns parent child-membership into the selector hrest condition us-
 ing root distance. Therefore `theorem49BoundaryRootNonemptyProjectedSynthesis_of_`
`closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_`
`via_boundaryFreeSelector` routes the same source-fixed parent hypotheses through
 the selector-to-annulus construction surface directly. This selector route now also
 has downstream route-facing forms: `theorem49BoundaryRawQuotientErrorPackage_of_`
`closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_`
`via_boundaryFreeSelector` exposes the raw Definition 4.8 quotient/error pack-
 age, and `exists_theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_`
`closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_`
`and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector` packages the
 graph-level source hypotheses with the local `HasUnblockedInteriorEndpoint` wit-
 ness directly. But under the current honest-source semantics that stronger
 endpoint-consuming shell is also vacuous: `not_hasUnblockedInteriorEndpoint_of_`
`closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover,`
`not_hasUnblockedInteriorEndpoint_of_boundaryReachabilityData_`
`and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_`
`boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover,` and
 the matching graph-level no-coexistence theorems `not_exists_embedding_`
`closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_`
`and_hasUnblockedInteriorEndpoint` and `not_exists_embedding_`
`boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_`
`selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_`
`and_hasUnblockedInteriorEndpoint` show that the stricter child-membership shell can-
 not actually coexist with the named local endpoint witness on the same honest source
 surfaces. So these selector/projected face-membership wrappers are presently route-
 diagnostic, not inhabited. More sharply, the generic converse theorems `not_forall_some_`
`closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_`
`of_exists_embedding_closedWalkAnnulusBoundarySource_and_hasUnblockedInteriorEndpoint`
`and_not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_`
`of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_`
`and_selectedBoundaryArc_and_hasUnblockedInteriorEndpoint,` together
 with the seed-free shared-interior-pair instances `not_forall_some_`
`closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_`
`of_closedWalkAnnulusBoundarySource_and_hasUnblockedInteriorEndpoint_`
`sharedInteriorPair` and `not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_`
`of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_`
`selectedBoundaryArc_and_hasUnblockedInteriorEndpoint_sharedInteriorPair,`
 show that no Tait coloring or exact v23 residual seed is needed to refute a uni-
 versal derivation of that stronger shell. The same shared-interior-pair bench-
 mark now also bundles the whole local selector/canonical-parent burden at once:
`not_forall_some_selectorOrCanonicalParentOrKnownSameEmbeddingGeometry_`
`of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_`
`hasUnblockedInteriorEndpoint_sharedInteriorPair` and `not_forall_`
`some_selectorOrCanonicalParentOrKnownSameEmbeddingGeometry_of_`
`boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_`
`selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_`

`sharedInteriorPair` show that honest source data or the live successor-cycle selected-arc shell, even after adding a Tait coloring, still do not universally force any of the local `BoundaryFreeIncidentFaceSelector`, raw canonical-parent shared-edge cover, stricter canonical-parent face-membership cover, or the previously known same-embedding residual, height-ordered, collar-layer, attached-certificate, well-founded, or annulus-collar geometry surfaces from `HasUnblockedInteriorEndpoint` alone. So this endpoint-only obstruction now blocks the whole local selector/parent ladder, not just one stronger shell at a time. And this is no longer only a benchmark-facing statement:

`not_forall_some_selectorOrCanonicalParentOrKnownSameEmbeddingGeometry_of_exists_embedding_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint` and `not_forall_some_selectorOrCanonicalParentOrKnownSameEmbeddingGeometry_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint` lift that whole bundled failure pattern to reusable witness-based converses, so any same-embedding example with honest-source or live successor-cycle data, a Tait coloring, and `HasUnblockedInteriorEndpoint` but without the local selector/canonical-parent/known-geometry package already refutes a universal derivation of that whole burden. The selector lane now also closes directly at full theorem 4.9 on its own construction surface:

`theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusConstruction_and_hasUnblockedInteriorEndpoint` and `BoundaryFreeIncidentFaceSelector`. `theorem49BoundaryRootSynthesis_of_strictDescent_and_hasUnblockedInteriorEndpoint` show that once an injective strict-descent selector and a local endpoint witness exist on an embedding, no further projected-synthesis packaging remains below the selector-built annulus construction. The actual successor-cycle boundary-order shell now has the matching raw-cover selector route too: `theorem49BoundaryRootNonemptyProjectedSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_via_boundaryFreeSelector`, `theorem49BoundaryRawQuotientErrorPackage_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_via_boundaryFreeSelector`, and `exists_theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector`. So once the live successor-cycle shell carries the raw canonical-parent shared-edge cover together with `HasUnblockedInteriorEndpoint`, it already reaches the selector-built annulus-construction lane directly, without first detouring through residual-boundary packaging; the remaining source-facing selector burden is only to derive that injective strict-descent selector package itself on the live shell. The same raw-cover selector lane now also exposes the route-facing terminal selected-face break itself: `exists_terminal_face_boundary_remainders_and_disjoint_of_interiorDualBoundaryRootParentPeelData_via_boundaryFreeSelector` lifts the generic selector terminal-face theorem to any canonical-parent package, and the source/raw/live-shell specializations `exists_terminal_face_boundary_remainders_and_disjoint_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_via_boundaryFreeSelector`, `exists_terminal_face_boundary_remainders_and_disjoint_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_via_boundaryFreeSelector` and `exists_terminal_face_boundary_remainders_and_disjoint_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and`

`selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_via_boundaryFreeSelector`, together with `exists_terminal_face_boundary_remainders_and_disjoint_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_via_boundaryFreeSelector` show that both the stricter canonical-parent face-membership shell and the raw canonical-parent cover shell, plus `HasUnblockedInteriorEndpoint`, already yield a peeled face whose non-witness remainders are ambient-boundary edges while its full face boundary remains disjoint from `selectedBoundarySupport`. Thus the live selector compatibility lane now reaches an explicit same-shell terminal boundary-break witness before any residual-boundary or collar-layer repackaging, even before discarding the child-membership refinement. This parent-oriented packet now composes all the way to the nonempty projected Theorem 4.9 endpoint: `theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_direct` is the fixed-embedding source-to-synthesis theorem, and `exists_theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_with_nonempty_selectedBoundaryInteriorEdgeEndpointVertices_direct` is its graph-level existential form. The wheel-with-inner-triangle regression now instantiates this source-fixed parent bridge on a concrete graph: `wheelWithInnerTriangle_theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkSourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover` specializes the theorem to the wheel embedding and its explicit Tait coloring. The companion theorem `wheelWithInnerTriangle_source_to_theorem49_and_height_cycle_contradiction_of_closedWalkSourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover` shows that the same source-fixed parent hypotheses lower to height-ordered witness-cover data and contradict the known wheel height-cycle obstruction, so `not_exists_wheelWithInnerTriangle_closedWalkSourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_and_theorem49BoundaryRootNonemptyProjectedSynthesis` blocks this projected endpoint on the wheel. This is the concrete soundness check for the new parent route: the wheel's honest source, Tait coloring, and unblocked endpoint alone do not make the parent bridge fire. The same development now separates the first of these obligations: the theorem `PlanarBoundaryClosedWalkAnnulusBoundarySource.rootSetCovers_boundaryFaceRoots_of_interiorDualBoundaryRootAdjDistancePeelData` shows that any generic boundary-root IDBRAD datum already makes the source boundary-face root set cover the interior-dual components. By contrast, `PlanarBoundaryClosedWalkAnnulusBoundarySource.not_rootSetSeparated_boundaryFaceRoots_of_reachable_outerBoundaryFace_innerBoundaryFace` shows that separation of all source boundary faces is refuted by any interior-dual reachability from an outer boundary face to an inner boundary face. This has now been lifted to the generic IDBRAD layer: `PlanarBoundaryClosedWalkAnnulusBoundarySource.not_reachable_outerBoundaryFace_innerBoundaryFace_of_interiorDualBoundaryRootAdjDistancePeelData` shows that any IDBRAD datum on the closed-walk source forces the extracted outer and inner boundary-face layers to be interior-dual reachability-separated, and `PlanarBoundaryClosedWalkAnnulusBoundarySource.not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_of_reachable_outerBoundaryFace_innerBoundaryFace` records the corresponding failed-converse form. The route-facing wrappers `not_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_and_reachable_outerBoundaryFace_innerBoundaryFace` and `not_forall_interiorDualBoundaryRootAdjDistancePeelData`

`of_exists_closedWalkAnnulusBoundarySource_and_reachable_outerBoundaryFace_innerBoundaryFace` package the same obstruction as a same-embedding no-coexistence theorem and as a failed universal derivation from closed-walk source data. This obstruction now already kills the source boundary-face separation premise itself on the live successor-cycle shell: `not_rootSetSeparated_boundaryFaceRoots_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_reachable_outerBoundaryFace_innerBoundaryFace` and `not_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_rootSetSeparated_boundaryFaceRoots_and_reachable_outerBoundaryFace_innerBoundaryFace` show that outer-to-inner interior-dual reachability blocks the source `hsepRoots` hypothesis before any rooted shared-edge cover or generic IDBRAD payload is added. This obstruction now also appears directly on the live successor-cycle shell: `not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_reachable_outerBoundaryFace_innerBoundaryFace` and `not_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData_and_reachable_outerBoundaryFace_innerBoundaryFace` show that if the source extracted from actual boundary reachability and successor-cycle data still has an interior-dual path from one outer boundary face to one inner boundary face, then the route already fails before the source-fixed rooted-cover IDBRAD bridge or any theorem-4.9 packaging is used. The concrete two-band triangular annulus benchmark now realizes this live successor-cycle failure on an actual dart-successor shell: `twoBandAnnulusDartSuccessorCycleGeometry` and `twoBandAnnulusDartSuccessorCycleGeometry_selectedBoundaryArcOnFace` instantiate the live surface, `twoBandDartSuccessorClosedWalkAnnulusBoundarySource_boundaryData_eq` and `twoBandDartSuccessorBoundaryFaceRoots_eq_univ` show that this live shell extracts the same annulus boundary split and the same full six-face boundary-root set as the closed-walk source, `rootSetCovers_twoBandDartSuccessorBoundaryFaceRoots` makes the coverage half explicit there too, and `not_rootSetSeparated_twoBandAnnulusBoundaryFaceRoots_via_dartSuccessor_source_reachability` and `not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_twoBandAnnulus_via_dartSuccessor_source_reachability` show that the same outer-to-inner interior-dual adjacency already kills only the source `hsepRoots` premise, and hence generic IDBRAD data, before any rooted-cover or theorem-4.9 packaging is introduced. The obstruction is now formalized one premise deeper on the live source-fixed rooted-cover shell as well: `false_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover_and_reachable_outerBoundaryFace_innerBoundaryFace` and `not_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover_and_reachable_outerBoundaryFace_innerBoundaryFace` show that if actual successor-cycle data still extract an outer boundary face reaching an inner boundary face, then no rooted shared-edge face-membership package can exist there at all. The same two-band benchmark now instantiates that sharper failure directly: `not_exists_twoBandAnnulus_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover_via_dartSuccessor_source_reachability` rules out every boundary-free peeled-face set, rooted shared-edge cover, and strict child-membership witness package on the actual successor-cycle shell. This obstruction is now instantiated on a concrete honest source in `Theorem49PlanarBoundaryAnnulusRootInteriorDualObstructionRegression.lean`. The

two-band triangular annulus `twoBandAnnulusGraph` has six quadrilateral faces, disjoint selected outer and inner boundary cycles, and the closed-walk source package `twoBandClosedWalkAnnulusBoundarySource`; nevertheless `twoBandAnnulus_interiorDual_adj_outer0_inner0` gives an interior-dual adjacency between an extracted outer boundary face and an extracted inner boundary face. The theorem `not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_twoBandAnnulus_via_source_reachability` therefore proves the concrete nonexistence of generic IDBRAD data on this embedding. The same concrete source now pins down the fixed-root constructor premise: `twoBandAnnulusBoundaryFaceRoots_eq_univ` proves that the source-extracted `boundaryFaceRoots` are all six faces, `rootSetCovers_twoBandAnnulusBoundaryFaceRoots` proves the coverage premise, and `not_rootSetSeparated_twoBandAnnulusBoundaryFaceRoots` refutes the separation premise using the shared middle edge `tbaM01`. The same concrete source also blocks the primitive boundary-free peel-cover premise: `twoBandAnnulus_peelFaces_eq_empty_of_boundaryFree` proves that every boundary-free peel-face set is empty, because every face touches selected boundary support, and `not_interiorEdgeSupport_subset_boundaryFreePeelImage_twoBandAnnulus` then uses the interior edge `tbaM01` to refute any interior-edge-support cover by an image of such faces. This local obstruction is now connected to the shared forcing-edge API by `twoBandAnnulusForcingInteriorEdgeWitness`, `not_nonempty_boundaryFreeIncidentFaceSelector_twoBandAnnulus`, and `not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_twoBandAnnulus_via_forcingInteriorEdgeWitness`. The forcing obstruction now also applies to the source-fixed parent route itself. Lean proves `everyInteriorEdgeHasBoundaryFreeIncidentFace_of_interiorDualBoundaryRootParentPeelData`, so any canonical-parent boundary-root package supplies a boundary-free incident face for every interior edge. Consequently the same two-band forcing witness proves `not_nonempty_interiorDualBoundaryRootParentPeelData_twoBandAnnulus_via_forcingInteriorEdgeWitness` and `not_nonempty_planarBoundaryAnnulusRootParentPeelData_twoBandAnnulus_via_forcingInteriorEdgeWitness`. The same forcing witness now blocks the live theorem-4.9 raw canonical-parent cover shell one layer earlier: `false_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_and_forcingInteriorEdgeWitness` and `not_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_and_forcingInteriorEdgeWitness` show that once a successor-cycle shell carries a forcing interior edge, no raw canonical-parent shared-edge cover package can exist there at all. The two-band benchmark now instantiates that sharper failure directly as `not_exists_twoBandAnnulus_boundaryFaceRootsCanonicalParentSharedEdgeCover_via_forcingInteriorEdgeWitness`. The same benchmark also kills the older raw carrier repair on that exact shell: `not_exists_twoBandAnnulus_boundaryFaceRootsCanonicalParentSharedEdgeCover_and_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport` shows that even endpoint-support separation plus a nonempty raw interior-edge endpoint carrier cannot rescue the live parent-cover lane on the actual successor-cycle benchmark. So the remaining source-side IDBRAD burden is a genuine separation principle plus a nontrivial boundary-free peel source, and the parent-oriented variant has the same no-forcing obligation; neither follows from the current closed-walk source package alone. The generic cycle obstruction now makes this requirement checkable: a source-level derivation must show that each apparent remainder-witness cycle is broken by selected boundary support, by witness identification, or by an actual well-founded

orientation. The list-level theorem `false_of_isChain_height_lt_and_getLast_lt_head` supplies the generic order obstruction used for arbitrary finite cycles. Once that datum is present, the remaining non-vacuity burden is the local unblocked endpoint. The direct well-founded lane now accepts the older carrier-side hypotheses without first repackaging them as that endpoint witness: `theorem49BoundaryRootNonemptyProjectedSynthesis_of_wellFoundedFacePeelWitnessData_and_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport` uses explicit `PlanarBoundaryWellFoundedFacePeelWitnessData`, endpoint-support separation, and nonempty raw `interiorEdgeEndpointSupport` to reach the projected synthesis endpoint, while the named well-founded route packages and their honest-source wrappers now also expose full theorem-4.9 through `Theorem49ClosedWalkAnnulusWellFoundedPositiveProjectedGeometry`. `boundaryRootSynthesis`, `Theorem49SuccessorCycleAnnulusWellFoundedPositiveProjectedGeometry`. `boundaryRootSynthesis`, and `theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySource_and_wellFoundedFacePeelWitnessData_and_hasUnblockedInteriorEndpoint`. The route-facing wrappers `theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySource_and_wellFoundedFacePeelWitnessData_and_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport` and `theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_wellFoundedFacePeelWitnessData_and_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport` expose the same bridge on the honest closed-walk and successor-cycle source surfaces. Thus bypassing the blocked interior-dual derivation by supplying a same-embedding well-founded witness-cover datum no longer leaves a separate endpoint- or theorem-4.9-packaging obligation; the remaining hard part is still to obtain that acyclic datum from genuine boundary-order geometry. `Theorem49PositiveGeometricSourceReplacementRoute.lean` now discharges that local endpoint witness at the same root-distance source layer from two older geometric surfaces: global endpoint-support separation plus raw `interiorEdgeEndpointSupport` nonemptiness, and peeled-face endpoint no-touch on the root-distance annulus construction plus the same raw nonemptiness. The closed-walk theorem names are `theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_and_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport` and `theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_of_rootDistance_peelFaces_endpoint_disjoint_selectedBoundarySupport`; the successor-cycle source fields have parallel projected-synthesis wrappers. The carrier burden has now been lowered one step further: `interiorEdgeEndpointSupport_nonempty_of_interiorEdgeSupport_nonempty` derives the raw endpoint carrier from a nonempty interior edge, and the closed-walk root-distance bridge exposes corresponding projected-synthesis variants with nonempty `interiorEdgeSupport` in place of pre-expanded endpoint support. `Theorem49PositiveGeometricSourceReplacementRoute.lean` then reconnects this direct replacement surface to the honest closed-walk annulus source shell and to the route-facing successor-cycle source shell. The closed-walk package `Theorem49ClosedWalkAnnulusCollarPositiveProjectedGeometry` carries an honest source, same-embedding annulus collar geometry, and the nonempty purified carrier, and it lowers to both replacement graph packages while now also exposing graph-level full theorem-4.9 synthesis directly, so the nonempty projected synthesis endpoint is no longer the top of that source-facing surface. The successor-cycle package `Theorem49SuccessorCycleAnnulusCollarPositiveProjectedGeometry` carries boundary reachability, successor-cycle facial data, selected-boundary arcs, same-embedding

annulus collar geometry, and the nonempty purified carrier. It lowers to the closed-walk route package for traceability and to both replacement graph packages for the constructive route, and it inherits the same graph-level full-synthesis closure. The current positive route is therefore reduced to producing this successor-cycle annulus-geometry-plus-carrier input, or its direct annulus-collar analogue, without passing back through the refuted canonical-choice or same-boundary one-collar surfaces. The route-facing file now also supplies direct closed-walk and successor-cycle full-synthesis constructors from `HasUnblockedInteriorEndpoint`, together with the matching graph-level existential closures from either a surviving carrier or one local endpoint witness, namely `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_hasUnblockedInteriorEndpoint_direct` and `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_annulusCollarGeometry_and_hasUnblockedInteriorEndpoint_direct`. Thus source-facing examples can provide the carrier side through a local endpoint witness while still keeping the annulus-collar geometry explicit. The older endpoint-separation route now feeds the same local witness without a separate purification step: `hasUnblockedInteriorEndpoint_of_endpointSupport_disjoint_and_nonempty` turns raw `interiorEdgeEndpointSupport` nonemptiness plus disjointness from the selected-boundary endpoint support into `HasUnblockedInteriorEndpoint`. The replacement and route files now expose the corresponding direct full-synthesis constructors, including `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport_direct` and `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_annulusCollarGeometry_and_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport_direct`. The same annulus-collar source lane now has the matching full-synthesis closures from canonical peeled-face endpoint no-touch as well. The route-facing package layer has matching constructors as well: `theorem49ClosedWalkAnnulusCollarPositiveProjectedGeometryOn_of_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport` and `theorem49SuccessorCycleAnnulusCollarPositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_annulusCollarGeometry_and_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport` inhabit the closed-walk and successor-cycle annulus-collar replacement packages before any Tait coloring is supplied. `Theorem49PositiveGeometricSourceReplacementRegression.lean` calibrates that sharpened target on the shared-interior-pair annulus. That model carries boundary reachability, successor-cycle selected arcs, a Tait coloring, and a nonempty purified carrier, but Lean proves `not_theorem49HeightOrderedPositiveProjectedGeometryOn_sharedInteriorPair` and `not_theorem49CollarLayerPositiveProjectedGeometryOn_sharedInteriorPair`. Consequently the theorem `not_forall_replacementPositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_sharedInteriorPair` rules out deriving the direct replacement packages merely from those boundary order, coloring, and carrier facts. The same regression now also proves `not_theorem49ClosedWalkAnnulusCollarPositiveProjectedGeometryOn_sharedInteriorPair`

and `not_theorem49SuccessorCycleAnnulusCollarPositiveProjectedGeometryOn_sharedInteriorPair`, with failed universal route-facing forms for the closed-walk and successor-cycle annulus-collar packages. Thus even the sharpened source-facing packages require genuine annulus-collar or witness-cover geometry, not just a repacking of the available boundary-order, coloring, and carrier facts. `Theorem49PositiveGeometricSourceConstruction.lean` now supplies the first positive inhabitant of the replacement surface. It reuses the two-collar annulus geometry benchmark and proves `counterEmbedding_selectedBoundaryInteriorEdgeEndpointVertices_nonempty`: the intermediate interior edge contributes endpoints which are not incident to the selected ambient boundary and therefore survive purification. With the new exact support lemmas `counterEmbedding_interiorEdgeSupport_eq`, `counterEmbedding_interiorEdgeEndpointSupport_eq`, `counterEmbedding_selectedBoundaryEndpointSupport_eq`, and `counterEmbedding_endpointSupport_disjoint_selectedBoundarySupport`, Lean now records that this survival is precisely endpoint-support disjointness for the shared intermediate edge. The same calculation gives the named local witness `counterEmbedding_hasUnblockedInteriorEndpoint`, matching the first-order carrier obligation used by the replacement route. With the existing annulus-collar conversions, Lean then constructs both `counterEmbedding_collarLayoutPositiveProjectedGeometryOn` and `counterEmbedding_heightOrderedPositiveProjectedGeometryOn`, together with the graph-level packages `counterGraph_collarLayoutPositiveProjectedGeometry` and `counterGraph_heightOrderedPositiveProjectedGeometry`. The same benchmark now carries an explicit constant nonzero Tait coloring `counterGraphTaitEdgeColoring`, and Lean proves `counterEmbedding_boundaryRootNonemptyProjectedSynthesis`; the concrete two-collar geometry therefore reaches the nonempty projected theorem-4.9 endpoint, not merely the precursor package. Lean now also records the same positive benchmark at the raw quotient/error endpoint: `counterEmbedding_collarLayoutGeometry_explicitTait_nonemptyCarrier_to_theorem49RawQuotientErrorPackage` bundles the actual collar geometry, explicit Tait coloring, nonempty purified carrier, nonempty projected synthesis package, and the raw Definition 4.8 representative plus selected-boundary-kernel error decomposition for every Kirchhoff chain on this embedding. The same module now also records the source-side limitation of this benchmark: `counterGraph_isAcyclic` proves that its graph is a three-edge matching, `not_nonempty_closedWalkAnnulusBoundarySource_counterEmbedding` blocks an honest closed-walk annulus source on the fixed embedding, and `counterEmbedding_rawQuotientErrorPackage_without_closedWalkAnnulusBoundarySource` packages the raw quotient/error endpoint together with that obstruction. The same benchmark now also records the repaired-witness calibration `counterEmbedding_previousBoundaryWitness_nonemptyProjectedSynthesis_without_parentWitnessOrientation`: the repaired previous-boundary witness geometry and explicit endpoint witness coexist with the nonempty projected endpoint, while the stricter canonical annulus-construction parent-witness orientation still fails on the same collar. Thus the replacement endpoint is not vacuous, and it should not be conflated with the stronger parent-orientation route; the remaining burden is to lift this positive construction from a benchmark to route-facing boundary-order hypotheses. The generic bridge has also been factored explicitly: an existential annulus collar geometry together with nonempty purified carrier now reaches `Theorem49BoundaryRootNonemptyProjectedSynthesis` for any supplied Tait coloring via `exists_theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_annulusCollarLayoutGeometry_and_nonempty_selectedBoundaryInteriorEdgeEndpointVertices_direct`. The same bridges now also accept the named local endpoint witness `HasUnblockedInteriorEndpoint`: the fixed-embedding constructors `theorem49CollarLayoutPositiveProjectedGeometryOn_of_annulusCollarLayoutGeometry_and_hasUnblockedInteriorEndpoint` and

`theorem49HeightOrderedPositiveProjectedGeometryOn_of_annulusCollarGeometry_`
`and_hasUnblockedInteriorEndpoint`, their graph-level package constructors, and `exists_`
`theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_annulusCollarGeometry_`
`and_hasUnblockedInteriorEndpoint_direct` let a future source proof supply one un-
blocked interior endpoint directly, without first repackaging the purified carrier set. The
same endpoint witness can now be obtained from the older endpoint-support separa-
tion condition: `raw interiorEdgeEndpointSupport` nonemptiness together with disjoint-
ness from selected-boundary endpoint support implies `HasUnblockedInteriorEndpoint`,
and the annulus-collar projected endpoint has direct closed-walk and successor-cycle
source wrappers for this form. This bridge now reaches the concrete collar geometry
layer as well: `PlanarBoundaryAnnulusCollarGeometry.hasUnblockedInteriorEndpoint_`
`of_peelFaces_endpoint_disjoint_selectedBoundarySupport` applies peeled-face end-
point no-touch on the canonical construction `planarBoundaryAnnulusConstruction_`
`of_annulusCollarGeometrydata` and `raw interiorEdgeEndpointSupport`
nonemptiness to produce `HasUnblockedInteriorEndpoint`. Consequently
`Theorem49PositiveGeometricSourceReplacement.lean` now exposes direct finite-collar,
height-ordered, and projected-synthesis constructors from annulus collar geometry plus that
peeled-face no-touch obligation, without an intermediate purified-carrier proof. The source-
facing file `Theorem49PositiveGeometricSourceReplacementRoute.lean` now exposes the same
peeled-face no-touch surface at the honest source layer: closed-walk and successor-cycle source
data, same-embedding annulus collar geometry, canonical peeled-face endpoint no-touch, and `raw`
`interiorEdgeEndpointSupport` nonemptiness directly inhabit the route-facing annulus-collar pos-
itive packages and reach the nonempty projected synthesis endpoint for any supplied Tait coloring.
The same source-layer wrappers now also return the graph-level finite-collar and height-ordered
replacement packages directly. Thus an honest closed-walk or successor-cycle caller carrying the
peeled-face no-touch input can consume the replacement API without first building the route-facing
annulus-collar package and then lowering it by a separate step. This has now been pushed one
step further in `Theorem49Synthesis.lean`. The theorem `theorem49BoundaryRootSynthesis_`
`of_planarBoundaryAnnulusConstruction_of_parentWitness` assembles the full purified
theorem-4.9 synthesis package directly from `PlanarBoundaryAnnulusConstruction` plus
`ParentWitnessOrientation`: besides the previously isolated span, separation, kernel, and finrank
consequences, it also supplies the corrected raw-source image identity, projected/raw finite gener-
ator representations, raw Kirchhoff representatives, and the boundary-kernel decomposition inside
one route-level object. The graph-level theorem `exists_theorem49BoundaryRootSynthesis_`
`of_admitsPlanarBoundaryAnnulusConstructionParentWitnessOrientation` shows that the
remaining constructive burden is now upstream of synthesis itself: once actual boundary-order ge-
ometry yields honest parent-witness orientation on an annulus construction, Lean already produces
the whole purified theorem-4.9 endpoint without passing back through the older interior-dual
wrapper stack.

This also records a precise blocker in Lean: `selectedBoundaryInteriorEdgeEndpointVertices_`
`ne_interiorEdgeEndpointSupport_of_endpoint_shared` says that if a face-incidence interior
edge and a selected-boundary edge share a primal endpoint, then `raw interior-edge endpoints`
cannot already satisfy the boundary-erasure vertex side condition. Thus edge-level disjointness of
`selectedBoundarySupport` and `interiorEdgeSupport` is not the same as the needed vertex-level
separation.

This is not just a theoretical concern. The module
`Theorem49InteriorVerticesEndpointObstruction.lean` builds a two-edge path incidence
model in which the selected-boundary support and interior-edge support are disjoint as edge

sets, but the selected-boundary edge and the interior edge share a primal endpoint. The theorem `edge_disjointness_does_not_imply_endpoint_separation_sharedEndpoint` therefore refutes the shortcut from incidence-count edge disjointness to endpoint separation, and `not_endpointSupport_disjoint_sharedEndpoint` records the same failure as non-disjointness of the finite endpoint supports. The theorem `selectedBoundaryInteriorEdgeEndpointVertices_ne_interiorEdgeEndpointSupport_sharedEndpoint` shows that the raw endpoint carrier is genuinely too large in this model. The same countermodel now also admits linear endpoint-sharing face-boundary data: `nonempty_orderedFaceBoundaryData_sharedEndpoint` and `nonempty_faceBoundaryRunGeometry_sharedEndpoint` build the ordered-run shell, while `orderedFaceBoundaryData_does_not_imply_endpointSupport_disjoint_sharedEndpoint` records that endpoint-support disjointness still fails. Thus a merely linear boundary-order/run interface cannot discharge the raw endpoint-separation burden; the route needs either an explicit endpoint-separation theorem or a stronger geometric source, such as honest cyclic/closed-walk data tied to the annular interior. The file now also records the complementary fact that this particular countermodel is only a linear-run obstruction. The theorem `not_nonempty_cyclicOrderedFaceBoundaryData_sharedEndpoint`, together with `not_nonempty_faceBoundaryClosedRunGeometry_sharedEndpoint` and `not_nonempty_faceBoundaryClosedWalkGeometry_sharedEndpoint`, shows that the singleton interior face prevents cyclic closure or honest closed-walk data. The combined theorem `linearRun_endpointSupport_obstruction_not_closedWalk_sharedEndpoint` therefore localizes the next viable target: cyclic/closed-walk geometry may still be the right source for the missing endpoint-separation proof, but the linear boundary-run shell definitely is not. The next regression closes the tempting cyclic-only shortcut as well. The two-triangle diamond model `diamondEmbedding` has cyclic closed face-boundary runs on both triangular faces, witnessed by `nonempty_faceBoundaryClosedRunGeometry_diamond` and the underlying `diamondCyclicOrderedFaceBoundaryData`. Nevertheless, its shared edge is an interior edge and shares a primal endpoint with the selected-boundary edge `d01`. The theorem `cyclicClosedRun_does_not_imply_endpointSupport_disjoint_diamond` records that cyclic closed-run data alone still does not imply endpoint-support disjointness. This obstruction now reaches the honest facial-walk interface as well: `diamondFaceBoundaryPureDartCycleGeometry` supplies oriented dart cycles for the two triangular faces and `nonempty_faceBoundaryClosedWalkGeometry_diamond` lowers them to actual facial closed walks, while `faceBoundaryClosedWalkGeometry_does_not_imply_endpointSupport_disjoint_diamond` still refutes endpoint-support disjointness. The remaining geometric burden is therefore not merely cyclic ordering or the existence of facial closed walks. The strengthened regression `closedWalkSelectedArc_does_not_imply_endpointSupport_disjoint_diamond` rotates the same diamond facial walks so that the selected-boundary edges form a contiguous selected arc on each induced linear face run, and endpoint-support disjointness still fails. Thus the missing ingredient is not just closed walks plus selected arcs: it is an annular no-touch/separation theorem, or an explicit purification statement, for the intended interior vertices. The latest regression lifts this warning to the actual closed-walk annulus boundary-source package. The model `diamondWithTriangleEmbedding` keeps the diamond endpoint failure in one selected-boundary component and adds a disjoint triangular selected-boundary component, so that the two-root shared-endpoint boundary-reachability data `diamondWithTriangleAnnulusBoundaryReachabilityData` is available. It also carries honest facial closed walks with selected arcs, packaged as `diamondWithTriangleClosedWalkAnnulusBoundarySource`. Nevertheless `closedWalkAnnulusBoundarySource_does_not_imply_endpointSupport_disjoint_`

`diamondWithTriangle` still refutes endpoint-support disjointness. This is intentionally not a claim against the later peeled-face no-touch bridge; it says that the closed-walk annulus boundary source alone cannot replace the missing endpoint-level separation hypothesis. That bridge is now explicit at the generic two-sided annulus-root surface: `Theorem49InteriorVerticesAnnulusConstruction.lean` defines `planarBoundaryAnnulusConstruction_of_planarBoundaryAnnulusRootAdjDistancePeelData` and proves the corresponding annulus-root lossless-purification theorem from peeled-face endpoint no-touch, while `Theorem49Synthesis.lean` proves `theorem49AnnulusRootNonemptySynthesis_of_planarBoundaryAnnulusRootAdjDistancePeelData_of_peelFaces_endpoint_disjoint_selectedBoundarySupport`. The closed-walk theorem `theorem49AnnulusRootNonemptySynthesis_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_of_peelFaces_endpoint_disjoint_selectedBoundarySupport` is now the source-facing specialization: it uses the canonical construction's peeled-face endpoint no-touch condition to derive global raw-carrier endpoint disjointness before invoking the non-vacuous synthesis wrapper. The abstract obstruction `false_of_interiorDualBoundaryRootAdjDistancePeelData_of_forcing_interiorEdge` now isolates the same mechanism at the peeled interior-dual layer: if some interior edge is forced to meet selected-boundary support on every incident face, then covering that edge would require a peeled face that violates the edge-disjoint `hpeelInterior` field. The calibration theorem `not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_diamondWithTriangle` is the concrete specialization of this criterion to the unique interior edge `dt12`. The direct incompatibility theorem `closedWalkAnnulusBoundarySource_does_not_imply_interiorDualBoundaryRootAdjDistancePeelData_diamondWithTriangle` packages this in the exact shape of the current positive route: on this embedding the honest closed-walk source exists, but the peeled interior-dual package does not. The same source model also survives the natural source-side strengthening to cyclic ordered face-arc data. The closed-walk witness upgrades to `diamondWithTriangleCyclicOrderedFaceArcEmbeddingData`, and the theorem `annulusBoundaryReachability_and_cyclicOrderedFaceArcEmbeddingData_does_not_imply_interiorDualBoundaryRootAdjDistancePeelData_diamondWithTriangle` shows that even after adjoining the live annulus boundary-reachability data and the stronger cyclic per-face boundary order with selected-boundary contiguity, the peeled interior-dual package is still absent. The same contradiction is now formalized one layer earlier, before any interior-dual data are introduced. The theorem `false_of_planarBoundaryAnnulusConstruction_and_hpeelInterior_diamondWithTriangle` shows that on the same embedding, any annulus construction with the usual peeled-interior disjointness field already fails: the unique interior edge `dt12` must be covered by some peeled witness face, but every face containing `dt12` also contains selected-boundary support. Hence even the weakest current construction shell `PlanarBoundaryAnnulusConstructionBoundarySupportFaceData` is impossible, as recorded by `not_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData_diamondWithTriangle` and the summary theorem `closedWalkAnnulusBoundarySource_does_not_imply_planarBoundaryAnnulusConstructionBoundarySupportFaceData_diamondWithTriangle`. The cyclic ordered face-arc strengthening still does not bypass this earlier geometric failure either: the theorem `annulusBoundaryReachability_and_cyclicOrderedFaceArcEmbeddingData_does_not_imply_planarBoundaryAnnulusConstructionBoundarySupportFaceData_diamondWithTriangle` shows that the same embedding has annulus roots and cyclic ordered face-arc data, yet still does not reach even the weakest peeled-interior construction shell. The same obstruction now survives after discarding the cyclic face order and retaining only the bundled per-face selected-boundary arc

witness. `diamondWithTriangleSelectedBoundaryArcGeometry` is extracted from the same embedding, and the theorems `annulusBoundaryReachability_and_selectedBoundaryArcGeometry_does_not_imply_interiorDualBoundaryRootAdjDistancePeelData_diamondWithTriangle` and `annulusBoundaryReachability_and_selectedBoundaryArcGeometry_does_not_imply_planarBoundaryAnnulusConstructionBoundarySupportFaceData_diamondWithTriangle` show that even annulus roots plus the weaker `PlanarBoundarySelectedBoundaryArcGeometry` shell still do not start the peeled interior-dual route or even its weakest construction shell on this model. There is now also a same-embedding regression strictly earlier in the source ladder. In `PlanarBoundaryFaceBoundaryRunRegression.lean`, the explicit two-face model `doubleStarEmbedding` carries both `doubleStarAnnulusBoundaryReachabilityData` and `doubleStarCyclicOrderedFaceArcEmbeddingData`, but `not_nonempty_planarBoundaryClosedWalkAnnulusBoundarySource_doubleStarEmbedding` shows that it still admits no honest closed-walk annulus source. The left ambient face has the three-star boundary, and any candidate facial closed walk there transfers to the acyclic graph `doubleStarLeftGraph`. So annulus boundary reachability plus cyclic ordered face-arc data do not yet recover the honest facial closed-walk source; this is summarized abstractly by `not_forall_nonempty_planarBoundaryClosedWalkAnnulusBoundarySource_of_nonempty_planarBoundaryAnnulusBoundaryReachabilityData_and_nonempty_planarBoundaryCyclicOrderedFaceArcEmbeddingData`. The same two-face model also carries the weaker bundled shell `PlanarBoundarySelectedBoundaryArcGeometry`, via `nonempty_planarBoundarySelectedBoundaryArcGeometry_doubleStarEmbedding`, and the theorems `exists_embedding_withPlanarBoundaryAnnulusBoundaryReachabilityData_andPlanarBoundarySelectedBoundaryArcGeometry_withoutClosedWalkAnnulusBoundarySource` and `not_forall_nonempty_planarBoundaryClosedWalkAnnulusBoundarySource_of_nonempty_planarBoundaryAnnulusBoundaryReachabilityData_and_nonempty_planarBoundarySelectedBoundaryArcGeometry` show that even annulus roots plus one contiguous selected-boundary arc on each ambient face still do not recover the honest closed-walk annulus source. The positive direction is now explicit too: the source file `PlanarBoundaryClosedWalkSource.lean` lowers any honest `PlanarBoundaryClosedWalkAnnulusBoundarySource` directly to both `AdmitsPlanarBoundaryAnnulusBoundaryReachabilityData` together with `AdmitsPlanarBoundaryCyclicOrderedFaceArcEmbeddingData`, and to `AdmitsPlanarBoundaryAnnulusBoundaryReachabilityData` together with `AdmitsPlanarBoundarySelectedBoundaryArcGeometry`. So the double-star and related regression theorems should be read as genuine failed converses of an available positive lowering, not merely as mismatched interfaces. This is now graph-level as well, not only same-embedding: the file proves `doubleStarGraph_isAcyclic`, then uses the reusable acyclic blocker to derive `not_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource_doubleStarGraph`. Consequently the new theorems `not_forall_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource_of_admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_cyclicOrderedFaceArcEmbeddingData` and `not_forall_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource_of_admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_selectedBoundaryArcGeometry` show that even after existentially quantifying over embeddings, the paired annulus-rooted shells still do not force the honest closed-walk annulus source. The same double-star witness now blocks rotation-side repairs too: `not_admitsPlanarBoundaryDartCycleEmbeddingData_doubleStarGraph`, `not_admitsPlanarBoundaryPureDartCycleEmbeddingData_doubleStarGraph`, and `not_admitsPlanarBoundaryDartSuccessorCycleEmbeddingData_doubleStarGraph` show that

the graph admits no dart-cycle, pure dart-cycle, or successor-cycle source on any embedding. The failed-converse wrappers `not_forall_admitsPlanarBoundaryDartCycleEmbeddingData_of_admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_selectedBoundaryArcGeometry`, `not_forall_admitsPlanarBoundaryPureDartCycleEmbeddingData_of_admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_selectedBoundaryArcGeometry`, and `not_forall_admitsPlanarBoundaryDartSuccessorCycleEmbeddingData_of_admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_selectedBoundaryArcGeometry` therefore rule out the obvious repair of replacing closed facial walks by those stronger rotation-side source packages. This is now abstracted in the source file itself, not just in the double-star regression: the generic theorems `admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_cyclicOrderedFaceArcEmbeddingData_withoutClosedWalkAnnulusBoundarySource_of_isAcyclic_of_nonempty_edgeSet`, `admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_selectedBoundaryArcGeometry_withoutClosedWalkAnnulusBoundarySource_of_isAcyclic_of_nonempty_edgeSet`, and their two existential failed-converse corollaries show that any acyclic positive-edge witness graph carrying one of these paired shells already blocks the route to the honest closed-walk annulus source. This obstruction is now also factored through one reusable acyclic theorem in the shared walk-embedding layer. The file `WalkPlanarEmbeddingWithBoundaryAcyclic.lean` proves `not_nonempty_faceBoundaryClosedWalkGeometry_of_isAcyclic_of_nonempty_edgeSet`: any acyclic graph with at least one edge admits no honest facial closed-walk geometry on any boundary-aware embedding, because some ambient face must exist and its nonempty closed trail would be a path, hence nil. The FourColor wrapper `not_nonempty_planarBoundaryClosedWalkEmbeddingData_of_isAcyclic_of_nonempty_edgeSet` exposes the same fact directly on the route-facing source interface. So the path, star, and peeled-endpoint-touch source impossibility lemmas are now instances of one reusable theorem rather than separate local arguments. This abstraction now reaches one layer higher as well: the route-facing theorems `not_admitsPlanarBoundaryDartCycleEmbeddingData_of_isAcyclic_of_nonempty_edgeSet`, `not_admitsPlanarBoundaryPureDartCycleEmbeddingData_of_isAcyclic_of_nonempty_edgeSet`, `not_admitsPlanarBoundaryDartSuccessorCycleEmbeddingData_of_isAcyclic_of_nonempty_edgeSet`, and `not_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource_of_isAcyclic_of_nonempty_edgeSet` show that every acyclic positive-edge model already blocks the stronger rotation-side and annulus-source interfaces before any model-specific geometry is used. The same witness pattern is now abstracted one step further at the failed-converse level: `not_forall_admitsPlanarBoundaryClosedWalkEmbeddingData_of_exists_isAcyclic_of_nonempty_edgeSet_and_property` and `not_forall_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource_of_exists_isAcyclic_of_nonempty_edgeSet_and_property` package the generic reason why any acyclic positive-edge witness carrying a shell P refutes a universal theorem “ P implies honest closed walks” or “ P implies the honest annulus source”. The star/path regression theorems for cyclic ordered arcs, selected-boundary arcs, cyclic face-boundary vertex walks, and ordered face arcs now instantiate these generic criteria directly. This is no longer just a one-off gadget computation: the file now isolates the reusable obstruction `false_of_planarBoundaryAnnulusConstruction_and_hpeelInterior_of_forcing_interiorEdge`, which says abstractly that if some interior edge is forced to meet selected-boundary support on every incident face, then the peeled-interior annulus shell is impossible. This forcing-edge mechanism is now factored into the smaller route-facing wrapper file `Theorem49ForcingInteriorEdgeObstruction.lean`. Its witness structure `ForcingInteriorEdgeWitness` packages exactly the charitable

repair burden: one designated interior edge together with a proof that every incident face already carries selected-boundary support. From that single witness, the new generic theorems `not_nonempty_interiorDualBoundaryRootAdjDistancePeelData`, `not_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData`, and `closedWalkAnnulusBoundarySource_does_not_imply_planarBoundaryAnnulusConstructionBoundarySupport` recover the obstruction directly at the route surface, without reusing the diamond specialization by hand. The construction-side positive invariant is now stated in `PlanarBoundaryAnnulusConstructionCore.lean` as `PlanarBoundaryAnnulusConstruction.exists_peelFace_witnessEdge_eq_and_faceBoundary_disjoint_selectedBoundarySupport_of_mem_interiorEdgeSupport`: any annulus construction whose peeled faces avoid selected-boundary support must assign every interior edge to a witness peeled face whose entire boundary avoids that support. The forcing-edge contradiction is now the direct negation of this invariant. This negation has also been isolated as the positive face-incidence predicate `EveryInteriorEdgeHasBoundaryFreeIncidentFace`: every interior edge has some incident ambient face whose whole boundary is disjoint from selected boundary support. The theorem `not_nonempty_forcingInteriorEdgeWitness_iff_everyInteriorEdgeHasBoundaryFreeIncidentFace` proves that this predicate is exactly the absence of a `ForcingInteriorEdgeWitness`, and `everyInteriorEdgeHasBoundaryFreeIncidentFace_of_interiorDualBoundaryRootAdjDistancePeelData` now proves that the interior-dual boundary-root adjacency-distance package itself supplies this predicate: its cover field assigns every interior edge to a peeled witness face, and its peeled-interior field makes that face disjoint from selected-boundary support. The corollary `not_nonempty_forcingInteriorEdgeWitness_of_interiorDualBoundaryRootAdjDistancePeelData` therefore records the source-side necessary condition for deriving `InteriorDualBoundaryRootAdjDistancePeelData`: forcing interior edges must already have been eliminated before any later annulus-construction or selector compatibility step is relevant. Separately, `everyInteriorEdgeHasBoundaryFreeIncidentFace_of_planarBoundaryAnnulusConstructionBoundarySupportFaceData` shows that the weak boundary-support construction shell supplies it. The same file now also exposes the data-valued companion `BoundaryFreeIncidentFaceSelector`, whose nonemptiness is equivalent to `EveryInteriorEdgeHasBoundaryFreeIncidentFace`; the noncomputable extraction `boundaryFreeIncidentFaceSelector_of_planarBoundaryAnnulusConstructionBoundarySupportFaceData` turns the weak construction shell into an actual per-interior-edge choice of a selected-boundary-free incident face. This selector is deliberately only a local face-choice layer: it does not yet assert compatibility among the chosen faces for distinct interior edges. The selector/forcing relationship is now also available without the intermediate predicate: `nonempty_boundaryFreeIncidentFaceSelector_iff_not_nonempty_forcingInteriorEdgeWitness` and `not_nonempty_boundaryFreeIncidentFaceSelector_iff_nonempty_forcingInteriorEdgeWitness` state that a boundary-free selector exists exactly when no forcing interior edge exists. That compatibility layer is now partially formalized in `Theorem49BoundaryFreeSelectorConstruction.lean`. The definition `BoundaryFreeIncidentFaceSelector.peelFaces` takes the finite image of the selected faces, and `BoundaryFreeIncidentFaceSelector.selectedWitnessEdge` inverts the selector on that image. If the selector is injective on interior edges and every non-witness remainder edge is either ambient-boundary support or the selected witness of a strictly deeper selected face, then `BoundaryFreeIncidentFaceSelector.toPlanarBoundaryAnnulusConstruction` builds a genuine `PlanarBoundaryAnnulusConstruction`. The companion lemma `BoundaryFreeIncidentFaceSelector.toPlanarBoundaryAnnulusConstruction_`

`hpeelInterior` proves that the constructed peeled faces are disjoint from selected-boundary support. The strict-descent compatibility condition also now has its own selector-level cycle test: `BoundaryFreeIncidentFaceSelector.remainerDependency` records when the selected witness of one peeled face appears as a non-witness, non-boundary remainder edge of another, and `BoundaryFreeIncidentFaceSelector.faceDistance_lt_of_remainerDependency` turns each such dependency into a strict face-distance inequality. The theorem `BoundaryFreeIncidentFaceSelector.false_of_remainerDependency_cycle` therefore rules out every nonempty finite closed selector-dependency chain under the same strict-distance hypothesis consumed by the construction bridge. The new comparison theorem `BoundaryFreeIncidentFaceSelector.toPlanarBoundaryAnnulusConstruction_remainerDependency_iff` shows that after lowering a strict-descent selector to a `PlanarBoundaryAnnulusConstruction`, this selector dependency relation is exactly the construction-level `PlanarBoundaryAnnulusConstruction.remainerDependency`; the strict face-distance field on the construction side is recovered from selector `hrest`. Thus selector-side cycle diagnostics transfer to the central annulus construction without a second dependency argument. The companion `BoundaryFreeIncidentFaceSelector.toPlanarBoundaryAnnulusConstruction_no_remainerDependency_iff` shows that terminal no-dependency is preserved exactly by the same lowering. The maximum-distance strengthening `BoundaryFreeIncidentFaceSelector.false_of_total_remainerDependency` rules out a nonempty finite selected-face set in which every selected face has an outgoing non-boundary remainder dependency, and `BoundaryFreeIncidentFaceSelector.exists_face_without_remainerDependency` extracts a terminal selected face with no such outgoing dependency. This is a more usable diagnostic than an explicit cycle enumeration: any successful source-to-selector construction must turn that terminal face into a real ambient-boundary break, or provide a separate well-founded orientation. The selector terminal condition now has the same local characterization as the construction terminal condition: `BoundaryFreeIncidentFaceSelector.no_remainerDependency_iff_boundary_remainders` states that, on a selected face, no outgoing selector dependency is equivalent to every non-witness remainder edge lying on the outer-or-inner ambient annulus boundary. The theorem `BoundaryFreeIncidentFaceSelector.exists_terminal_face_boundary_remainders` now states that boundary break directly: for some selected face, every non-witness remainder edge is already in the outer-or-inner ambient annulus boundary. This is the local form of the selector construction burden, separated from the later non-peeled-face and frontier obligations. The variants `BoundaryFreeIncidentFaceSelector.peelFaces_nonempty_of_mem_interiorEdgeSupport`, `BoundaryFreeIncidentFaceSelector.exists_terminal_face_boundary_remainders_of_mem_interiorEdgeSupport`, and `BoundaryFreeIncidentFaceSelector.exists_terminal_face_boundary_remainders_of_interiorEdgeSupport_nonempty` remove the separate selected-face nonemptiness premise as soon as a live interior edge is present. The same maximum-distance principle is now available at the central construction surface as `PlanarBoundaryAnnulusConstruction.exists_terminal_peelFace_boundary_remainders`: any nonempty BFS-stratified annulus construction has a maximum-distance peeled face whose every non-witness remainder edge is already on the ambient outer or inner boundary split. This theorem now factors through the construction-level strict dependency relation `PlanarBoundaryAnnulusConstruction.remainerDependency`. Non-boundary remainders yield such dependencies by `PlanarBoundaryAnnulusConstruction.exists_remainerDependency_of_nonboundary_remainer`, closed finite dependency chains are ruled out by `PlanarBoundaryAnnulusConstruction.false_of_remainerDependency_cycle`, and the total-outgoing form `PlanarBoundaryAnnulusConstruction.false_of_total_`

`remainderDependency` produces `PlanarBoundaryAnnulusConstruction.exists_terminal_peelFace_no_remainderDependency`. The local terminal condition is now characterized exactly by `PlanarBoundaryAnnulusConstruction.no_remainderDependency_iff_boundary_remainders`: for a peeled face, no outgoing construction dependency is equivalent to all non-witness remainders lying on the ambient annulus boundary. Its forward direction is also available as `PlanarBoundaryAnnulusConstruction.boundary_remainders_of_no_remainderDependency`. Thus the terminal boundary break is now a named finite strict-descent obstruction on the reusable annulus-construction surface. The nonemptiness side condition has now been pushed back to the construction cover itself: `PlanarBoundaryAnnulusConstruction.peelFaces_nonempty_of_mem_interiorEdgeSupport` shows that any live interior edge covered by the construction forces a peeled face, and `PlanarBoundaryAnnulusConstruction.exists_terminal_peelFace_boundary_remainders_of_mem_interiorEdgeSupport` packages the resulting terminal boundary break directly. Thus the terminal boundary break is not just a selector artifact; it is a direct consequence of the construction's strict-descent `hrest` field as soon as the construction actually covers a live interior edge. `Theorem49PlanarBoundaryAnnulusConstructionOrientationObstruction.lean` now also records the corresponding negative calibration: `not_forall_live_planarBoundaryAnnulusConstruction_terminalNoDependency_implies_parentWitnessOrientation` shows that live interior-edge support together with a terminal peeled face with no outgoing construction-level `remainderDependency` still does not force `PlanarBoundaryAnnulusConstruction.ParentWitnessOrientation`. The terminal dependency break is therefore a diagnostic for unbroken strict remainder cycles, not the missing well-founded parent-witness orientation itself. The same finite counter-construction also lowers to `PlanarBoundaryAnnulusConstructionPositiveFrontierData` and `PlanarBoundaryAnnulusConstructionBoundarySupportFaceData` while still refuting parent orientation, as recorded by `not_forall_planarBoundaryAnnulusConstructionPositiveFrontierData_implies_parentWitnessOrientation` and `not_forall_planarBoundaryAnnulusConstructionBoundarySupportFaceData_implies_parentWitnessOrientation`. Thus selected-boundary contact and positive current-frontier bookkeeping do not by themselves supply the missing well-founded parent witness. The construction-level obstruction is sharper than merely pointing at the zero stratum: `not_forall_planarBoundaryAnnulusConstruction_positive_faceDistance_implies_parentWitnessOrientation` uses a variant in which every peeled face has positive stored distance, but the bad witness still has no incident face of strictly smaller distance. At the same time, this positivity condition cannot be imposed as a repair on the positive-frontier shell itself: `PlanarBoundaryAnnulusConstructionPositiveFrontierData.exists_peelFace_faceDistance_zero` and `PlanarBoundaryAnnulusConstructionPositiveFrontierData.not_forall_peelFaces_positive` show that positive-frontier data necessarily contains a zero-distance peeled stratum. The construction core now factors this as a stratum-level clash: `PlanarBoundaryAnnulusConstruction.not_forall_stratumFaces_nonempty_of_parentWitness` shows that parent orientation makes the zero stratum empty, while the current positive-frontier axioms inhabit every stratum. The core construction file now records the stronger incompatibility as `PlanarBoundaryAnnulusConstructionPositiveFrontierData.not_parentWitnessOrientation`, with corresponding face-partition and boundary-support variants. Thus the live parent-orientation burden is genuine lower-distance incidence, not scalar positivity of the stored distances or the current zero-based positive-frontier shell. `PlanarBoundaryAnnulusConstruction.lean` now makes the source-side mismatch explicit as `not_exists_planarBoundaryAnnulusConstructionPositiveFrontierData_with_rootDistanceConstruction_of_boundaryReachabilityData_and`

`interiorDualBoundaryRootAdjDistancePeelData`: the parent-oriented root-distance construction produced from boundary reachability plus interior-dual boundary-root adjacency-distance data cannot be wrapped in the current positive-frontier shell. Thus the root-distance parent-orientation route and the zero-based frontier bookkeeping route are different construction surfaces, not interchangeable views of one object. The same construction file now records the exact reindexing: `rootDistance_faceDistance_succ_filter_eq_shifted_faceDistance_filter_of_boundaryReachabilityData_and_interiorDualBoundaryRootAdjDistancePeelData` says that root-distance layer $n + 1$ on peeled faces is exactly shifted construction layer n , and `rootDistance_faceDistance_one_filter_eq_shifted_faceDistance_zero_filter_of_boundaryReachabilityData_and_interiorDualBoundaryRootAdjDistancePeelData` specializes this to root-distance layer 1 versus shifted layer 0. `rootDistance_faceDistance_one_filter_nonempty_iff_not_shifted_parentWitnessOrientation_of_boundaryReachabilityData_and_interiorDualBoundaryRootAdjDistancePeelData` then sharpens the interpretation: nonempty root-distance layer 1 is exactly failure of parent-witness orientation for the shifted construction. The stronger bridge `BoundaryFreeIncidentFaceSelector.toPlanarBoundaryAnnulusConstructionBoundarySupportFaceData` then reaches `PlanarBoundaryAnnulusConstructionBoundarySupportFaceData` once the remaining annulus-side hypotheses are supplied explicitly: every non-peeled face touches selected-boundary support, outer and inner boundary faces are separated, and the positive current outer frontiers are nonempty and pairwise disjoint. Thus the selector supplies the construction cover and peeled-face selected-boundary avoidance, while the real unresolved geometric burden remains visible as face/frontier data rather than hidden in the witness-choice layer. The theorem `BoundaryFreeIncidentFaceSelector.not_nonempty_forcingInteriorEdgeWitness_of_boundarySupportFaceData_conditions` packages this stronger bridge with the forcing-edge obstruction. The positive endpoint of this selector lane is now recorded in `Theorem49BoundaryFreeSelectorPositiveRoute.lean`, and it now factors through a reusable construction-level bridge in `Theorem49PositiveGeometricSourceReplacement.lean`. The theorem `theorem49HeightOrderedPositiveProjectedGeometryOn_of_planarBoundaryAnnulusConstruction` shows that any `PlanarBoundaryAnnulusConstruction`, once paired with a live purified carrier on the same embedding, already gives `Theorem49HeightOrderedPositiveProjectedGeometryOn`; the companion `theorem49BoundaryRootNonemptyProjectedSynthesis_of_planarBoundaryAnnulusConstruction` adds a Tait coloring and reaches the corrected nonempty projected synthesis endpoint. The selector theorem `BoundaryFreeIncidentFaceSelector.theorem49HeightOrderedPositiveProjectedGeometryOn_of_strictDescent` shows that a boundary-free selector, an annulus boundary split, injectivity over interior edges, the strict-descent remainder condition, and a nonempty purified selected-boundary carrier supply such a construction-level bridge. With a supplied Tait coloring, the companion theorem `BoundaryFreeIncidentFaceSelector.theorem49BoundaryRootNonemptyProjectedSynthesis_of_strictDescent` reaches the nonempty projected boundary-root synthesis endpoint. This does not derive the selector or its compatibility from source data; it isolates the precise annulus-construction positive target that a source-facing boundary-order argument must now inhabit. The same construction and live carrier hypotheses also expose the terminal boundary-break face through `theorem49TerminalPeelFaceBoundaryRemainders_of_planarBoundaryAnnulusConstruction`, so the positive endpoint and the local maximum-distance boundary break now live on the same construction surface. The construction surface also has the matching local-endpoint entry points `theorem49HeightOrderedPositiveProjectedGeometryOn_of_planarBoundaryAnnulusConstruction_and_hasUnblockedInteriorEndpoint`,

`theorem49BoundaryRootNonemptyProjectedSynthesis_of_planarBoundaryAnnulusConstruction_and_hasUnblockedInteriorEndpoint`, and `theorem49TerminalPeelFaceBoundaryRemainders_of_planarBoundaryAnnulusConstruction_and_hasUnblockedInteriorEndpoint`, so a source-side `HasUnblockedInteriorEndpoint` witness can be consumed without first being repackaged as raw carrier nonemptiness. The selector positive route now uses this local endpoint language as well: `BoundaryFreeIncidentFaceSelector.theorem49HeightOrderedPositiveProjectedGeometryOn_of_strictDescent_and_hasUnblockedInteriorEndpoint` and `BoundaryFreeIncidentFaceSelector.theorem49BoundaryRootNonemptyProjectedSynthesis_of_strictDescent_and_hasUnblockedInteriorEndpoint` consume the endpoint witness directly, while `BoundaryFreeIncidentFaceSelector.exists_terminal_face_boundary_remainders_of_strictDescent_and_hasUnblockedInteriorEndpoint` exposes the terminal boundary-break face from the same witness. The small bridge `interiorEdgeSupport_nonempty_of_hasUnblockedInteriorEndpoint` is now the only conversion needed to use selector terminal-break theorems that expect live interior-edge support. The sharpened selector diagnostic `BoundaryFreeIncidentFaceSelector.exists_terminal_face_boundary_remainders_and_disjoint_of_strictDescent_and_hasUnblockedInteriorEndpoint` keeps the two pieces on the same selected face: every non-witness remainder is ambient-boundary support, and the whole face boundary is still disjoint from selected-boundary support. The same forcing obstruction has now been calibrated on the wheel-with-inner-triangle source benchmark as well. In `Theorem49ForcingInteriorEdgeObstructionRegression.lean`, `not_nonempty_boundaryFreeIncidentFaceSelector_wheelWithInnerTriangle` shows that the honest-source, Tait-colorable wheel model with nonempty purified carrier cannot even carry a boundary-free incident-face selector; this now follows through the direct selector/forcing duality rather than a separate face-choice argument. The theorem `wheelWithInnerTriangle_closedWalkSource_tait_hasUnblockedEndpoint_without_interiorDualBoundaryRootAdjDistancePeelData` puts the same obstruction in the exact raw source-IDBRAD bridge shape: source, coloring, and ‘`HasUnblockedInteriorEndpoint`’ coexist, but the generic interior-dual datum does not. The theorem `not_forall_some_boundaryFreeSelector_or_acyclicEndpoint_of_closedWalkSource_tait_carrier_wheel` then refutes the source-level universal claim that honest closed-walk annulus source data, a Tait coloring, and a nonempty purified carrier force either such a selector or any of the current acyclic annulus endpoints: well-founded witness-cover data, height-ordered witness-cover data, finite collar-layer data, or weak annulus-collar geometry. The obstruction file also exposes the universal failed-converse form `not_forall_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData_of_exists_embedding_with_property_and_forcingInteriorEdgeWitness` and its honest-source specialization `not_forall_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData_of_exists_embedding_closedWalkAnnulusBoundarySource_and_forcingInteriorEdgeWitness`, so a single same-embedding forcing edge blocks any theorem deriving the weakest peeled-interior construction shell from the source property alone. The same abstract witness now also covers the weaker paired source shell `annulusBoundaryReachability_and_selectedBoundaryArcGeometry_does_not_imply_interiorDualBoundaryRootAdjDistancePeelData` and its construction-side companion `annulusBoundaryReachability_and_selectedBoundaryArcGeometry_does_not_imply_planarBoundaryAnnulusConstructionBoundarySupportFaceData`, as well as the reduced construction companion `annulusBoundaryReachability_and_selectedBoundaryArcGeometry_does_not_imply_planarBoundaryAnnulusConstructionPositiveFrontierData`, so the selected-boundary-arc weakening on the diamond witness is no longer a separate local argument.

The accompanying regression file `Theorem49ForcingInteriorEdgeObstructionRegression.lean` reconstructs the earlier diamond-with-triangle failures, including the selected-boundary-arc weakening, the reduced positive-frontier shell, and the concrete `PlanarBoundaryAnnulusConstructionFaceLayerData` shell that feeds the annulus decomposition route, from this abstract witness alone. The same file now also lifts these statements one level up to graph-level same-embedding existential form: on `diamondWithTriangleGraph` there exists an embedding carrying the honest source shell or the weaker annulus-rooted shells together with a forcing interior-edge witness, but no embedding of that same graph can carry the witness together with the peeled interior-dual package, the reduced positive-frontier shell, or the concrete construction face-layer shell. The obstruction file now also states the generic exact-shell converse failures `not_forall_nonempty_planarBoundaryAnnulusConstructionFaceLayerData_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness` and `not_forall_nonempty_planarBoundaryAnnulusConstructionFaceLayerData_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness`: any explicit same-embedding exact one-collar/v23 witness carrying a forcing interior edge already refutes a universal derivation of the concrete construction face-layer shell. The shared-interior-pair exact-shell failures are therefore one concrete instance of a generic route-level false burden. The obstruction file also states the exact local burden already on the one-collar v23/v23.5 shell: `not_forall_not_nonempty_forcingInteriorEdgeWitness_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` and the successor-cycle counterpart `not_forall_not_nonempty_forcingInteriorEdgeWitness_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` show that even this exact one-collar source package still does not universally exclude forcing interior edges. So any sound repair on that shell must add or derive a precise local no-forcing hypothesis before it can reach the downstream construction surface. The obstruction file now also states the forward repaired burden directly on that downstream shell: `everyInteriorEdgeHasBoundaryFreeIncidentFace_of_planarBoundaryAnnulusConstructionFaceLayerData`, `boundaryFreeIncidentFaceSelector_of_planarBoundaryAnnulusConstructionFaceLayerData`, and `not_nonempty_forcingInteriorEdgeWitness_of_planarBoundaryAnnulusConstructionFaceLayerData` show that any theorem reaching `PlanarBoundaryAnnulusConstructionFaceLayerData` must already build the local boundary-free selector, equivalently exclude forcing interior edges, before any later annulus-decomposition lowering. The repaired positive direction is now stated on that exact shell too. `Theorem49ResidualBoundaryRoute.lean` proves `theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_hasUnblockedInteriorEndpoint` and the actual boundary-order counterpart `theorem49BoundaryRootNonemptyProjectedSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and`

`planarBoundaryAnnulusRootParentPeelData_and_hasUnblockedInteriorEndpoint`. So the current strongest sound `v23/v23.5` repair is explicit: if the exact one-collar shell already carries annulus-root parent peel data on the same annulus boundary split, then the local endpoint witness closes the route all the way to `Theorem49BoundaryRootNonemptyProjectedSynthesis`. The remaining geometric burden is therefore the same-split parent-peel certificate itself, not additional projection symmetry. The constructive selector/construction lane now reaches one real shell beyond that point. `Theorem49BoundaryFreeSelectorConstruction`. `lean` proves `planarBoundaryAnnulusConstructionBoundarySupportFaceData_of_boundaryReachabilityData_and_closedWalkEmbeddingData_and_selectedBoundaryArcOnFace_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary` and the successor-cycle variant `planarBoundaryAnnulusConstructionBoundarySupportFaceData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary`. So same-split annulus-root parent peel data already lowers to the real `PlanarBoundaryAnnulusConstructionBoundarySupportFaceData` shell once the exact local gap is discharged: peeled faces missed by the selector image must still meet selected-boundary support, and the positive `currentOuterBoundary` layers of the resulting construction must be nonempty and pairwise disjoint. This identifies the remaining parent-peel-to-construction burden precisely. It is not a generic non-peeled-face condition, but the smaller selector-deficit peeled-face requirement. Lean now also proves `selectorDeficitSelectedBoundary_of_planarBoundaryAnnulusRootParentPeelData_and_nonPeelSelectedBoundary`, so whenever an older route already gives the stronger hypothesis that every non-peeled face of the same parent-peel construction touches selected-boundary support, this repaired selector-deficit premise follows immediately. The same exact one-collar/`v23` shell now also has generic forcing-edge converse failures already at that stronger construction surface and the next two intermediate construction shells: `not_forall_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness` and `not_forall_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness`. `not_forall_nonempty_planarBoundaryAnnulusConstructionFacePartitionData_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness`, `not_forall_nonempty_planarBoundaryAnnulusConstructionFacePartitionData_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness`, `not_forall_nonempty_planarBoundaryAnnulusConstructionPositiveFrontierData_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and`

`v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness`, and `not_forall_nonempty_planarBoundaryAnnulusConstructionPositiveFrontierData_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness` now show the same obstruction one layer later inside the construction lane. So any explicit same-embedding exact-shell witness carrying a forcing interior edge already refutes universal derivations of `PlanarBoundaryAnnulusConstructionBoundarySupportFaceData`, `PlanarBoundaryAnnulusConstructionFacePartitionData`, and `PlanarBoundaryAnnulusConstructionPositiveFrontierData`. The live gap therefore persists before the final lowering to face-layer data. That same repaired exact-shell package now reaches the downstream construction face-layer shell used by annulus decomposition. `Theorem49ResidualBoundaryRoute.lean` proves `nonempty_planarBoundaryAnnulusConstructionFaceLayerData_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary` and `nonempty_planarBoundaryAnnulusConstructionFaceLayerData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary`. So once the exact one-collar shell carries same-split parent peel data together with the selector-deficit selected-boundary condition and the positive `currentOuterBoundary` side conditions, the route no longer stops at selected-boundary support data: it already produces real `PlanarBoundaryAnnulusConstructionFaceLayerData` on that same embedding. So the remaining issue is not any further lowering below parent-peel itself, but whether the current exact shell forces that package at all. That same repaired exact-shell package now also reaches a real annulus decomposition on the same embedding. `Theorem49ResidualBoundaryRoute.lean` proves `nonempty_planarBoundaryAnnulusDecomposition_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary` and `nonempty_planarBoundaryAnnulusDecomposition_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary`. So once the exact shell carries the same-split parent-peel and selector-deficit package, the route already passes through annulus decomposition itself. The remaining burden is therefore not another decomposition-side shell, but whatever stronger geometry or witness data is needed downstream of that repaired local package. Lean now also proves `nonempty_planarBoundaryAnnulusDecomposition_of_planarBoundaryAnnulusConstructionFaceLayerData`, `theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusConstructionFaceLayerData_and_resultingDecompositionWitnessAssignment`, `theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusConstructionFaceLayerData_and_inducedDecompositionWitnessAssignment`, `nonempty_planarBoundaryAnnulusDecomposition_of_planarBoundaryAnnulusConstructionBoundarySupportFaceData`, and `theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusConstructionBoundarySupportFaceData_and_inducedDecompositionWitnessAssignment`, `theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusConstructionBoundarySupportFaceData_and`

resultingDecompositionWitnessAssignment, so once the route reaches PlanarBoundaryAnnulusConstructionFaceLayerData, this decomposition step is automatic and the full Theorem 4.9 endpoint reduces exactly to witness ownership on the exact annulus decomposition canonically induced by that shell. The stronger selected-boundary PlanarBoundaryAnnulusConstructionBoundarySupportFaceData shell is now only an upstream route into that face-layer package, not part of the downstream closure theorem itself. The same file now also proves the graph-level route wrappers exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_planarBoundaryAnnulusConstructionFaceLayerData_and_resultingDecompositionWitnessAssignment_direct, the stronger selected-boundary-support analogue, and the matching successor-cycle pair. So once the honest-source or successor-cycle route reaches either same-embedding construction shell and the resulting decomposition carries witness ownership, full Theorem49BoundaryRootSynthesis already follows directly at graph level; no further repackaging step remains below that shell. Theorem49PositiveGeometricSourceReplacement.lean now also proves theorem49BoundaryRootNonemptyProjectedSynthesis_of_planarBoundaryAnnulusConstructionFaceLayerData, theorem49BoundaryRawQuotientErrorPackage_of_planarBoundaryAnnulusConstructionFaceLayerData, theorem49BoundaryRootNonemptyProjectedSynthesis_of_planarBoundaryAnnulusConstructionBoundarySupportFaceData, and theorem49BoundaryRawQuotientErrorPackage_of_planarBoundaryAnnulusConstructionBoundarySupportFaceData, so once either construction shell is paired with a surviving purified carrier on the same embedding, the current projected endpoint and its raw quotient/error package are automatic as well. The remaining burden above those shells is therefore to preserve the purified carrier, or else derive stronger earlier geometry, not to find another replacement-side lowering. Theorem49PositiveGeometricSourceReplacementRoute.lean now also proves the graph-package methods Theorem49ClosedWalkAnnulusConstructionFaceLayerPositiveProjectedGeometry_boundaryRootSynthesis, Theorem49ClosedWalkAnnulusConstructionFaceLayerPositiveProjectedGeometry_exists_boundaryRootSynthesis, and the matching successor-cycle pair. So once those named construction-face-layer route packages themselves carry witness ownership on the same derived annulus decomposition, full Theorem49BoundaryRootSynthesis follows immediately; there is no remaining repackaging gap between those graph-level route objects and the Theorem 4.9 endpoint. Theorem49PositiveGeometricSourceReplacementRoute.lean now also packages the stronger selected-boundary-contact shell itself as the fixed-embedding predicates Theorem49ClosedWalkAnnulusConstructionBoundarySupportPositiveProjectedGeometryOn and Theorem49SuccessorCycleAnnulusConstructionBoundarySupportPositiveProjectedGeometryOn, together with the graph-level structures Theorem49ClosedWalkAnnulusConstructionBoundarySupportPositiveProjectedGeometry and Theorem49SuccessorCycleAnnulusConstructionBoundarySupportPositiveProjectedGeometry. Those graph packages already expose boundaryRootSynthesis and exists_boundaryRootSynthesis on the induced annulus decomposition, so callers no longer need to first repackage this shell as construction-face-layer route data just to state the direct Theorem 4.9 closure. The same file still packages the derived face-layer closure as the fixed-embedding predicates Theorem49ClosedWalkAnnulusConstructionFaceLayerPositiveProjectedGeometryOn and Theorem49SuccessorCycleAnnulusConstructionFaceLayerPositiveProjectedGeometryOn, together with the graph-level structures Theorem49ClosedWalkAnnulusConstructionFaceLayerPositiveProjectedGeometry and Theorem49SuccessorCycleAnnulusConstructionFaceLayerPositiveProjectedGeometry. The stronger selected-boundary-contact shell still lowers directly into those derived face-layer route surfaces via PlanarBoundaryAnnulusConstructionBoundarySupportFaceData.toPlanarBoundaryAnnulusConstructionFaceLayerData. The same file now also proves not_theorem49ClosedWalkAnnulusConstructionFaceLayerPositiveProjectedGeometryOn_of_

`facewiseAtMostOneInteriorEdge`, `not_theorem49SuccessorCycleAnnulusConstructionFaceLayerPositiveP`
`of_facewiseAtMostOneInteriorEdge`, and `exists_two_distinct_interior_edges_on_`
`faceBoundary_of_successorCycleAnnulusConstructionFaceLayerPositiveProjectedGeometryOn`.
 So the stronger source-facing construction face-layer shell itself already sits outside the
 canonical at-most-one branch, and the successor-cycle shell now directly exposes the same
 local two-interior-edge obstruction there too. So once the honest-source or successor-
 cycle route reaches same-embedding `PlanarBoundaryAnnulusConstructionFaceLayerData`
 and still carries `HasUnblockedInteriorEndpoint`, the current projected The-
 orem 4.9 endpoint, the raw quotient/error package, and the replacement-
 positive height/collar packages already follow directly, without any further collar-
 geometry or well-founded-witness detour. That diagnosis is now repaired again.
`Theorem49BoundaryFreeSelectorConstruction.lean` proves `parentWitnessOrientation_of_`
`boundaryData_and_interiorDualBoundaryRootParentPeelData`, `parentWitnessOrientation_`
`of_planarBoundaryAnnulusRootParentPeelData`, and `not_exists_`
`planarBoundaryAnnulusConstructionBoundarySupportFaceData_with_planarBoundaryAnnulusRootParentP`
 So the exact selector-driven parent-peel construction is already `PlanarBoundaryAnnulusConstruction`.
`ParentWitnessOrientation` on the same embedding, while any same-construction
 selected-boundary-support shell still forbids parent orientation. The same con-
 struction file now isolates the resulting contradiction at the actual shell that cre-
 ates it: `false_of_boundaryReachabilityData_and_closedWalkEmbeddingData_`
`and_selectedBoundaryArcOnFace_and_planarBoundaryAnnulusRootParentPeelData_`
`and_selectorDeficitSelectedBoundary_and_currentOuterBoundary` and `false_`
`of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_`
`selectedBoundaryArc_and_planarBoundaryAnnulusRootParentPeelData_and_`
`selectorDeficitSelectedBoundary_and_currentOuterBoundary`. The same con-
 struction file now also packages this as raw boundary-order graph-level nonexistence
 theorems: `not_exists_boundaryReachabilityData_and_closedWalkEmbeddingData_`
`and_selectedBoundaryArcOnFace_and_planarBoundaryAnnulusRootParentPeelData_`
`and_selectorDeficitSelectedBoundary_and_currentOuterBoundary` and `not_`
`exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_`
`selectedBoundaryArc_and_planarBoundaryAnnulusRootParentPeelData_and_`
`selectorDeficitSelectedBoundary_and_currentOuterBoundary`. It also now gives the fixed-
 embedding local nonexistence forms `not_exists_planarBoundaryAnnulusRootParentPeelData_`
`and_selectorDeficitSelectedBoundary_and_currentOuterBoundary_of_boundaryReachabilityData_`
`and_closedWalkEmbeddingData_and_selectedBoundaryArcOnFace` and `not_exists_`
`planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_`
`currentOuterBoundary_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_`
`and_selectedBoundaryArc`, which are the direct reusable statements downstream route ar-
 guments should consume. So the repaired selector-deficit plus positive-current-boundary
 package is already globally impossible before any exact one-collar or exact-v23 residual hy-
 potheses are added. `Theorem49ResidualBoundaryRoute.lean` then re-exports this at the
 repaired exact one-collar shell itself with `false_of_closedWalkAnnulusBoundarySource_`
`and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_`
`planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBo`
`and false_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryA`
`and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundary`
`and_selectorDeficitSelectedBoundary_and_currentOuterBoundary`. So the old posi-
 tive `currentOuterBoundary` side conditions are not merely another local burden to derive

on the same parent-peel construction: they are incompatible with that construction. The earlier exact-shell face-layer, decomposition, and synthesis theorems with those hypotheses remain valid but vacuous. The same route file now also states that the stronger older non-peeled-face selected-boundary-contact repair collapses for the same reason: `false_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_nonPeelSelectedBoundary_and_currentOuterBoundary` and `false_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_nonPeelSelectedBoundary_and_currentOuterBoundary`. So the stronger `nonPeelSelectedBoundary` branch is not merely reducible to the selector-deficit branch inside later synthesis theorems; it is already directly inconsistent with the same old positive current-boundary side conditions. `Theorem49ResidualBoundaryRoute.lean` now also packages this as direct graph-level nonexistence theorems: `not_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary` and `not_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary`. So this repaired selector-deficit plus positive-current-boundary shell is not merely unforced by the live route: even the stricter exact-shell witness form is globally impossible, as an immediate corollary of the boundary-order contradiction above. The matching stronger non-peeled-face graph-level nonexistence theorems are now present too: `not_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_nonPeelSelectedBoundary_and_currentOuterBoundary` and `not_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_nonPeelSelectedBoundary_and_currentOuterBoundary`. So even that stronger repair branch is globally impossible on the exact shell. `Theorem49ResidualBoundaryObstruction.lean` now upgrades this to direct exact-shell failed universals: `not_forall_exists_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` and `not_forall_exists_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair`, with the matching forcing-edge abstract converse failures. It now also exposes the exact-shell counterexamples themselves as `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary_sharedInteriorPair` and `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary_sharedInteriorPair`. So even the full repaired selector-deficit plus positive-current-boundary package is not forced by the live exact shell on the shared-interior-pair benchmark. The same obstruction file now upgrades the stronger older non-peeled-face branch in the same way: `not_forall_exists_planarBoundaryAnnulusRootParentPeelData_and_nonPeelSelectedBoundary_and_currentOuterBoundary_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair`.

`and_currentOuterBoundary_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` and `not_forall_exists_planarBoundaryAnnulusRootParentPeelData_and_nonPeelSelectedBoundary_and_currentOuterBoundary_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair`, with the matching exact-shell counterexamples `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusRootParentPeelData_and_nonPeelSelectedBoundary_and_currentOuterBoundary_sharedInteriorPair` and `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusRootParentPeelData_and_nonPeelSelectedBoundary_and_currentOuterBoundary_sharedInteriorPair`. So even that stronger older repair is not forced by the live exact shell on the same benchmark. The live route must therefore avoid that positive-frontier shell entirely and continue through the already parent-oriented construction / well-founded-parent lane. `Theorem49ResidualBoundaryRoute.lean` now also proves the direct exact-shell endpoint theorems `theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData` and `theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData`. So once the exact one-collar shell carries same-split annulus-root parent peel data at all, the full Theorem 4.9 synthesis endpoint is already immediate on that embedding; the selector-deficit and positive-current-boundary detour is not just unnecessary but inconsistent on that same construction. The live burden is therefore sharper once again: derive the actual `PlanarBoundaryAnnulusRootParentPeelData` package from the residual shell, rather than any construction-shell refinement beyond it. The same file now also proves the even earlier exact-shell theorems `theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_boundaryFaceRootsCanonicalParentSharedEdgeCover` and `theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_boundaryFaceRootsCanonicalParentSharedEdgeCover`, together with the matching graph-level direct existence forms. So once the exact one-collar shell already carries the raw source-fixed canonical-parent shared-edge cover on the extracted boundary-face roots, full `Theorem49BoundaryRootSynthesis` is already immediate on that embedding, with no `HasUnblockedInteriorEndpoint` and no separate parent-peel repackaging step. The live exact-shell burden is therefore sharper again: force that concrete source-fixed shared-edge cover itself, or prove that the repaired shell cannot support it. The same file now also proves the exact-shell selector-side lowerings `theorem49HeightOrderedPositiveProjectedGeometryOn_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_via_boundaryFreeSelector` and `theorem49HeightOrderedPositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_via_boundaryFreeSelector`, together with the matching graph-level existential forms. So once that same exact shell carries the raw source-fixed canonical-parent shared-edge cover together with `HasUnblockedInteriorEndpoint` it already lands on the named `Theorem49HeightOrderedPositiveProjectedGeometryOn` surface before any residual-boundary repackaging. The same file now also lowers the sharper rooted-cover shell plus

`HasUnblockedInteriorEndpoint` to the named closed-walk and successor-cycle root-distance positive route surfaces, and from there to the projected nonempty and raw quotient/error theorem-4.9 endpoints on the same embedding. So the exact one-collar/v23 rooted-cover shell now enters the modular replacement-positive pipeline itself, not only as a direct full-synthesis shortcut. The same file now also proves the direct exact-shell height-ordered lowerings `theorem49HeightOrderedPositiveProjectedGeometryOn_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBound` and `boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover_and_hasUnblockedInteriorEndp` and `theorem49HeightOrderedPositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData` and `v23ResidualBoundaryInitialState_and_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembersh` and `_hasUnblockedInteriorEndpoint`, together with the matching graph-level existential forms. So once that same exact shell carries the concrete rooted shared-edge face-membership cover together with `HasUnblockedInteriorEndpoint`, it already lands on the named `Theorem49HeightOrderedPositiveProjectedGe` surface itself, not just on the intermediate annulus root-distance positive package. The same file now also proves the direct exact-shell finite-collar lowerings `theorem49CollarLayerPositiveProjectedGeometryOn_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBound` and `boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover_and_hasUnblockedInteriorEndp` and `theorem49CollarLayerPositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData` and `v23ResidualBoundaryInitialState_and_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembersh` and `_hasUnblockedInteriorEndpoint`, together with the matching graph-level existential forms. So that same exact shell now also lands directly on the named `Theorem49CollarLayerPositiveProjectedGeometryOn` surface itself, not only after passing through the height-ordered route surface or the later residual-boundary positive interface. The same file now also lowers those named root-distance positive route surfaces to `Theorem49Res` and the associated current-boundary remainder witness interface, so the exact one-collar/v23 rooted-cover shell reaches the residual-boundary positive lane by composition too. `Theorem49ResidualBoundaryPeeling`. `lean` now also proves the same residual bridge directly from the rooted shared-edge face-membership cover itself on both the honest closed-walk and successor-cycle shells, and `Theorem49ResidualBoundaryRoute`. `lean` adds the matching exact one-collar/v23 wrappers. So the exact rooted-cover shell no longer needs an explicit detour through a named root-distance package just to expose residual- positive geometry or a current-boundary remainder witness; those surfaces are available there directly as well. The same route file now also packages these direct exact-shell residual consequences at graph level: from existential one-collar/v23 rooted-cover data plus `HasUnblockedInteriorEndpoint`, it now directly derives existential residual-positive geometry and existential current-boundary remainder witnesses on both the honest closed-walk and successor-cycle shells. So downstream route consumers no longer need to rebuild that existential residual plumbing by hand once they start from the live exact shell. The same exact rooted-cover shell now also has matching graph-level theorem-4.9 endpoint packaging: existential one-collar/v23 rooted-cover data plus `HasUnblockedInteriorEndpoint` directly yields existential `Theorem49BoundaryRootNonemptyPro` and existential `Theorem49BoundaryRawQuotientErrorPackage` on both the honest closed-walk and successor-cycle shells. So the exact rooted-cover lane now exports both the residual interface and the theorem-4.9 endpoint interface without further existential repackaging. At the more generic residual layer- face witness level, the peeling and route files now also expose the raw quotient/error theorem-4.9 endpoint directly: same-embedding witness data plus `HasUnblockedInteriorEndpoint` yields `Theorem49BoundaryRawQuotie` and the honest closed-walk / successor-cycle existential shells now package that endpoint at graph level too. So the residual witness layer itself now exports the full theorem-4.9 endpoint pair, not only the root-synthesis and projected-nonempty half. The selector shell now matches that route-facing endpoint coverage too. Starting from honest closed-walk or successor-cycle source data on a fixed embedding, same-embedding `ResidualBoundarySelectorData` now exports `Theorem49BoundaryRootSynthesis`, `Theorem49Bounda`

and `Theorem49BoundaryRawQuotientErrorPackage`; and the route file now packages all three at graph level under the corresponding existential closed- walk / successor-cycle selector shells. So downstream work can now use the selector lane as a full theorem-4.9 endpoint interface rather than only as a source of root synthesis. The residual interface is now exposed at that same boundary-order layer too. Honest closed-walk or successor-cycle source data paired with either same-embedding residual witness data or same-embedding `ResidualBoundarySelectorData`, together with `HasUnblockedInteriorEndpoint`, now directly exports `Theorem49ResidualBoundaryPositiveProjectedGeometryOn` and the corresponding current-boundary remainder witness on the same embedding; and the route file packages both consequences at graph level for all four shells. So both the residual witness lane and the selector lane now expose not only theorem-4.9 endpoints but also the upstream residual-positive interface used elsewhere in the live route. The same graph-level residual export is now available one layer more generally too: once an exact one-collar/v23 shell already carries same-embedding `InteriorDualBoundaryRootAdjDistancePeelData` together with `HasUnblockedInteriorEndpoint`, the route file now directly packages the same existential residual-positive geometry and current-boundary remainder witnesses on both the honest closed-walk and successor-cycle shells. So any later proof of the generic adj-distance package on a live exact-shell witness can enter the residual interface without further existential repackaging. The same file now also proves the exact-shell theorems `theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_interiorDualBoundaryRootAdjDistancePeelData` and `theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_interiorDualBoundaryRootAdjDistancePeelData`. So on the exact one-collar v23/v23.5 shell, even the annulus-root parent-peel package is no longer the minimal positive burden: it already suffices to derive generic `InteriorDualBoundaryRootAdjDistancePeelData` on the same embedding, or any equivalent cycle-breaking parent witness strong enough to construct it. The same file now also proves the exact-shell route-surface lowerings `theorem49ClosedWalkAnnulusRootParentPositiveProjectedGeometryOn_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryRootParentPeelData_and_hasUnblockedInteriorEndpoint`, `theorem49SuccessorCycleAnnulusRootParentPositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryRootParentPeelData_and_hasUnblockedInteriorEndpoint`, `theorem49ClosedWalkAnnulusRootAdjDistancePositiveProjectedGeometryOn_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryRootAdjDistancePeelData_and_hasUnblockedInteriorEndpoint`, and `theorem49SuccessorCycleAnnulusRootAdjDistancePositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_interiorDualBoundaryRootAdjDistancePeelData_and_hasUnblockedInteriorEndpoint`. So the live exact shell no longer merely has endpoint corollaries: once it carries the parent-peel or generic adj-distance package together with `HasUnblockedInteriorEndpoint`, it already lands on the named root-parent or root-distance positive route surfaces used elsewhere in the replacement pipeline. The same file now also packages the corresponding graph-level existential forms through `nonempty_theorem49ClosedWalkAnnulusRootParentPositiveProjectedGeometry_of_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryRootParentPeelData_and_hasUnblockedInteriorEndpoint`, its successor-cycle analogue, and the matching root-distance pair. So downstream graph-level route consumers can now use the live exact shell directly, without rebuilding those positive packages by hand. `Theorem49PositiveGeometricSourceReplacementJointObstruction` lean now also gives those four graph-level positive route packages their own `boundaryRootSynthesis` and `exists_boundaryRootSynthesis` methods, so once the exact shell has lowered into the named root-parent or root-distance surfaces, no further manual descent through collar-layer packages is needed to recover full Theorem 4.9 synthesis. `Theorem49PositiveGeometricSourceReplacementJointObstruction`.

lean now adds a durable joint burden theorem on these same four named surfaces: each of the closed-walk/successor-cycle root-parent and root-distance positive route packages simultaneously forces a face boundary with two distinct interior edges and excludes forcing interior-edge witnesses on the same embedding. So the constructive parent-oriented lane is no longer tracked only by endpoint closure; it now carries an explicit combined local profile that any live derivation into that lane must satisfy. The same file now closes one step lower still: it proves the direct exact-shell graph-level endpoint and raw quotient/error wrappers `exists_theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_closedWalkAnnulusBoundaryData_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_direct`, its successor-cycle analogue, the matching generic adj-distance pair, and the four corresponding `exists_theorem49BoundaryRawQuotientErrorPackage`. . . theorems. So once the exact shell already carries parent-peel or generic adj-distance data together with `HasUnblockedInteriorEndpoint`, the route no longer needs even an explicit graph-level positive-surface unpacking step before reaching the current projected endpoint or raw quotient/error package. The same file now also proves the direct graph-level full-synthesis closures `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_direct`, its successor-cycle analogue, and the matching generic adj-distance pair. So once the live exact shell already carries same-embedding parent-peel data or generic adj-distance data at all, the route now reaches full `Theorem49BoundaryRootSynthesis` directly; even the projected-endpoint or positive-surface packaging steps are no longer downstream burdens below that shell. The residual interface now closes one step lower too. `Theorem49ResidualBoundaryPeeling.lean` proves `theorem49BoundaryRootSynthesis_of_residualBoundarySelectorData`, `theorem49BoundaryRootSynthesis_of_residualBoundaryLayerFacePeelData` and `theorem49BoundaryRootSynthesis_of_residualBoundaryPositiveProjectedGeometryOn`, while `Theorem49ResidualBoundaryRoute.lean` proves `theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundaryData_and_residualBoundarySelectorData` and `theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundaryData_and_residualBoundaryLayerFacePeelWitnessData`. So once the route reaches residual/current-boundary selector data itself, the full Theorem 4.9 synthesis endpoint is already immediate on that embedding; the lower residual witness package and `HasUnblockedInteriorEndpoint` are no longer downstream burdens beyond that surface. `Theorem49ResidualBoundaryPeeling.lean` now also proves `nonempty_residualBoundaryData_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData` and `nonempty_residualBoundarySelectorData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData`. So the residual selector surface is no longer the first unresolved layer above the live parent-oriented shell: generic `InteriorDualBoundaryData` already lowers to that surface on the same embedding. The live constructive burden therefore sharpens again to deriving that adjacency-distance package, or any equivalent cycle-breaking parent witness strong enough to build residual/current-boundary selector data. `Theorem49ResidualBoundaryPeeling.lean` now also proves `theorem49ResidualBoundaryPositiveProjectedGeometryOn_of_boundaryData_and_interiorDualBoundaryRootAdjDistancePeelData_and_hasUnblockedInteriorEndpoint` and `theorem49CurrentBoundaryRemainders_of_boundaryData_and_interiorDualBoundaryRootAdjDistancePeelData_and_hasUnblockedInteriorEndpoint`, while `Theorem49ResidualBoundaryRoute.lean` adds the exact one-collar closed-walk and successor-cycle wrappers. So once the same embedding carries generic `InteriorDualBoundaryData` together with `HasUnblockedInteriorEndpoint`, the route already recovers both the residual/current-boundary positive projected geometry surface and a concrete layer-face current-boundary remainder witness there. These lower residual surfaces therefore are not any additional constructive burden below the adjacency-distance package; they are already downstream consequences of it. The same route file now also exposes those residual/current-boundary consequences one layer higher on the exact same-split parent-peel shell itself, together with matching graph-level existential closures. So once the live exact shell reaches `PlanarBoundaryAnnulusRootParentPeelData` together with `HasUnblockedInteriorEndpoint`, the

residual interface is already present there as well; downstream users no longer need to unpack the intermediate annulus-parent positive package or rebuild existential residual witnesses by hand. `Theorem49ResidualBound` now calibrates that same live burden directly on the exact one-collar/v23 shared-interior-pair shell. It proves `sharedInteriorPair_closedWalkSource_tait_hasUnblockedInteriorEndpoint_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_without_interiorDualBoundaryRootAdjDistancePeelData` and `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_interiorDualBoundaryRootAdjDistancePeelData` for `sharedInteriorPair`, together with the closed-walk and successor-cycle failed-universal forms. So the current live adjacency-distance burden is not merely unproved on that benchmark: even exact one-collar current-boundary data, a real Tait coloring, `HasUnblockedInteriorEndpoint`, and the exact v23 residual initial state still do not force any same-embedding generic `InteriorDualBoundaryRootAdjDistancePeelData` there. More sharply, the same exact shell also fails to force weak annulus-collar geometry itself: the new one-collar theorems `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair`, `not_forall_nonempty_planarBoundaryAnnulusCollarGeometry_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair`, `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair`, and `not_forall_nonempty_planarBoundaryAnnulusCollarGeometry_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` show that even this single route-facing same-embedding geometry shell is not recovered there. The same exact shell also fails at the central acyclic positive burden itself: the one-collar theorems `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryWellFoundedFacePeelWitnessData_sharedInteriorPair`, `not_forall_nonempty_planarBoundaryWellFoundedFacePeelWitnessData_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair`, `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryWellFoundedFacePeelWitnessData_sharedInteriorPair`, and `not_forall_nonempty_planarBoundaryWellFoundedFacePeelWitnessData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` show that even same-embedding well-founded parent-peeling is not recovered there. More primitively, the same exact shell also fails at the earlier child-membership face-membership shell itself: the new one-collar theorems `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_sharedInteriorPair`, `not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair`, `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint`

`and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsCanonicalParentSharedEdgeFaceMemberships`
`sharedInteriorPair`, and `not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentSharedEdgeFaceMemberships`
`of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` show that even exact one-collar current-boundary data preserving the extracted boundary split still does not recover that stronger source-preserving shell on the same embedding. The same file now also lifts this face-membership obstruction to generic exact-shell converse failures: `not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMemberships`
`of_exists_embedding_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState`, `not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMemberships`
`of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState`, `not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentSharedEdgeFaceMemberships`
`of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState` and `not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentSharedEdgeFaceMemberships`
`of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState`. So any same-embedding exact-shell witness with `HasUnblockedInteriorEndpoint` already refutes a universal derivation of that stricter face-membership shell; the shared-interior-pair examples are just concrete instantiations. More primitively, the same exact shell also fails at the earlier raw-cover and degenerate boundary-face-root shells themselves: the new one-collar theorems `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsCanonicalParentSharedEdgeCover_sharedInteriorPair`, `not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_sharedInteriorPair`
`of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair`, `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsCanonicalParentSharedEdgeCover_sharedInteriorPair`, `not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentSharedEdgeCover_sharedInteriorPair`
`of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair`, and the matching `NoInteriorEdges` quartet show that even exact one-collar current-boundary data preserving the extracted boundary split still does not recover those earlier route shells on the same embedding. The same file now also lifts the raw-cover half of this diagnosis to generic exact-shell converse failures: `not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_of_exists_embedding_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState`, `not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState`, `not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentSharedEdgeCover_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState`, and `not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentSharedEdgeCover_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and`

`hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState`. So any same-embedding exact-shell witness with `HasUnblockedInteriorEndpoint` already refutes a universal derivation of that raw canonical-parent shell; the shared-interior-pair examples are just concrete instantiations. The same exact shell also fails to repair the entire currently known same-embedding geometry burden in generic form: `not_forall_some_knownSameEmbeddingGeometry_of_exists_embedding_closedWalkAnnulusBoundaryS` and `_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState` and `not_forall_some_knownSameEmbeddingGeometry_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState` show that any explicit exact-shell witness carrying simultaneous failure of the currently known same-embedding geometry surfaces already refutes a universal derivation of that whole disjunctive burden. The same file now also proves `not_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_sharedInteriorPair, sharedInteriorPair_closedWalkSource_tait_hasUnblockedInteriorEndpoint_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusRootAdjDistancePeelData, exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusRootAdjDistancePeelData, sharedInteriorPair`, and the matching closed-walk / successor-cycle failed-universal theorems. The same obstruction is now lifted to reusable forcing-edge converse failures too: `not_forall_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness` and `not_forall_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness`. So the same repaired exact one-collar/v23 shell does not merely fail to force generic interior-dual peel data: it already fails to force the source-preserving two-sided annulus-root adjacency-distance route package on the same embedding. The same file now also proves `not_nonempty_planarBoundaryAnnulusRootParentPeelData_sharedInteriorPair, sharedInteriorPair_closedWalkSource_tait_hasUnblockedInteriorEndpoint_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusRootParentPeelData, exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusRootParentPeelData_sharedInteriorPair`, and the matching closed-walk / successor-cycle failed-universal theorems. So the same repaired exact one-collar/v23 shell does not merely fail at the stronger generic adjacency-distance package: it already fails to force the actual same-embedding `PlanarBoundaryAnnulusRootParentPeelData` route target itself. The live diagnosis is therefore sharper than “derive parent-peel from the residual shell”: that implication is false on the current shell, so the route must either strengthen the upstream same-embedding hypotheses or turn the residual shell into an impossibility theorem instead of continuing to lower downstream. The same obstruction is now lifted from the benchmark instance to reusable route-level failed-universal theorems too: `not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_exists_embedding_closedWalkAnnulusBoundaryS` and `_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness` and `not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState`

and_forcingInteriorEdgeWitness. So any same-embedding exact-shell witness carrying a forcing interior edge now formally refutes the parent-peel converse; the obstruction is no longer stated only as a single shared-interior-pair counterexample. Theorem49InteriorVerticesEndpointObstruction.lean now exposes the opposite concrete calibration: nonempty_interiorDualBoundaryRootAdjDistancePeelData_peeledEndpointTouch and admitsPlanarBoundaryInteriorDualBoundaryRootAdjDistancePeelData_withoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph show that generic boundary-root adjacency-distance data is itself realizable on a concrete benchmark, but only off the live source-preserving route. The endpoint-touch graph is acyclic, so it cannot satisfy the honest closed-walk annulus source at all. Thus the shared-interior-pair exact-shell failure is a source-compatible obstruction on the live route, not an absolute impossibility of InteriorDualBoundaryRootAdjDistancePeelData itself. Theorem49ForcingInteriorEdgeWitness.lean now strengthens that diagnosis on an honest source-compatible benchmark too. The chained-diamonds source already inhabits the stronger local package exists_closedWalkSource_with_atMostOneInteriorEdgePerFace_chainedDiamondsWithTriangleGraph, which is enough elsewhere in the development to derive canonical witness choice and weak collar geometry. But the new theorems not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_of_closedWalkSource_with_atMostOneInteriorEdgePerFace_chainedDiamondsWithTriangle_via_forcingInteriorEdgeWitness and not_forall_nonempty_interiorDualBoundaryRootAdjDistancePeelData_of_closedWalkSource_with_atMostOneInteriorEdgePerFace_chainedDiamondsWithTriangleGraph show that even this stronger honest closed-walk source package still does not force generic boundary-root adjacency-distance data on the same embedding. So the adjacency-distance obstruction is not peculiar to the repaired exact one-collar/v23 shell; it already appears higher up on a source-compatible local source surface. The same file now also proves not_nonempty_planarBoundaryAnnulusRootParentPeelData_of_closedWalkSource_with_atMostOneInteriorEdgePerFace_chainedDiamondsWithTriangle_via_forcingInteriorEdgeWitness and not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_closedWalkSource_with_atMostOneInteriorEdgePerFace_chainedDiamondsWithTriangleGraph, so even that stronger honest-source package still does not force the current parent-oriented annulus-root target on the same embedding. This sharpens the live burden again: the route now needs genuinely stronger source-side structure than the chained-diamonds at-most-one package, not just more downstream lowering from it. That same repaired exact-shell package now also has an honest full Theorem 4.9 synthesis endpoint once the exact construction induced by that package carries witness ownership. Theorem49ResidualBoundaryRoute.lean proves theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_inducedDecompositionWitnessAssignment, theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_inducedDecompositionWitnessAssignment, and also the older chosen-decomposition forms theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_resultingDecompositionWitnessAssignment and theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_nonPeelSelectedBoundary_and_resultingDecompositionWitnessAssignment. So once the exact shell carries the same-split parent-peel and selector-deficit package, no further Theorem 4.9 algebra remains after the derived decomposition acquires witness ownership. More sharply, the live downstream burden is now to build PlanarBoundaryAnnulusRootParentPeelData on the exact same-embedding annulus decomposition canonically induced by that repaired parent-peel construction, rather than on an arbitrary chosen decomposition or through another selector wrapper. The new theorem theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_nonPeelSelectedBoundary_and_resultingDecompositionWitnessAssignment.

packages this downward derivation on the live successor-cycle shell itself: under the older stronger non-peeled-face contact hypothesis, the selector-deficit burden is discharged internally and the remaining exact-shell work is to reach that boundary-support shell and then supply witness ownership. The live wheel benchmark now shows that this burden is genuinely new. `Theorem49ForcingInteriorEdgeObstructionRegression`. `lean proves not_exists_planarBoundaryAnnulusDecomposition_and_witnessAssignment_wheelWithInnerTriangle`. `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_without_any_planarBoundaryAnnulusWitnessAssignment_wheelWithInnerTriangle`, and `not_forall_exists_planarBoundaryAnnulusDecomposition_and_witnessAssignment_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle`. So even the actual successor-cycle one-collar exact-v23 shell on the live wheel carrier still does not force any same-embedding annulus decomposition carrying `PlanarBoundaryAnnulusWitnessAssignment`. The missing repaired burden is therefore honestly witness ownership itself, not any further decomposition-side wrapper. That same repaired local package now already discharges the forcing-edge obstruction on the exact one-collar v23/v23.5 shell itself. `Theorem49ResidualBoundaryRoute`. `lean proves not_nonempty_forcingInteriorEdgeWitness_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary` and `not_nonempty_forcingInteriorEdgeWitness_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary`. So once the exact shell carries same-split parent peel data together with the selector-deficit selected-boundary hypothesis and the positive `currentOuterBoundary` side conditions, forcing interior edges are excluded on that same embedding. The remaining exact-shell burden is therefore no longer an unspecified no-forcing theorem, but precisely to derive this explicit selector-deficit construction package from the actual residual shell. Lean now also records that this burden can be stated one layer more directly on the actual boundary-order selector package itself: `not_nonempty_forcingInteriorEdgeWitness_of_boundaryReachabilityData_and_closedWalkEmbeddingData_and_selectedBoundaryArcOnFace_and_boundaryFreeIncidentFaceSelector_and_boundarySupportFaceData_conditions` and `theorem49BoundaryRoute`. `of_boundaryReachabilityData_and_closedWalkEmbeddingData_and_selectedBoundaryArcOnFace_and_boundaryFreeIncidentFaceSelector_and_boundarySupportFaceData_conditions_and_inducedDecompositionWitnessAssignment`. The same route is now also named directly on the actual successor-cycle shell by `not_nonempty_forcingInteriorEdgeWitness_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFreeIncidentFaceSelector_and_boundarySupportFaceData_conditions`, `theorem49BoundaryRoute`. `of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFreeIncidentFaceSelector_and_boundarySupportFaceData_conditions_and_inducedDecompositionWitnessAssignment` and the graph-level closure `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFreeIncidentFaceSelector_and_boundarySupportFaceData_conditions_and_inducedDecompositionWitnessAssignment_direct`. So once boundary reachability together with facewise selected-boundary arcs already supports a boundary-free selector whose nonselected faces meet selected-boundary support and whose induced positive `currentOuterBoundary` layers are nonempty and pairwise disjoint, both the forcing-edge obstruction and the remaining Theorem 4.9 algebra disappear on that same embedding. More sharply, the remaining burden is witness ownership on the exact selector-induced annulus decomposition, not merely on an unspecified chosen decomposition. Thus the live exact-shell burden is better stated directly at this selector-side local package, and then at that induced decomposition, rather than at another intermediate construction shell. Once that exact successor-cycle selector package is available at graph level, no further route-level repackaging step

remains before full `Theorem49BoundaryRootSynthesis`. This forcing-edge route now also blocks the current two-sided annulus-root lane itself. `Theorem49ForcingInteriorEdgeObstruction.lean` now proves `closedWalkAnnulusBoundarySource_does_not_imply_planarBoundaryAnnulusRootAdjDistancePeelData` and the same-embedding existential form `not_exists_embedding_closedWalkAnnulusBoundarySource_and_forcingInteriorEdgeWitness_and_planarBoundaryAnnulusRootAdjDistancePeelData`, with the diamond-with-triangle regression instantiated by `closedWalkAnnulusBoundarySource_does_not_imply_planarBoundaryAnnulusRootAdjDistancePeelData_diamondWithTriangle_via_forcingInteriorEdgeWitness`. So a forcing interior edge does not merely obstruct later annulus construction or synthesis shells; it already prevents any two-sided annulus-root repackaging on that same honest-source embedding. This has now been pushed one step earlier in the boundary-order pipeline as well. The same file proves `annulusBoundaryReachability_and_cyclicOrderedFaceArcEmbeddingData_does_not_imply_planarBoundaryAnnulusRootAdjDistancePeelData` and `annulusBoundaryReachability_and_selectedBoundaryArcGeometry_does_not_imply_planarBoundaryAnnulusRootAdjDistancePeelData` plus the corresponding same-embedding existential obstructions. The regression file instantiates both on the diamond-with-triangle witness. So even the stronger cyclic boundary-order shells still do not rule out the forcing-edge pathology needed to reach two-sided annulus-root data. This forcing-edge calibration is now also packaged at the right theorem surface. `Theorem49ForcingInteriorEdgeObstruction.lean` proves the generic failed-converse theorem `not_forall_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_of_exists_embedding_with_property_and_forcingInteriorEdgeWitness`: any explicit embedding carrying a property P together with a forcing interior-edge witness refutes a universal same-embedding theorem from P to `PlanarBoundaryAnnulusRootAdjDistancePeelData`. The same file records the route-facing specializations where P is the honest closed-walk annulus source shell, the reachability-plus-cyclic shell, and the reachability-plus-selected-arc shell; the regression file instantiates all three on `diamondWithTriangleGraph`. So the forcing-edge blocker is no longer only a list of local witness failures. It is now a reusable same-embedding no-universal-converse theorem surface for the current annulus-root route. The missing burden is now stated even more directly. The same file proves the generic theorem `not_forall_not_nonempty_forcingInteriorEdgeWitness_of_exists_embedding_with_property_and_forcingInteriorEdgeWitness`: any explicit embedding carrying a property P together with a forcing interior-edge witness refutes a universal same-embedding theorem saying that P excludes forcing interior edges. The route-facing specializations cover the honest closed-walk annulus source shell, the reachability-plus-cyclic shell, and the reachability-plus-selected-arc shell, with the regression file again instantiating all three on `diamondWithTriangleGraph`. So the remaining geometric burden is explicit in Lean: current source-side boundary order data does not yet rule out the forcing-edge pathology that blocks the surviving annulus-root lane. The attempted face-local strengthening is now also refuted. The obstruction file proves `not_forall_not_nonempty_forcingInteriorEdgeWitness_of_exists_embedding_reachability_and_closedWalkEmbedding_and_atMostOneInteriorEdge_and_forcingInteriorEdgeWitness`, and the regression file instantiates it through `exists_embedding_reachability_and_closedWalkEmbedding_and_atMostOneInteriorEdge_and_forcingInteriorEdgeWitness_diamondWithTriangleGraph` and `not_forall_not_nonempty_forcingInteriorEdgeWitness_of_reachability_and_closedWalkEmbedding_and_atMostOneInteriorEdge_diamondWithTriangle`. Thus, on `diamondWithTriangleEmbedding`, annulus boundary reachability, honest facial closed walks, and the condition that every face has at most one interior boundary edge coexist with the forcing interior edge `dt12`. The positive exclusion burden therefore cannot be met by this at-most-one condition alone; it needs a genuinely stronger collar/peel ownership principle or a different geometric exclusion hypothesis. That calibration is now sharpened on the honest source shell itself. `PlanarBoundaryClosedWalkSource.lean` now proves that on a fixed embedding `Nonempty(PlanarBoundaryClosedWalkAnnulusBoundarySourceemb)` already lowers to both `Nonempty(PlanarBoundaryCyclicOrderedFaceArcEmbeddingDataemb)` and `Nonempty(PlanarBoundarySelectedBoundaryArcGeometryemb)`. `Theorem49ForcingInteriorEdgeObstruction.lean` then adds `closedWalkAnnulusBoundarySource_and_cyclicOrderedFaceArcEmbeddingData_does_not_imply_planarBoundaryAnnulusRootAdjDistancePeelData` and its selected-arc analogue,

while the regression file realizes both on `diamondWithTriangleEmbedding`. So adjoining the stronger cyclic or bundled selected-boundary-arc shells on top of the honest closed-walk source still does not rescue the annulus-root route from the forcing-edge blocker. The same strengthened honest-source surfaces are now blocked at graph level too. The obstruction file adds `not_exists_embedding_closedWalkAnnulusBoundary_and_cyclicOrderedFaceArcEmbeddingData_and_forcingInteriorEdgeWitness_and_planarBoundaryAnnulus` and the selected-arc analogue, with the regression file instantiating both on `diamondWithTriangleGraph`. So even the conjunction “honest source + honest cyclic boundary-order shell” is now formally separated from the surviving annulus-root lane by the forcing-edge witness. So the honest closed-walk annulus-boundary source still does not supply the peeled-interior geometric burden needed to start the current annulus construction route. A sharper regression now reaches the peeled edge-disjoint data itself. The minimal model `peeledEndpointTouchEmbedding` admits `peeledEndpointTouchOuterBoundaryRootAdjDistancePeelData`: its peeled face contains only the shared interior edge, so the current `hpeelInterior` edge-disjoint field is satisfied. But that interior edge shares an endpoint with a selected-boundary edge on the root face, and `outerBoundaryRootAdjDistancePeelData_does_not_imply_endpointSupport_disjoint_peeledEndpointTouch` therefore refutes endpoint-support disjointness even under the current outer-boundary-rooted adjacency-distance peel package. The extra peeled-face endpoint no-touch hypothesis is not cosmetic; it is strictly stronger than the current edge-disjoint annulus data. The direct theorem `peelFaces_endpoint_disjoint_selectedBoundarySupport_fails_peeledEndpointTouch` checks the exact construction-level obligation: the peeled singleton face and the selected boundary support share a primal endpoint, so the no-touch bridge cannot be recovered from the present peeled package by unpacking definitions. The theorem `planarBoundaryAnnulusConstruction_edge_disjoint_peelFaces_does_not_imply_endpoint_noTouch_peeledEndpointTouch` packages the same separation at the bare annulus-construction surface: its peeled faces are edge-disjoint from the selected boundary support, but the endpoint no-touch condition still fails. The negated universal theorem `not_forall_planarBoundaryAnnulusConstruction_edge_disjoint_implies_endpoint_noTouch_peeledEndpointTouch` rules out deriving the endpoint no-touch condition from annulus-construction data plus peeled-face edge disjointness, even on this fixed embedding. The no-touch bridge is now also factored through the aggregate finite carrier `peelFaceBoundaryEndpointSupport`. The lemma `interiorEdgeEndpointSupport_subset_peelFaceBoundaryEndpointSupport` says that every raw interior-edge endpoint lies on this peeled-face boundary endpoint carrier, and `peelFacesEndpointDisjointness_iff` identifies aggregate carrier disjointness with the previous per-peeled-face no-touch schema. The endpoint-touch model refutes this aggregate repair as well: `planarBoundaryAnnulusConstruction_edge_disjoint_peelFaces_does_not_imply_aggregate_endpoint_noTouch_peeledEndpointTouch` and `not_forall_planarBoundaryAnnulusConstruction_edge_disjoint_implies_aggregate_endpoint_noTouch_peeledEndpointTouch` show that peeled-face edge disjointness does not imply disjointness of the peeled-face endpoint carrier from selected-boundary endpoints. At the outer-boundary-rooted route surface, the file also proves `outerBoundaryRootAdjDistancePeelData_does_not_imply_withEndpointSeparation_peeledEndpointTouch`: the same embedding admits the current peeled route package, but admits no `PlanarBoundaryOuterBoundaryRootAdjDistancePeelDataWithEndpointSeparation` package because raw endpoint support disjointness is false. The companion theorem `selectedBoundaryInteriorEdgeEndpointSupport_eq_empty_peeledEndpointTouch` records the positive alternative on the same model: the selected-boundary purified endpoint carrier is exactly empty. The raw interior edge contributes vertices 1 and 2, but `pet01` removes vertex 1 and `pet23` removes vertex 2. The summary theorem `peeledEndpointTouch_rawCarrierRepair01` packages the strongest current conclusion in one Lean statement. This should be read as a high-confidence crux-block for one precise repair strategy: deriving the raw `interiorEdgeEndpointSupport` carrier, or the peeled-face endpoint no-touch premise needed for it, from the current outer-boundary-rooted peeled annulus data and peeled-face edge disjointness. It should not be overstated as a high-confidence block of every possible Theorem 4.9 continuation. Two honest continuations remain open in the formalization: add and prove a genuinely stronger geometric endpoint no-touch hypothesis, or keep the selected-boundary

purification explicit, as in `Theorem49OuterBoundaryRootSynthesis`. The theorem `peeledEndpointTouch_emptyCarrier_synthesis` makes the fork even more explicit on the same fixed model: for any Tait coloring, the current synthesis package is available from `peeledEndpointTouchOuterBoundaryRootAdjDistancePeel`, but its selected-boundary endpoint carrier is exactly $\emptyset$. Thus the positive synthesis theorem and the raw-carrier obstruction are not contradictory; they prove that this data package supports the purified statement while carrying no non-vacuous raw endpoint information in the endpoint-touch model. The Lean surface now separates this explicitly: `Theorem49OuterBoundaryRootNonemptySynthesis` is the same synthesis package plus the additional proof that the purified carrier is nonempty. The endpoint-touch model has nonempty raw carrier, certified by `interiorEdgeEndpointSupport_nonempty_peeledEndpointTouch`, but `not_theorem49OuterBoundaryRootNonemptySynthesis_peeledEndpointTouch` shows that it cannot satisfy this non-vacuous selected-boundary synthesis surface. This blocks the tempting interpretation that the purified endpoint theorem has preserved a genuine annular interior vertex in this model. The same issue now survives on the newer two-sided annulus-root synthesis lane too. The endpoint-touch model carries `peeledEndpointTouchAnnulusRootAdjDistancePeelData`, and `peeledEndpointTouch_emptyCarrier_annulusSynthesis` shows that the ordinary `Theorem49AnnulusRootSynthesis` package is still available there for any Tait coloring, but only over the empty purified carrier. Likewise `not_theorem49AnnulusRootNonemptySynthesis_peeledEndpointTouch` and `peeledEndpointTouch_annulusNonvac` show that abandoning the all-outer-root condition does not by itself repair the non-vacuity problem: the selected-boundary purification can still collapse the endpoint carrier exactly on the two-sided annulus-root surface. This collapse already survives one further step upstream on the direct boundary height-ordered witness-cover lane. The endpoint-touch model now carries `peeledEndpointTouchPlanarBoundaryHeightOrderedFacePeelWitnessData_does_not_imply_nonempty_selectedBoundaryInteriorEdgeEndpointVertices_peeledEndpointTouch` shows that even this earlier planarity-side witness endpoint can coexist with a nonempty raw interior-edge endpoint carrier and an empty purified selected-boundary carrier. The packaged theorem `peeledEndpointTouch_heightOrderedWitnessCarrierCruxObstructionSummary` records that the graph already admits `AdmitsPlanarBoundaryHeightOrderedFacePeelWitnessData` while the purified carrier is still exactly $\emptyset$, and `admitsPlanarBoundaryHeightOrderedFacePeelWitnessData_withoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph` shows that this earlier witness-cover endpoint still does not recover the honest closed-walk annulus source on the same acyclic graph. This collapse already survives one further step upstream on the direct well-founded witness-cover lane. The endpoint-touch model now carries `peeledEndpointTouchPlanarBoundaryWellFoundedFacePeelWitnessData_does_not_imply_nonempty_selectedBoundaryInteriorEdgeEndpointVertices_peeledEndpointTouch` shows that even this earlier planarity-side witness endpoint can coexist with a nonempty raw interior-edge endpoint carrier and an empty purified selected-boundary carrier. The packaged theorem `peeledEndpointTouch_wellFoundedWitnessCarrierCruxObstructionSummary` records that the graph already admits `AdmitsPlanarBoundaryWellFoundedFacePeelWitnessData` while the purified carrier is still exactly $\emptyset$, and `admitsPlanarBoundaryWellFoundedFacePeelWitnessData_withoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph` shows that this earlier witness-cover endpoint still does not recover the honest closed-walk annulus source on the same acyclic graph. This collapse also survives on the direct attached-certificate surface. The endpoint-touch model now carries the explicit graph-level certificate `peeledEndpointTouchAttachedCertificate_emptyCarrier` but `peeledEndpointTouch_attachedCertificate_emptyCarrier` shows that its own purified selected-boundary endpoint carrier is still exactly $\emptyset$. The theorem `planarBoundaryAttachedRootedFacePeelCertificate_does_not_imply_nonempty_selectedBoundaryInteriorEdgeEndpointVertices_peeledEndpointTouch` packages the same separation in route-facing form: the raw interior-edge endpoint carrier is nonempty, yet the attached-certificate endpoint alone does not force a nonempty purified carrier. And `admitsPlanarBoundaryWellFoundedFacePeelWitnessData_withoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph` shows that even the graph-level attached-certificate endpoint still does not recover the honest closed-walk annulus source on this acyclic model. The crux-directed summary theorem `peeledEndpointTouch_nonvacuousPurifiedSynthesisCruxObstructionSummary`

packages this in the form most relevant to the proof route: the current peeled annulus data exists and the raw interior-edge endpoint carrier is nonempty, but the purified carrier is empty; consequently every non-vacuous purified synthesis package is impossible on this model, even though the ordinary purified synthesis package remains available for any Tait coloring. This is a robust obstruction to any continuation that silently reads the selected-boundary purified endpoint theorem as a theorem about a nonempty annular interior carrier. The next crux certificate blocks a broader carrier workaround. The theorem `peeledEndpointTouch_arbitraryCarrierCruxObstructionSummary` shows that, in the same model, selected-boundary purification empties not only the interior-edge endpoint carrier but also the canonical all-graph-endpoint carrier `selectedBoundaryGraphInteriorVertices`. More generally, every finite candidate carrier contained in `graphEdgeEndpointSupport` purifies to $\emptyset$. Thus this countermodel is not repaired by replacing the raw interior-edge endpoint carrier with an arbitrary subcarrier of graph-edge endpoints; any non-vacuous route must supply genuinely new geometry that produces vertices outside this selected-boundary-covered endpoint set, or must change the proof strategy. This has now been sharpened to the arbitrary-carrier level for the same fixed model. The theorem `peeledEndpointTouch_arbitraryCarrierCruxObstructionSummary` proves that `graphEdgeEndpointSupport` is all of `Fin 4`, and that the selected-boundary endpoint support is also all of `Fin 4`. Hence for every finite carrier `vertices:Finset (Fin 4)`, the purified set `selectedBoundaryGraphInteriorVertices` is $\emptyset$. This closes carrier-only repairs inside the endpoint-touch model: a non-vacuous continuation must either add a hypothesis excluding this model, such as a genuine endpoint no-touch condition, or abandon this local carrier strategy. This endpoint-touch model is now also calibrated against the honest boundary-order source. Because `peeledEndpointTouchGraph` is just the three-edge path, it is acyclic; the theorem `not_nonempty_planarBoundaryClosedWalkEmbeddingData_peeledEndpointTouchGraph` shows that no embedding of it can carry honest facial closed-walk geometry, hence `not_admitsPlanarBoundaryClosedWalkEmbeddingData_peeledEndpointTouchGraph`. At the same time `admitsPlanarBoundaryOuterBoundaryRootAdjDistancePeelData_peeledEndpointTouchGraph` still holds, and `admitsPlanarBoundaryOuterBoundaryRootAdjDistancePeelData_withoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph`, `admitsPlanarBoundaryHeightOrderWithoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph`, `admitsPlanarBoundaryWellFoundedWithoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph`, and `admitsPlanarBoundaryAttachedWithoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph` make the calibration explicit: the endpoint-touch countermodel blocks later peeled-route and certificate consequences, but it is not itself a same-model counterexample to the honest closed-walk source route.

The raw carrier is now connected as far as the current route can honestly take it. The lemma `interiorEdgeEndpointSupport_avoids_selectedBoundarySupport_of_no_endpoint_shared` turns endpoint-level separation into the boundary-erasure side condition, and `theorem49BoundaryZeroKirchhoffSubspace_interiorEdgeEndpointSupport_image_and_separation_of_planarBoundaryOuterBoundaryRootAdjDistancePeelData` then gives the boundary-erasure image identity and projected-generator separation theorem for the unpurified `interiorEdgeEndpointSupport`. The graph-level wrapper `exists_theorem49BoundaryZeroKirchhoffSubspace_interiorEdgeEndpointSupport_image_and_separation_of_admitsPlanarBoundaryOuterBoundaryRootAdjDistancePeelData` exposes the same statement from admitted outer-boundary-rooted annulus data, parameterized by the missing endpoint-separation proof.

The same route is now stated in the cleaner finite-support language. The theorem `selectedBoundaryInteriorEdgeEndpointVertices_eq_interiorEdgeEndpointSupport_iff_endpointSupport_disjoint` identifies the raw-carrier condition with disjointness between `interiorEdgeEndpointSupport` and the endpoint support of `selectedBoundarySupport`. The route theorem `theorem49BoundaryZeroKirchhoffSubspace_interiorEdgeEndpointSupport_image_and_separation_of_planarBoundaryOuterBoundaryRootAdjDistancePeelData_of_endpointSupport_disjoint` uses this endpoint-support disjointness directly, with graph-level

wrapper exists_theorem49BoundaryZeroKirchhoffSubspace_interiorEdgeEndpointSupport_image_and_separation_of_admitsPlanarBoundaryOuterBoundaryRootAdjDistancePeelData_of_endpointSupport_disjoint.

The endpoint-separation obligation is now also present as a named annulus-route data surface rather than only as an external parameter. The structure PlanarBoundaryOuterBoundaryRootAdjDistancePeelDataWithEndpointSeparation extends the outer-boundary-rooted annulus data by exactly the finite endpoint-support disjointness needed for the raw interiorEdgeEndpointSupport carrier. The theorem theorem49BoundaryZeroKirchhoffSubspace_interiorEdgeEndpointSupport_image_and_separation_of_planarBoundaryOuterBoundaryRootAdjDistancePeelDataWithEndpointSeparation then gives the raw-carrier image identity and projected-generator separation from this packaged route data, with a matching graph-level wrapper exists_theorem49BoundaryZeroKirchhoffSubspace_interiorEdgeEndpointSupport_image_and_separation_of_admitsPlanarBoundaryOuterBoundaryRootAdjDistancePeelDataWithEndpointSeparation. This package now also yields the first honest positive non-vacuity repair on the endpoint side. The theorem PlanarBoundaryOuterBoundaryRootAdjDistancePeelDataWithEndpointSeparation.selectedBoundaryInteriorEdgeEndpointVertices_eq_interiorEdgeEndpointSupport proves that the packaged endpoint-separation field makes selected-boundary purification lossless on the raw interior-edge endpoint carrier. Therefore theorem49OuterBoundaryRootNonemptySynthesis_of_planarBoundaryOuterBoundaryRootAdjDistancePeelDataWithEndpointSeparation shows that once the same data also has a nonempty raw interiorEdgeEndpointSupport, the current outer-boundary-rooted route really does close to the non-vacuous theorem-4.9 synthesis endpoint. The graph-level wrapper exists_theorem49OuterBoundaryRootNonemptySynthesis_of_exists_planarBoundaryOuterBoundaryRootAdjDistancePeelDataWithEndpointSeparation_with_nonempty_interiorEdgeEndpointSupport makes the repaired target explicit.

There is now a first construction-level no-touch bridge into this endpoint condition. In Theorem49InteriorVerticesAnnulusConstruction.lean, the theorem PlanarBoundaryAnnulusConstruction.endpointSupport_disjoint_of_peelFaces_endpoint_disjoint_selectedBoundarySupport uses the annulus construction's cover of interiorEdgeSupport by peeled-face witness edges: if every peeled face boundary is endpoint-disjoint from the selected-boundary support, then interiorEdgeEndpointSupport is endpoint-disjoint from the selected-boundary endpoint support. The theorem PlanarBoundaryOuterBoundaryRootAdjDistancePeelData.selectedBoundaryInteriorEdgeEndpointVertices_eq_interiorEdgeEndpointSupport_of_peelFaces_endpoint_disjoint_selectedBoundarySupport lifts the same fact to the outer-boundary-rooted annulus data surface, and exists_selectedBoundaryInteriorEdgeEndpointVertices_eq_interiorEdgeEndpointSupport_of_admitsPlanarBoundaryOuterBoundaryRootAdjDistancePeelData_of_peelFaces_endpoint_disjoint_selectedBoundarySupport records the graph-level lossless-purification consequence. The theorem PlanarBoundaryOuterBoundaryRootAdjDistancePeelData.withEndpointSeparationOfPeelFacesEndpointDisjoint then packages this construction-level no-touch proof as PlanarBoundaryOuterBoundaryRootAdjDistancePeelDataWithEndpointSeparation, with graph-level constructor admitsPlanarBoundaryOuterBoundaryRootAdjDistancePeelDataWithEndpointSep of_admitsPlanarBoundaryOuterBoundaryRootAdjDistancePeelData_of_peelFaces_endpoint_disjoint_selectedBoundarySupport. Finally, theorem49BoundaryZeroKirchhoffSubspace_interiorEdgeEndpointSupport_image_and_separation_of_planarBoundaryOuterBoundaryRootAdjDistancePeelData_of_peelFaces_endpoint_disjoint_selectedBoundarySupport therefore gives the raw-carrier $W_0(H)$ image

identity and separation endpoint directly from a peeled-face no-touch obligation, and the graph-level wrapper `exists_theorem49BoundaryZeroKirchhoffSubspace_interiorEdgeEndpointSupport_image_and_separation_of_admitsPlanarBoundaryOuterBoundaryRootAdjDistancePeelData_of_peelFaces_endpoint_disjoint_selectedBoundarySupport` exposes the same endpoint from admitted annulus data plus the per-datum no-touch proof. This does not yet prove the manuscript’s geometry; it identifies the concrete geometric condition that would discharge the raw carrier without falling back to selected-boundary purification.

The positive route has now been pushed one layer further. In `Theorem49Synthesis.lean`, the theorem `theorem49OuterBoundaryRootNonemptySynthesis_of_planarBoundaryOuterBoundaryRootAdjDistancePeelData_of_peelFaces_endpoint_disjoint_selectedBoundarySupport` shows that the same peeled-face endpoint no-touch hypothesis, together with a nonempty raw `interiorEdgeEndpointSupport`, already yields the non-vacuous outer-boundary-root Theorem 4.9 synthesis package. Then `PlanarBoundaryClosedWalkSourceRoute.lean` adds the honest source-facing wrapper `theorem49OuterBoundaryRootNonemptySynthesis_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_of_rootsOuterAmbientBoundary_and_peelFaces_endpoint_disjoint_selectedBoundarySupport` and its graph-level existential form, so the current positive burden is now precise at the closed-walk source surface: prove root localization to the outer ambient boundary, peeled-face endpoint no-touch, and raw-carrier nonemptiness on one embedding.

That positive bridge is now calibrated honestly as well. The same file proves `not_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_and_rootsOuterAmbientBoundary_and_peelFaces_endpoint_disjoint_selectedBoundarySupport_and_nonempty_interiorEdgeEndpointSupport` and `not_exists_closedWalkAnnulusBoundarySource_and_theorem49OuterBoundaryRootNonemptySynthesis_sameBoundaryData`. So on the honest same-boundary split extracted from the closed-walk source, the new outer-root non-vacuity theorem is formally vacuous: the required all-roots-on-the-extracted-outer-boundary witness already contradicts the closed-walk source’s boundary separation, and therefore the resulting outer-boundary-root non-vacuous synthesis endpoint cannot coexist with that same split either.

The honest positive route therefore shifts to the two-sided annulus-root surface. `Theorem49Synthesis.lean` now proves `theorem49AnnulusRootNonemptySynthesis_of_planarBoundaryAnnulusRootAdjDistancePeelData_of_endpointSupport_disjoint` and the graph-level wrapper `exists_theorem49AnnulusRootNonemptySynthesis_of_exists_planarBoundaryAnnulusRootAdjDistancePeelData_with_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport`: for annulus-root data, non-vacuity already follows from nonempty raw `interiorEdgeEndpointSupport` plus endpoint-support disjointness from the selected boundary. Then `PlanarBoundaryClosedWalkSourceRoute.lean` adds the honest source-facing theorem `theorem49AnnulusRootNonemptySynthesis_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_of_endpointSupport_disjoint` and its graph-level existential form. This is the first non-vacuous theorem on a boundary-root surface that survives on the honest closed-walk source shell without any all-roots-on-the-outer-boundary hypothesis.

That annulus-root repair theorem is now calibrated on its own route surface as well. `Theorem49InteriorVerticesEndpointObstruction.lean` proves `annulusRootAdjDistancePeelData_does_not_imply_endpointSupport_disjoint_peeledEndpointTouch` and the fixed-embedding universal form `not_forall_annulusRootAdjDistancePeelData_implies_endpointSupport_disjoint`

peeledEndpointTouch. So the new two-sided annulus-root bridge is an honest conditional repair, not something already forced by arbitrary current annulus-root data.

The same missing burden now has an honest geometric calibration too. `Theorem49PlanarBoundaryAnnulusHonestGeometryRegression.lean` proves `closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_attachedRootedFacePeelCertificate_with_nonempty_interiorEdgeEndpointSupport_without_endpointSupport_disjoint_diamondWithTriangle`: on the genuine cyclic source model `diamondWithTriangleEmbedding`, the honest closed-walk annulus source, a weak one-collar annulus geometry, the attached rooted face-peel certificate endpoint, and a nonempty raw `interiorEdgeEndpointSupport` all coexist, yet endpoint-support disjointness from the selected boundary still fails. So the present endpoint-separation repair lies strictly later than the current honest annulus-geometry and certificate lane as well; it is not already forced by reaching those earlier geometric surfaces. The same file now strengthens this to the repaired geometry surface itself: `closedWalkAnnulusBoundarySource_and_previousBoundaryWitnessGeometry_and_wellFoundedFacePeelWitnessData_with_nonempty_interiorEdgeEndpointSupport_without_endpointSupport_disjoint_diamondWithTriangle`. So even the explicit previous-boundary witness placement needed to feed the boundary-aware well-founded peel endpoint still does not force the endpoint separation required by the non-vacuous annulus-root repair. More sharply, the same honest source model already satisfies `closedWalkAnnulusBoundarySource_and_previousBoundaryWitnessGeometry_and_wellFoundedFacePeelWitnessData_with_nonempty_interiorEdgeEndpointSupport_and_empty_selectedBoundaryInteriorEdgeEndpointVertices_diamondWithTriangle`: the raw endpoint carrier is nonempty, but the purified carrier `selectedBoundaryInteriorEdgeEndpointVertices` is exactly empty. So the current non-vacuous repair is not merely missing a disjointness lemma; it is missing a real geometric exclusion principle on an honest cyclic source model. This now persists one layer higher as well: the same file explicitly builds the collar-layer witness package `diamondWithTriangleOneCollarLayerFacePeelWitnessData`, lowers it to the height-ordered peel witness `diamondWithTriangleOneCollarHeightOrderedFacePeelWitnessData`, and proves `closedWalkAnnulusBoundarySource_and_previousBoundaryWitnessGeometry_and_heightOrderedFacePeelWitnessData_with_nonempty_interiorEdgeEndpointSupport_and_empty_selectedBoundaryInteriorEdgeEndpointVertices_diamondWithTriangle`. So even the explicit layered peel surface still does not force a nonempty purified endpoint carrier on an honest cyclic source model.

The remaining honest burden is therefore sharper: either prove the manuscript's intended annular interior vertices avoid selected-boundary endpoints, or make the purification step explicit in the paper-facing statement.

Independently of that non-vacuity calibration, the positive Theorem 4.9 synthesis layer has now been pushed earlier and made route-independent above the boundary-aware certificate surface. In `Theorem49Synthesis.lean`, the theorem `theorem49BoundaryRootSynthesis_of_boundaryZeroAnnihilatorTrivial` packages the full purified synthesis endpoint from the single algebraic hypothesis that every selected-boundary-zero annihilator of the Definition 4.8 family vanishes. The same file then proves `theorem49BoundaryRootSynthesis_of_planarBoundaryAttachedRootedFacePeelCertificate`, `theorem49BoundaryRootSynthesis_of_planarBoundaryWellFoundedFacePeelWitnessData`, and the graph-level wrappers `exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryAttachedRootedFacePeelCertificate` and `exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryWellFoundedFacePeelWitnessData`.

The older interior-dual and annulus-construction synthesis theorems now factor through this same generic bridge. So once any honest route reaches the attached certificate or well-founded peel witness surface, no further Theorem 4.9 algebra remains; the live burden is entirely upstream geometric construction of that certificate data. More sharply, the same file now proves `theorem49BoundaryRootSynthesis_of_planarBoundaryHeightOrderedFacePeelWitnessData`, `theorem49BoundaryRootSynthesis_of_planarBoundaryCollarLayerFacePeelWitnessData`, `theorem49BoundaryRootSynthesis_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData`, `theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusWitnessAssignment`, `exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryHeightOrderedFacePeelWitnessData`, `exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryCollarLayerFacePeelWitnessData`, `exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData`, and `exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryAnnulusWitnessAssignment`, so theorem-4.9 synthesis already starts at the boundary-aware height-ordered witness-cover surface and even at the weaker finite collar-layer witness-cover surface, hence also at the explicit local remainder annulus surface and at the split “annulus decomposition + witness assignment” surface. `Theorem49PositiveGeometricSourceReplacement.lean` now also proves `theorem49BoundaryRootSynthesis_of_collarLayerPositiveProjectedGeometryOn`, `Theorem49HeightOrderedPositiveProjectedGeometry.boundaryRootSynthesis`, `Theorem49HeightOrderedPositiveProjectedGeometry.exists_boundaryRootSynthesis`, and the matching collar-layer graph-package pair. So once the route already reaches either of those named positive witness-cover packages, full `Theorem49BoundaryRootSynthesis` is immediate; the projected endpoint and raw quotient/error package are now explicit projections of that stronger endpoint rather than the top of those surfaces. `admitsPlanarBoundaryAttachedRootedFacePeelCertificate_of_admitsPlanarBoundaryAnnulusConstructionParentWitnessOrientation`, so the root-distance parent-witness surface already lands on the certificate surface explicitly. The honest closed-walk and successor-cycle source corollaries in `PlanarBoundaryClosedWalkSourceRoute.lean` now factor their direct certificate and synthesis endpoints through that route instead of dropping straight from interior-dual admits to synthesis. The same source file now also proves `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_heightOrderedFacePeelWitnessData_direct`, `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_collarLayerFacePeelWitnessData_direct`, `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_annulusWitnessAssignment_direct` and `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_heightOrderedFacePeelWitnessData_direct`, `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_collarLayerFacePeelWitnessData_direct`, `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_annulusWitnessAssignment_direct`, so the honest source shell already reaches the full synthesis endpoint as soon as the same embedding carries either boundary-aware height-sorted witness-cover data, the weaker finite collar-layer witness-cover package, or the later pure annulus decomposition together with witness-edge ownership. The same route file now also proves the matching direct Step 2 vanishing corollaries `zero_on_allEdges_of_exists_closedWalkAnnulusBoundarySource_and_heightOrderedFacePeelWitnessData_direct_of_annihilatesKempeClosureGeneratorFamily`, `zero_on_allEdges_of`

`exists_closedWalkAnnulusBoundarySource_and_collarLayerFacePeelWitnessData_`
`direct_of_annihilatesKempeClosureGeneratorFamily,` `zero_on_allEdges_of_`
`exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_`
`and_selectedBoundaryArc_and_heightOrderedFacePeelWitnessData_direct_`
`of_annihilatesKempeClosureGeneratorFamily,` `and` `zero_on_allEdges_of_`
`exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_`
`and_selectedBoundaryArc_and_collarLayerFacePeelWitnessData_direct_of_`
`annihilatesKempeClosureGeneratorFamily.` So the positive algebraic boundary now
 matches the positive synthesis boundary: once an honest source shell reaches boundary-
 aware height-sorted witness-cover data, or even just the finite collar-layer witness-
 cover package, no later annulus-geometry packaging is needed either for Step 2 van-
 ishing or for the purified theorem-4.9 endpoint. `theorem49BoundaryRootSynthesis_`
`of_planarBoundaryAnnulusCollarGeometry,` `theorem49BoundaryRootSynthesis_`
`of_planarBoundaryAnnulusPreviousBoundaryWitnessGeometry,` `and` `the`
`graph-level` `wrappers` `exists_theorem49BoundaryRootSynthesis_of_`
`admitsPlanarBoundaryAnnulusCollarGeometry,` `exists_theorem49BoundaryRootSynthesis_`
`of_admitsPlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry,` `and` `exists_`
`theorem49BoundaryRootSynthesis_of_exists_planarBoundaryAnnulusCollarGeometry_`
`and_witnessOnCurrentBoundary.` So for synthesis itself, the burden has now moved even
 earlier than the weak bundled collar geometry surface: once the honest route can produce
 the boundary-aware height-ordered witness-cover surface, or even the weaker finite collar-layer
 witness-cover package, the boundary-aware certificate and the full current Theorem 4.9 end-
 point are automatic. The live positive burden is now strictly upstream of explicit remainder
 bookkeeping and of annulus witness ownership themselves. That upstream geometric bur-
 den is now exposed directly in Lean. `Theorem49PlanarBoundaryAnnulusGeometry.lean` adds
`PlanarBoundaryAnnulusCollarGeometry.toPlanarBoundaryCollarLayerFacePeelWitnessData,`
`PlanarBoundaryAnnulusCollarGeometry.toPlanarBoundaryHeightOrderedFacePeelWitnessData,`
`admitsPlanarBoundaryCollarLayerFacePeelWitnessData_of_admitsPlanarBoundaryAnnulusCollarGeometry,`
`and` `admitsPlanarBoundaryHeightOrderedFacePeelWitnessData_of_`
`admitsPlanarBoundaryAnnulusCollarGeometry,` so weak annulus collar geom-
 etry canonically constructs the current witness-cover surfaces. The same
 file also adds `PlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry.`
`toPlanarBoundaryHeightOrderedFacePeelWitnessData,` `admitsPlanarBoundaryHeightOrderedFacePeelWitn`
`of_admitsPlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry,`
`and` `admitsPlanarBoundaryWellFoundedFacePeelWitnessData_of_exists_`
`planarBoundaryAnnulusCollarGeometry_and_witnessOnCurrentBoundary,` so the re-
 paired witness-on-current-boundary hypothesis lands directly on the well-founded peel
 lane. `PlanarBoundaryClosedWalkSourceRoute.lean` then exposes these bridges on the
 honest source shells through `admitsPlanarBoundaryCollarLayerFacePeelWitnessData_`
`of_exists_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_`
`direct,` `admitsPlanarBoundaryHeightOrderedFacePeelWitnessData_of_`
`exists_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_`
`direct,` `admitsPlanarBoundaryWellFoundedFacePeelWitnessData_of_`
`exists_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_`
`and_witnessOnCurrentBoundary_direct,` `and` `the` `successor-cycle` `compan-`
`ions` `admitsPlanarBoundaryCollarLayerFacePeelWitnessData_of_exists_`
`boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_`
`selectedBoundaryArc_and_annulusCollarGeometry_direct,` `admitsPlanarBoundaryHeightOrderedFacePeel`

`of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_`
`and_selectedBoundaryArc_and_annulusCollarGeometry_direct,` and
`admitsPlanarBoundaryWellFoundedFacePeelWitnessData_of_exists_`
`boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_`
`selectedBoundaryArc_and_annulusCollarGeometry_and_witnessOnCurrentBoundary_direct.`
 So the remaining live burden is now stated cleanly as a geometric one: derive same-
 embedding annulus collar geometry from honest source hypotheses, and, if one wants
 the repaired well-founded lane, also derive the current-boundary witness placement.
 Lean now packages one concrete sufficient criterion for that first geometric step on
 the canonical one-collar route. `Theorem49PlanarBoundaryAnnulusWitness.lean` de-
 fines `PlanarBoundaryCanonicalWitnessChoiceboundaryData`: for each ambient face of
 a fixed annulus boundary split, choose one boundary edge as witness so that all inte-
 rior edges are chosen by some face and every non-witness boundary edge already lies on
 the outer or inner ambient annulus boundary. `PlanarBoundaryCanonicalWitnessChoice.`
`toPlanarBoundaryAnnulusWitnessAssignment` then upgrades the canonical decomposition
`planarBoundaryAnnulusDecomposition_of_boundaryDataboundaryData` to a full annulus
 witness assignment on the same embedding. `PlanarBoundaryClosedWalkSourceRoute.lean`
 packages the source-facing consequence as `admitsPlanarBoundaryAnnulusCollarGeometry_`
`of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_direct.` So
 a surviving honest one-collar route no longer needs an opaque annulus geometry black box:
 it is enough to construct this explicit facewise canonical witness choice on the source bound-
 ary split. This has now been upgraded from graph-level admissibility to actual one-collar
 geometry on the same embedding. In `Theorem49PlanarBoundaryAnnulusGeometry.lean`,
`PlanarBoundaryCanonicalWitnessChoice.toPlanarBoundaryAnnulusCollarGeometry`
 constructs a `PlanarBoundaryAnnulusCollarGeometry` directly from any canon-
 ical witness choice on any annulus boundary split. The companion lemmas
`PlanarBoundaryCanonicalWitnessChoice.toPlanarBoundaryAnnulusCollarGeometry_`
`numCollars` and `PlanarBoundaryCanonicalWitnessChoice.toPlanarBoundaryAnnulusCollarGeometry_`
`boundaryData` prove that the constructed geometry has exactly one collar and preserves the
 original boundary data. The honest-source wrapper `exists_oneCollarAnnulusCollarGeometry_`
`with_sourceBoundaryData_of_exists_closedWalkAnnulusBoundarySource_and_`
`canonicalWitnessChoice` therefore proves one-collar geometry not merely on the diamond example
 but on the full class of honest closed-walk sources whose extracted boundary split carries a canonical
 facewise witness choice. The canonical witness choice itself now has a checked local sufficient con-
 dition. `PlanarBoundaryCanonicalWitnessChoice.of_atMostOneInteriorEdgePerFace` chooses,
 on each face, the unique interior boundary edge when one exists and a supplied fallback boundary
 edge otherwise. Its hypotheses are exactly the facewise geometry needed for the canonical choice: at
 most one interior edge on each face, and every non-interior edge on a face boundary lies on the outer
 or inner ambient annulus boundary. Lean now proves the second clause directly from the source's
 annulus boundary split: `PlanarBoundaryAnnulusBoundaryData.mem_outerAmbientBoundary_`
`union_innerAmbientBoundary_of_mem_faceBoundary_of_not_mem_interiorEdgeSupport` uses
 the incidence partition of ambient face-boundary edges into selected boundary support and
 interior support. Lean now also discharges the fallback edge from the honest closed-walk field
 itself: `PlaneEmbeddingWithBoundary.FaceBoundaryClosedWalk.faceBoundary_nonempty`
 shows that an honest nonempty facial boundary walk has a nonempty face-boundary
 edge set, and `PlanarBoundaryClosedWalkAnnulusBoundarySource.fallbackEdge` chooses
 such an edge with its membership proof. Consequently the new geometry wrap-
 per `exists_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_of_exists_`

`closedWalkAnnulusBoundarySource_and_facewiseUniqueInteriorEdge` constructs same-embedding one-collar collar geometry from an honest source plus only the equality-style facewise uniqueness of interior boundary edges. The source-route cardinal variant `exists_planarBoundaryCanonicalWitnessChoice_of_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge` packages the more convenient finite-cardinality at-most-one hypothesis into the same canonical witness-choice surface, while the older `exists_planarBoundaryCanonicalWitnessChoice_of_closedWalkAnnulusBoundarySource_and_atMostOneInteriorEdgePerFace` records the fully explicit fallback/count/boundary-rest package. The geometry file now carries the corresponding fixed-embedding collar result: `exists_oneCollarAnnulusCollarGeometry_of_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge` constructs a source-preserving one-collar `PlanarBoundaryAnnulusCollarGeometry` directly from an honest source and the facewise cardinal at-most-one hypothesis. No separate fallback or non-interior boundary-rest hypothesis remains at this layer; those are supplied by the honest source and its extracted annulus boundary split. Lean also exposes the graph-level source-facing endpoint `exists_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_of_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdgeCardinality`, which starts from an existential honest closed-walk annulus source plus the same facewise cardinal bound and returns a one-collar collar geometry preserving that source boundary split on the chosen embedding. The one-collar branch now also reaches the repaired previous-boundary-witness surface. The generic theorem `admitsPlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry_of_exists_oneCollarAnnulusCollarGeometry` packages the fact that the positive-collar witness-placement obligation is vacuous when there is only one collar. The honest-source geometry theorem `admitsPlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry_of_exists_closedWalkAnnulusBoundarySource_and_facewiseUniqueInteriorEdge` and the cardinal route theorem `admitsPlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry_of_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge_direct` therefore reduce the surviving source-tied one-collar burden to the at-most-one/unique-interior-edge condition alone. Thus the at-most-one branch no longer stops at weak collar geometry; it can feed the corrected well-founded, height-ordered, and collar-layer witness surfaces. Lean now also exposes this repair at the primitive canonical-choice layer. `PlanarBoundaryCanonicalWitnessChoice.toPlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry` turns a facewise canonical witness choice on an annulus boundary split directly into `PlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry`; internally it uses the canonical one-collar geometry, so the positive-collar witness condition is empty. The source-facing theorem `exists_planarBoundaryAnnulusPreviousBoundaryWitnessGeometry_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice` records the corresponding honest closed-walk source package. This is a positive statement about the canonical one-collar geometry built from the source boundary split, not a theorem that an arbitrary same-embedding weak collar satisfies the repaired placement condition; the latter is still exactly the explicit `WitnessOnCurrentBoundary` obligation. The route-facing admission theorem `admitsPlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_direct` now uses this repaired canonical-choice geometry theorem directly, rather than reconstructing the same landing through the intermediate one-collar existence surface. The same local criterion now reaches the theorem-4.9 synthesis endpoint directly. In `PlanarBoundaryClosedWalkSourceRoute.lean`, `exists_planarBoundaryCanonicalWitnessChoice_of_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge` factors the cardinal at-most-one proof into

the source route without caller fallback or boundary-rest assumptions, and `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge_direct` composes it with the canonical witness-choice route to produce `Theorem49BoundaryRootSynthesis` for any Tait coloring. The nonempty-carrier version is now isolated at the same route surface in the same caller-clean cardinal form: `theorem49BoundaryRootNonemptySynthesis_of_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge_direct` adds only the explicit hypothesis `(selectedBoundaryInteriorEdgeEndpointVerticesemb).Nonempty,` and `exists_theorem49BoundaryRootNonemptySynthesis_of_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge_with_nonempty_selectedBoundaryInteriorEdgeEndpointVertices_direct` packages the corresponding graph-level endpoint. Thus the current positive nonempty synthesis burden is exactly the honest source, facewise cardinal at-most-one, and a surviving selected-boundary-purified carrier; no fallback edge or boundary-rest datum remains caller supplied. The same source package is now connected to the canonical Definition 4.8 spanning bridge, not only to the projected synthesis endpoint. The direct wrappers `exists_theorem49_kempeGeneratorSubspace_spanning_of_exists_closedWalkAnnulusBoundarySource_and_atMostOneInteriorEdgePerFace_heightOrdered_direct` and `exists_theorem49_kempeGeneratorSubspace_spanning_of_exists_closedWalkAnnulusBoundarySource_and_atMostOneInteriorEdgePerFace_collarLayout_direct` compose the repaired one-collar source route with the canonical `Theorem49SubmoduleDualityData` constructor from `Theorem49KempeGeneratorSpan.lean`. Consequently, after the local at-most-one source package is established, the remaining spanning obligations are exactly the named target subspace, containment of the raw Definition 4.8 generator span, and the target's selected-boundary-zero interpretation. The reverse extraction from a stronger one-collar geometry surface is now also formalized. In `Theorem49PlanarBoundaryAnnulusGeometry.lean`, `PlanarBoundaryAnnulusCollarGeometry.toPlanarBoundaryCanonicalWitnessChoice_of_numCollars_eq_one` shows that any one-collar weak annulus geometry whose boundary split is a fixed annulus boundary split already contains the corresponding `PlanarBoundaryCanonicalWitnessChoice`; with only one collar there is no intermediate frontier, so every non-witness edge is forced onto the outer or inner ambient boundary. The source-facing wrapper `exists_planarBoundaryCanonicalWitnessChoice_of_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData` records the exact honest-source form: one-collar collar geometry preserving the boundary split extracted from the source supplies the canonical witness-choice package needed by the direct route theorem. `PlanarBoundaryClosedWalkSourceRoute.lean` now carries this all the way to the theorem-4.9 synthesis endpoint: `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_direct` uses the canonical witness choice to build the split annulus witness assignment consumed by the synthesis layer, and `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData` first extracts that canonical witness choice from same-boundary one-collar geometry. Thus the source-facing positive burden can be stated either as a canonical facewise witness choice or as a stronger one-collar collar geometry preserving the source boundary split; either package now reaches the current formal theorem-4.9 endpoint directly. The source route now pushes these two primitive one-collar surfaces into the canonical Definition 4.8 spanning lane as well. The direct wrappers `exists_theorem49_kempeGeneratorSubspace_spanning_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_heightOrdered_direct` and `exists_theorem49_kempeGeneratorSubspace_spanning_of_exists_closedWalkAnnulusBoundarySource_and_collarLayout_heightOrdered_direct` compose the repaired one-collar source route with the canonical `Theorem49SubmoduleDualityData` constructor from `Theorem49KempeGeneratorSpan.lean`.

`kempeGeneratorSubspace_spanning_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_collarLayout_direct` compose canonical source witness choice with the repaired previous-boundary surface and the height-ordered/collar-layer generator-span bridge. The corresponding one-collar geometry wrappers `exists_theorem49_kempeGeneratorSubspace_spanning_of_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_heightOrdered_direct` and `exists_theorem49_kempeGeneratorSubspace_spanning_of_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_collarLayout_direct` do the same when the source already carries one-collar geometry preserving its boundary split. Thus the theorem-4.9 submodule-duality data no longer has to be reconstructed separately for any of the source-side one-collar packages: the remaining spanning inputs are precisely the target subspace, raw-generator containment, and boundary-zero interpretation. `PlanarBoundaryClosedWalkSourceProjection.lean` records the corrected boundary-zero version of this source bridge. Since the raw Definition 4.8 generator span is not, in general, a subspace of the selected boundary-zero chains, the new wrappers use the boundary-erased generator source `projectedKempeClosureGeneratorSubspace`. Canonical witness choice, the local at-most-one source package, and same-boundary one-collar geometry now each imply both the subspace identity

$$\text{projectedKempeClosureGeneratorSubspace} = \text{planarBoundaryZeroSubmodule}$$

and the explicit-family span version, through either the height-ordered or finite collar-layer witness surface. This exposes a concrete source-side spanning conclusion with no caller-supplied target subspace or raw-generator containment hypothesis. The same projected boundary-zero endpoint has now been lifted to the route-facing successor-cycle source shell. The lowering lemmas in the same file convert successor-cycle boundary data plus selected arc contiguity into the honest closed-walk source package while preserving canonical witness choice, the local at-most-one package, or same-boundary one-collar geometry. The resulting successor-cycle theorems, such as `exists_span_projectedKempeClosureGeneratorFamily_eq_planarBoundaryZeroSubmodule_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_atMostOneInteriorEdgePerFace_heightOrdered_direct` and its collar-layer/one-collar/canonical-choice variants, show that the actual boundary-order source used in the route reaches the corrected projected span conclusion without reintroducing an arbitrary target subspace. `PlanarBoundaryClosedWalkSourceRoute.lean` also carries the same successor-cycle lowering through the raw Definition 4.8 submodule-duality bridge: canonical witness choice, the local at-most-one package, and same-boundary one-collar geometry each have height-ordered and finite-collar wrappers such as `exists_theorem49_kempeGeneratorSubspace_spanning_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_canonicalWitnessChoice_heightOrdered_direct`. These theorems deliberately keep the target subspace, raw-generator containment, and selected-boundary-zero interpretation as hypotheses, marking the unprojected `Theorem49SubmoduleDualityData` obligations rather than hiding them behind the projected boundary-zero endpoint. `PlanarBoundaryClosedWalkSourceRoute.lean` now records the corresponding full synthesis endpoints on the same route-facing source shell. Successor-cycle boundary data plus selected-arc contiguity and canonical witness choice reaches `Theorem49BoundaryRootSynthesis` directly, and with `nonempty_selectedBoundaryInteriorEdgeEndpointVertices` it now also reaches `Theorem49BoundaryRootNonemptySynthesis`. The local at-most-one package has the same synthesis endpoint in the facewise cardinal form `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_selectedBoundaryArc_and_atMostOneInteriorEdgePerFace_heightOrdered_direct`.

`and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_facewiseAtMostOneInteriorEdge_direct`, and with the same nonempty carrier reaches the non-vacuous package via `exists_theorem49BoundaryRootNonemptySynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_facewiseAtMostOneInteriorEdge_with_nonempty_selectedBoundaryInteriorEdgeEndpointVertices_direct`. A same-boundary one-collar annulus geometry on the successor-cycle source also has both the direct synthesis wrapper and, with the nonempty carrier hypothesis, a nonempty-synthesis wrapper. Thus the current route-facing positive endpoint is no longer merely a projected span statement: under the same source-side cardinal or one-collar hypotheses, the full synthesis package is available without a separate caller-supplied closed-walk source, fallback edge, or boundary-rest proof. This positive construction is now witnessed on a genuine honest source model. In `Theorem49PlanarBoundaryAnnulusHonestGeometryRegression`. `lean`, `diamondWithTriangleClosedWalkCanonicalWitnessChoice` constructs `PlanarBoundaryCanonicalWitnessChoice` for the boundary split extracted from `diamondWithTriangleClosedWalkAnnulusBoundarySource`: the shared diamond edge witnesses the two diamond faces and one inner triangle edge witnesses the separate triangle face. The file packages the source-level existence theorem `exists_embedding_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_diamondWithTriangleGraph` and the downstream one-collar witness endpoint `diamondWithTriangleGraph_admitsPlanarBoundaryAnnulusWitnessAssignment_via_closedWalkCanonicalWitnessChoice`. So the canonical-witness burden is no longer only a negatively calibrated hypothesis; it has a checked finite positive source instance. The same finite instance is now non-vacuous at the coloring layer as well: `diamondWithTriangleTaitEdgeColoring` gives an explicit proper nonzero edge coloring of `diamondWithTriangleGraph` by the three nonzero F_2^2 colors, and `diamondWithTriangleTaitEdgeColoring_isTait` verifies the Tait condition. Consequently `diamondWithTriangleEmbedding_theorem49BoundaryRootSynthesis_for_explicitTaitColoring` is a concrete source-to-synthesis theorem for an actual named Tait coloring, not merely a theorem parameterized by a possibly empty coloring class. The diamond benchmark now also reaches that endpoint through the generic at-most-one criterion itself: `exists_closedWalkSource_with_atMostOneInteriorEdgePerFace_diamondWithTriangleGraph` packages the local fallback/count/boundary-rest data, `diamondWithTriangleClosedWalkAnnulusBoundarySource_hasCanonicalWitnessChoice_via_atMostOneInteriorEdgePerFace` reconstructs canonical witness choice from the route theorem, and `diamondWithTriangleEmbedding_theorem49BoundaryRootSynthesis_for_explicitTaitColoring_via_atMostOneInteriorEdgePerFace` proves the same explicit-Tait synthesis endpoint through this local criterion. Lean now also exposes the endpoint as a raw theorem-4.9 quotient/decomposition package: `diamondWithTriangle_closedWalkSource_explicitTait_atMostOne_to_theorem49RawQuotientErrorPackage` bundles the honest source, explicit coloring, and at-most-one route with the raw Definition 4.8 representative and selected-boundary-kernel error decomposition for every Kirchhoff chain on the same embedding. The status theorem `diamondWithTriangle_explicitTait_atMostOne_synthesis_has_empty_selectedBoundaryInteriorCarrier` records that this at-most-one path still does not solve the stronger nonempty target on the diamond benchmark: selected-boundary purification deletes the interior-edge endpoint carrier. Lean now packages the whole currently sufficient same-embedding diamond stack as `diamondWithTriangle_explicitTait_currentSufficientSameEmbeddingGeometry`: the honest source, canonical witness choice, explicit Tait coloring, one-collar geometry, collar-layer, height-ordered, well-founded peel data, attached rooted certificate, and current theorem-4.9 boundary-root synthesis all coexist on the same embedding. The same file also

proves the stronger limitation `diamondWithTriangle_explicitTait_synthesis_has_empty_selectedBoundaryInteriorCarrier`. Thus this benchmark is genuinely positive for the current synthesis surface but still fails `Theorem49BoundaryRootNonemptySynthesis`, because its purified selected-boundary interior carrier is empty. The next positive construction burden is therefore a Tait-colorable honest source with a nonempty purified selected-boundary interior carrier, not another proof of the existing diamond synthesis endpoint. Lean now lifts this limitation to the current projected endpoint itself: `not_theorem49BoundaryRootNonemptyProjectedSynthesis_diamondWithTriangle` shows that no coloring on the diamond-with-triangle embedding can inhabit `Theorem49BoundaryRootNonemptyProjectedSynthesis`, because that package contains the nonempty synthesis carrier obligation. The status theorem `diamondWithTriangle_explicitTait_synthesis_blocks_nonemptyProjectedSynthesis` combines the explicit Tait-colored ordinary synthesis endpoint with this projected-endpoint failure. Thus the projected Kempe-span equalities do not repair an empty selected-boundary interior carrier. The carrier side of that burden is now an exact endpoint-witness condition: `HasUnblockedInteriorEndpoint` names the local condition, and `selectedBoundaryInteriorEdgeEndpointVertices_nonempty_iff_exists_unblocked_interior_endpoint` and `hasUnblockedInteriorEndpoint_iff_selectedBoundaryInteriorEdgeEndpointVertices_nonempty` state that the purified carrier is nonempty exactly when some interior face-incidence edge has an endpoint incident to no selected ambient-boundary edge. This gives future positive examples a small local target for the non-vacuity part, separate from the annulus-collar or witness-cover geometry. Endpoint-separated annulus constructions can now prove that local target by supplying a raw interior endpoint carrier and endpoint-support disjointness, which is the same condition already used by the annulus-root purification bridge. That carrier burden is now separated from the geometry burden by the shared-interior-pair source. Lean proves `selectedBoundaryInteriorEdgeEndpointVertices_nonempty_sharedInteriorPair` using vertex 1, which is incident to the interior shared edge `sip01` but to no selected-boundary edge. Combined with `sharedInteriorPairTaitEdgeColoring_isTait`, this gives a concrete Tait-colorable honest source with nonempty purified carrier; the strengthened regression now also packages this carrier as the local endpoint witness `hasUnblockedInteriorEndpoint_sharedInteriorPair`. The universal obstruction `not_forall_any_replacementPositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` shows that boundary reachability, successor-cycle selected arcs, a Tait coloring, and explicit `HasUnblockedInteriorEndpoint` data still do not force any direct or route-facing replacement package. The obstruction has also been lifted to the honest closed-walk source surface itself: `not_forall_any_replacementPositiveProjectedGeometryOn_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` shows that an honest closed-walk annulus boundary source, a Tait coloring, and the same explicit endpoint witness still do not produce any of the direct or route-facing replacement packages. The companion theorem `not_forall_some_witnessCoverData_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` pushes the obstruction below the package boundary: the same source, coloring, and endpoint hypotheses also do not universally produce either raw height-ordered or collar-layer witness-cover datum. The bundled theorem `not_forall_some_currentSufficientSameEmbeddingGeometry_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` extends the failure to every current same-embedding endpoint known to be sufficient

for synthesis: height-ordered data, collar-layer data, the raw attached certificate, well-founded witness data, and annulus-collar geometry. The exact one-collar current-boundary shell already fails at this lower layer too: `not_forall_some_witnessCoverData_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` and its successor-cycle analogue show that even after adding exact one-collar current-boundary data the shared-interior-pair benchmark still does not force raw height-ordered/collar-layer witness-cover data. The same exact one-collar/v23 shell now also has generic converse failures at the direct-or-route-facing replacement-positive layer: `not_forall_any_replacementPositiveProjectedGeometryOn_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState` and `not_forall_any_replacementPositiveProjectedGeometryOn_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState`. So any explicit same-embedding exact-shell witness carrying simultaneous failure of the current direct and route-facing replacement-positive endpoints already refutes a universal derivation of that whole positive disjunction; the shared-interior-pair shell is just one concrete instance. The same shared-interior-pair benchmark now also blocks the natural residual same-embedding interface on that exact one-collar/v23 shell: `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_naturalResidualSameEmbeddingGeometry_sharedInteriorPair`, `not_forall_some_naturalResidualSameEmbeddingGeometry_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair`, and the successor-cycle analogues show that even after adjoining a real exact-v23 residual initial state, exact one-collar current-boundary data, a Tait coloring, and `HasUnblockedInteriorEndpoint` still do not force residual-boundary witness data, residual selector data, or even raw height-ordered/collar-layer witness-cover data on the same embedding. The same shared-interior-pair shell now also fails the earliest selector target directly: `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFreeIncidentFaceSelector_sharedInteriorPair`, `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFreeIncidentFaceSelector_sharedInteriorPair`, `not_forall_nonempty_boundaryFreeIncidentFaceSelector_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair`, and `not_forall_nonempty_boundaryFreeIncidentFaceSelector_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair`. So on the exact one-collar/v23 shell the first same-embedding selector burden al-

ready fails directly, not only as one branch of a broader bundled obstruction. The same exact-shell shared-interior-pair benchmark now also fails the next construction surface directly: `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusConstructionBoundarySupportFaceData_sharedInteriorPair,` `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusConstructionBoundarySupportFaceData_sharedInteriorPair,` `not_forall_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair,` `and` `not_forall_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair.` So the exact one-collar/v23 shell now fails not only the earliest selector package but also the selected-boundary-contact construction package consumed by the later synthesis route. The earlier route package `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_without_currentSufficientSameEmbeddingGeometry_sharedInteriorPair` shows at the same time that this still does not provide the witness/certificate geometry needed for the current synthesis route. A second carrier-calibration benchmark now avoids leaning on the shared-interior-pair counterexample itself. The new module `Theorem49PlanarBoundaryAnnulusWheelWitnessRegression`.lean defines `wheelWithInnerTriangleGraph`, with three interior spokes meeting a central vertex and a separate triangular selected-boundary component. Lean proves `wheelWithInnerTriangleTaitEdgeColoring_isTait` and `selectedBoundaryInteriorEdgeEndpointVertices_nonempty_wheelWithInnerTriangle,` with the carrier proof now routed through the same unblocked-endpoint criterion via the central spoke, then packages them as `wheelWithInnerTriangle_tait_and_nonempty_selectedBoundaryInteriorCarrier`. The same module now constructs an honest closed-walk annulus boundary source, `wheelWithInnerTriangleClosedWalkAnnulusBoundarySource`, with nonempty wrapper `nonempty_wheelWithInnerTriangleClosedWalkAnnulusBoundarySource`. It is therefore a source-bearing Tait-plus-carrier benchmark, but Lean proves that it cannot be promoted to the canonical one-collar route: `exists_two_distinct_interior_edges_on_wheelWithInnerTriangle_boundary` exhibits a face with two distinct interior spokes, `not_nonempty_planarBoundaryCanonicalWitnessChoice_wheelWithInnerTriangle` rules out canonical witness choice for every boundary split on this embedding, and `not_exists_oneCollar_planarBoundaryAnnulusCollarGeometry_wheelWithInnerTriangle` rules out one-collar collar geometry. The status theorem `wheelWithInnerTriangle_tait_and_nonempty_carrier_without_canonical_or_oneCollar` keeps the nonempty-carrier/coloring burden distinct from the stronger same-embedding witness-ownership burden. The same benchmark has now been calibrated against the face-local at-most-one sufficient condition. Lean proves

`wheelWithInnerTriangle_face0_interiorEdgeFilter_card`: on face 0, the interior-edge filter has cardinality 2. Hence `not_wheelWithInnerTriangle_atMostOneInteriorEdgePerFace` blocks the local premise used by `PlanarBoundaryCanonicalWitnessChoice.of_atMostOneInteriorEdgePerFace`, and `not_exists_closedWalkSource_with_atMostOneInteriorEdgePerFace_wheelWithInnerTriangleEmbedding` shows that no honest closed-walk source on this fixed embedding can satisfy the full fallback/at-most-one/boundary-rest package. The sharper bundle `wheelWithInnerTriangle_tait_and_nonempty_carrier_without_atMostOne_or_canonical_or_oneCollar` therefore records that Tait colorability and nonempty purified carrier still do not imply the at-most-one local geometry, canonical witness choice, or one-collar geometry. The same fixed wheel embedding now blocks the raw replacement witness-cover surface as well. Lean proves `not_nonempty_planarBoundaryHeightOrderedFacePeelWitnessData_wheelWithInnerTriangle`: covering one central spoke forces the remaining spoke on that triangular face to be witnessed at a strictly later height, and following the three wheel faces forces a strict cycle through `wit01`, `wit02`, and `wit03`. The same dependency cycle now blocks the acyclic parent-peeling surface: `Theorem49PlanarBoundaryPeeling`. Lean proves `PlanarBoundaryWellFoundedFacePeelWitnessData.toPlanarBoundaryHeightOrderedFacePeelWitnessData` using the canonical parent-height rank, and the wheel regression applies this as `not_nonempty_planarBoundaryWellFoundedFacePeelWitnessData_wheelWithInnerTriangle`. This blocker is now factored out of the wheel proof itself. The same peeling file proves that, under injective witness choices on peeled faces, a non-boundary remainder edge equal to another peeled face's witness edge forces a parent dependency in the well-founded package and a strict inequality in the height-ordered package. The list-level order lemma `false_of_isChain_height_lt_and_getLast_lt_head` and the two `false_of_witnessRemainderDependency_cycle_of_injective_witness` lemmas therefore turn any nonempty closed non-boundary remainder-witness dependency chain into a formal blocker, independent of the particular wheel names. The wheel regression now uses the same list-level obstruction directly in its height-ordered no-data proof, applying it to the two possible three-face cycles around the spokes instead of closing those branches by a bespoke transitive-irreflexive calculation on natural-number heights. The well-founded parent side now also has the parent-map analogue `false_of_isChain_parentRel_and_getLast_parent_head`: a nonempty finite closed chain in `ParentRel` contradicts the well-founded parent map directly. Accordingly, the well-founded witness-cover finite-cycle blocker now first extracts actual parent edges and applies this parent-chain theorem, instead of rebuilding the obstruction locally as a parent-height cycle. The finite collar-layer version is ruled out by `not_nonempty_planarBoundaryCollarLayerFacePeelWitnessData_wheelWithInnerTriangle`, since collar-layer data lowers to height-ordered witness-cover data. The obstruction also reaches weak annulus-collar geometry itself: `not_nonempty_planarBoundaryAnnulusCollarGeometry_wheelWithInnerTriangle` uses the canonical collar-to-height lowering, and `not_forall_some_acyclicWitnessCover_or_annulusCollarGeometry_of_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle` packages the failed universal from Tait colorability and nonempty purified carrier to any current acyclic witness-cover or weak collar endpoint. Thus the missing positive construction is not merely a nonempty carrier or Tait coloring condition; it must also break or orient the interior-face dependency cycle before even the well-founded/height/collar witness surfaces, and hence weak annulus-collar geometry, can be expected. The source-bearing form is now checked as well: `wheelWithInnerTriangle_closedWalkSource_tait_and_nonempty_carrier_without_acyclicWitnessCover_or_annulusCollarGeometry` bundles the honest closed-walk source, Tait

coloring, nonempty purified carrier, and failure of all current acyclic witness-cover and weak annulus-collar endpoints. Its universal obstruction `not_forall_some_acyclicWitnessCover_or_annulusCollarGeometry_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle` shows that adding the honest closed-walk annulus source to the wheel hypotheses still does not construct the positive geometry; the missing input has to break or orient the dependency cycle. The same wheel model now also has an explicit forcing-edge calibration. `Theorem49ForcingInteriorEdgeObstructionRegression.lean` proves `wheelWithInnerTriangleForcingInteriorEdgeWitness`: the central spoke `wit01` is an interior edge, and every face incident to it already contains selected-boundary support, namely `wit12` on one wheel face and `wit31` on the other. The generic forcing-edge obstruction then yields `not_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_wheelWithInnerTriangle_via_forcingInteriorEdgeWitness` and `not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_wheelWithInnerTriangle_via_forcingInteriorEdgeWitness`, as well as `not_nonempty_planarBoundaryAnnulusConstructionFaceLayerData_wheelWithInnerTriangle_via_forcingInteriorEdgeWitness`. The bundled theorem `wheelWithInnerTriangle_tait_nonempty_carrier_and_forcing_obstruction` records the sharper status: the benchmark has an explicit Tait coloring and nonempty purified carrier, but it carries the same obstruction that blocks the current annulus-root and face-layer construction routes on the fixed embedding. The same source-side obstruction is now visible without choosing a benchmark. In `Theorem49PlanarBoundaryPeeling.lean`, `PlanarBoundaryHeightOrderedFacePeelWitnessData.exists_terminal_face_selectedBoundary_remainders` extracts a terminal peeled face from any nonempty height-ordered witness-cover package: all non-witness remainders on that face already lie on selected boundary support. For `InteriorDualBoundaryRootAdjDistancePeelData`, peeled faces are disjoint from selected-boundary support, so `InteriorDualBoundaryRootAdjDistancePeelData.exists_terminal_peelFace_witness_erase_eq_empty_of_mem_interiorEdgeSupport` forces that terminal face to have no non-witness boundary edges. The corollary `InteriorDualBoundaryRootAdjDistancePeelData.exists_peelFace_card_le_one_of_mem_interiorEdgeSupport` therefore says that any IDBRAD package covering even one interior edge contains a peeled face with boundary cardinality at most one. This is a direct source-side necessary condition: a proposed boundary-order derivation of IDBRAD must explain the terminal singleton-edge face rather than only supplying coverage and root distances. The wheel-with-inner-triangle regression now instantiates this criterion on a concrete source. Lean proves `wheelWithInnerTriangleFaceBoundary_card_eq_three` and `wheelWithInnerTriangle_two_le_faceBoundary_card`, so every ambient face in that benchmark has at least two boundary edges. With `wit01_mem_interiorEdgeSupport`, the theorem `not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_wheelWithInnerTriangle_via_terminal_card_lower_bound` refutes IDBRAD without appealing to the forcing-edge selector interface. The source-facing package `wheelWithInnerTriangle_closedWalkSource_tait_hasUnblockedEndpoint_without_interiorDualBoundaryRootAdjDistancePeelData_via_terminal_card` keeps the honest closed-walk source, Tait coloring, and local unblocked endpoint on the same embedding while deriving the contradiction through the terminal-card obstruction. The same concrete source is now connected to the actual projected Theorem 4.9 surface. Lean adds `wheelWithInnerTriangleGraph_edgeSet_fintype` and `hasUnblockedInteriorEndpoint_wheelWithInnerTriangle`, then proves `wheelWithInnerTriangle_theorem49BoundaryRootNonemptyProjectedSynthesis`

of_interiorDualBoundaryRootAdjDistancePeelData: if the wheel embedding had a same-embedding IDBRAD datum, the existing source bridge would yield Theorem49BoundaryRootNonemptyProjectedSynthesis for the concrete wheel Tait coloring. The raw Definition 4.8 endpoint is also instantiated as wheelWithInnerTriangle_theorem49BoundaryRawQuotientErrorPackage_of_interiorDualBoundaryRootAdjDistancePeelData, and wheelWithInnerTriangle_source_to_theorem49_and_terminal_card_contradiction_of_interiorDualBoundaryRootAdjDistancePeelData records that this same hypothetical datum is impossible by the terminal-card criterion. The regression now states the same obstruction directly at the blocked projected endpoint: not_exists_wheelWithInnerTriangle_closedWalkSource_and_interiorDualBoundaryRootAdjDistancePeelData_and_theorem49BoundaryRootNonemptyProjectedSynthesis rules out coexistence of the closed-walk source, a same-embedding IDBRAD datum, and the projected Theorem 4.9 endpoint on the wheel embedding. The bundled source-facing theorem wheelWithInnerTriangle_closedWalkSource_tait_hasUnblockedEndpoint_without_interiorDualBoundaryRootAdjDistancePeelData_based_projectedSynthesis keeps the honest source, concrete Tait coloring, and unblocked endpoint while ruling out an IDBRAD-based projected-endpoint instantiation. The same wheel regression now adds the parent-route counterpart wheelWithInnerTriangle_closedWalkSource_tait_hasUnblockedEndpoint_without_boundaryFaceRootsCanonicalParent_based_projectedSynthesis: the positive source, Tait, and endpoint facts remain present, but no projected endpoint can be obtained from source boundary-face roots plus canonical parent child-membership on this embedding. The cyclic-source derivation file has now been repaired into the corresponding checked failed universal. In Theorem49PlanarBoundaryAnnulusHonestWitnessRegression.lean, Lean proves sharedInteriorPair_sipFace0_interiorEdgeFilter_card: on the honest sharedInteriorPairAnnulusEmbedding, the face sipFace0 filters to the two interior edges sip01 and sip12. Hence not_sharedInteriorPair_atMostOneInteriorEdgePerFace. The route-facing module Theorem49CyclicSourceAtMostOneDerivation.lean now proves route_impossibility_at_most_one_not_derivable_from_cyclic_source and not_forall_closedWalkSource_implies_atMostOneInteriorEdgePerFace: an honest closed-walk annulus boundary source can exist while the face-local at-most-one premise fails. Thus the at-most-one condition remains an independent sufficient hypothesis, not a theorem of cyclic or closed-walk boundary-order source data. The strengthened-condition derivation file now makes the same point on the route-facing successor-cycle surface. In Theorem49StrengthenedConditionAtMostOneDerivation.lean, the theorem exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_without_atMostOneInteriorEdgePerFace_sharedInteriorPair packages the shared-interior-pair embedding with annulus boundary reachability, dart-successor facial geometry, and selected-boundary arcs, while not_forall_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryComponentConstant_implies_atMostOneInteriorEdgePerFace_sharedInteriorPair adds the boundary-component constancy normally obtained from the honest source fields. The same ‘sipFace0’ two-interior-edge obstruction still refutes the at-most-one conclusion. The route summary file now packages these separate calibrations as the checked theorem theorem49_route_obstruction_matrix in Theorem49DerivationImpossibilitySummary.lean. Its four components record the cyclic-source counterexample to at-most-one, the chained-diamonds counterexample to deriving a nonempty purified carrier from source plus at-most-one, the strengthened successor-cycle failed universal with boundary-component constancy, and the

wheel-with-inner-triangle Tait/nonempty-carrier model carrying a forcing interior edge that blocks the current annulus-root and face-layer construction routes. This has now been extended to a second, strictly larger positive model rather than another constructor theorem. The same regression file defines `chainedDiamondsWithTriangleGraph`, whose outer boundary component contains two chained diamond cells and whose inner boundary component is a separate triangle. The explicit source `chainedDiamondsClosedWalkAnnulusBoundarySource` supplies honest facial closed walks and selected-boundary arcs on all five faces. The direct finite witness `chainedDiamondsClosedWalkCanonicalWitnessChoice` chooses the two distinct interior shared edges `cdt12` and `cdt45` on their respective diamond face pairs and a boundary edge on the inner triangle. Lean now proves `chainedDiamondsWithTriangle_interiorEdgeSupport_eq`: the interior edge support is exactly $\{\text{cdt12}, \text{cdt45}\}$, and `chainedDiamondsWithTriangle_atMostOneInteriorEdgePerFace`: each face has at most one interior edge on its boundary, so this model satisfies the facewise local condition enabling the canonical-choice constructor. The fallback and boundary-rest obligations are now explicit as `chainedDiamondsCanonicalWitnessEdge_mem` and `chainedDiamonds_nonInteriorOnAmbient`, and the full generic premise is packaged as `exists_closedWalkSource_with_atMostOneInteriorEdgePerFace_chainedDiamondsWithTriangleGraph`. Lean reconstructs the canonical witness choice through the generic constructor as `chainedDiamondsClosedWalkAnnulusBoundarySource_hasCanonicalWitnessChoice_via_atMostOneInteriorEdgePerFace` and lowers it to `chainedDiamondsWithTriangleGraph_admitsPlanarBoundaryAnnulusWitnessAssignment_via_atMostOneInteriorEdgePerFace` and `chainedDiamondsWithTriangleGraph_admitsPlanarBoundaryAnnulusCollarGeometry_via_atMostOneInteriorEdgePerFace`. The endpoint bridge also applies to this larger benchmark: `chainedDiamondsWithTriangleEmbedding_theorem49BoundaryRootSynthesis_via_atMostOneInteriorEdgePerFace` and `exists_theorem49BoundaryRootSynthesis_chainedDiamondsWithTriangleGraph_via_atMostOneInteriorEdgePerFace` show that, conditional on a Tait coloring, the local criterion reaches the same theorem-4.9 synthesis surface as the named one-collar geometry. The same benchmark is now calibrated against the stronger nonempty-carrier surface: `selectedBoundaryInteriorEdgeEndpointVertices_eq_empty_chainedDiamondsWithTriangle` and `not_selectedBoundaryInteriorEdgeEndpointVertices_nonempty_chainedDiamondsWithTriangle` show that the purified carrier is empty, because every endpoint of `cdt12` and `cdt45` is incident to a selected-boundary edge. Together with `not_exists_taitEdgeColoring_chainedDiamondsWithTriangleGraph`, the status theorem `chainedDiamonds_atMostOne_source_has_empty_carrier_and_no_tait` records that this model supplies at-most-one source geometry but neither a surviving carrier nor a Tait coloring. The route-impossibility module now makes this obstruction a checked failed universal, not just a benchmark note: `route_impossibility_at_most_one_does_not_imply_nonempty_carrier` gives the concrete `Fin10` counterexample, `not_forall_atMostOneInteriorEdgePerFace_implies_nonempty_carrier` rules out deriving nonempty purified carrier from at-most-one alone, and `not_forall_closedWalkSource_and_atMostOneInteriorEdgePerFace_implies_nonempty_carrier` shows that adding the honest closed-walk annulus boundary source still does not recover nonemptiness. The stronger route-facing successor-cycle source shell now gives a sharper diagnosis. `Theorem49AtMostOneNonemptyCarrierImpossibility.lean` constructs explicit facial successor maps on the diamond-with-triangle model in `diamondWithTriangleDartSuccessorCycleGeometry`, proves selected-arc contiguity by `diamondWithTriangleDartSuccessorCycleGeometry_selectedBoundaryArcOnFace`, and packages the counterexample as `successor_cycle_at_most_one_and_empty_carrier_independent`. The failed universal `not_forall_boundaryReachabilityData_`

`and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_atMostOneInteriorEdgePerFace_implies_nonempty_carrier` is now explained by a universal obstruction: `selectedBoundaryInteriorEdgeEndpointVertices_eq_empty_of_closedWalkEmbeddingData_and_facewiseAtMostOneInteriorEdge` proves that honest facial closed-walk data plus facewise at-most-one forces the purified interior-edge endpoint carrier to be empty on any embedding. The closed-walk contradiction forms `not_exists_closedWalkEmbeddingData_and_facewiseAtMostOneInteriorEdge_and_nonempty_selectedBoundaryInteriorEdgeEndpointVertices` and `not_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge_and_nonempty_selectedBoundaryInteriorEdgeEndpointVertices` rule out the current honest-source at-most-one non-vacuous endpoint outright. The same obstruction is now available in constructive contrapositive form: `exists_two_distinct_interior_edges_on_faceBoundary_of_closedWalkEmbeddingData_and_nonempty_selectedBoundaryInteriorEdgeEndpointVertices` and the route-facing `exists_two_distinct_interior_edges_on_faceBoundary_of_closedWalkAnnulusBoundarySource_and_hasUnblockedInteriorEndpoint` show that any honest-source endpoint witness for the current carrier must exhibit a face boundary with two distinct interior edges. The positive annulus-collar replacement surface now exposes this obstruction directly: `exists_two_distinct_interior_edges_on_faceBoundary_of_closedWalkAnnulusCollarPositiveProjectedGeometryOn` consumes `Theorem49ClosedWalkAnnulusCollarPositiveProjectedGeometryOn`, so that route package is explicitly outside the facewise at-most-one branch before any theorem-4.9 synthesis endpoint is invoked. The same route file now also proves `exists_two_distinct_interior_edges_on_faceBoundary_of_successorCycleAnnulusCollarPositiveProjectedGeometryOn`, so the successor-cycle annulus-collar shell carries that same two-interior-edge obstruction directly rather than only through silent lowering to closed-walk data. Correspondingly, `not_theorem49ClosedWalkAnnulusCollarPositiveProjectedGeometryOn_of_facewiseAtMostOneInteriorEdge` and `not_theorem49SuccessorCycleAnnulusCollarPositiveProjectedGeometryOn_of_facewiseAtMostOneInteriorEdge` show that neither route-facing annulus-collar positive package can inhabit the canonical facewise-at-most-one branch. The stronger selected-boundary-contact construction shell is now blocked there as well: `not_theorem49ClosedWalkAnnulusConstructionBoundarySupportPositiveProjectedGeometryOn_of_facewiseAtMostOneInteriorEdge` and `not_theorem49SuccessorCycleAnnulusConstructionBoundarySupportPositiveProjectedGeometryOn_of_facewiseAtMostOneInteriorEdge` show that even that stronger source-facing package cannot inhabit the same branch, while `exists_two_distinct_interior_edges_on_faceBoundary_of_successorCycleAnnulusConstructionBoundarySupportPositiveProjectedGeometryOn` exposes the same local two-interior-edge obstruction directly on the successor-cycle selected-boundary-contact shell. The older successor-cycle theorem `selectedBoundaryInteriorEdgeEndpointVertices_eq_empty_of_dartSuccessorCycleEmbeddingData_and_facewiseAtMostOneInteriorEdge` and its contradiction form `not_exists_dartSuccessorCycleEmbeddingData_and_facewiseAtMostOneInteriorEdge_and_nonempty_selectedBoundaryInteriorEdgeEndpointVertices` shows that the current successor-cycle at-most-one branch cannot be made non-vacuous by merely adding the existing carrier nonemptiness hypothesis. The older direct package remains available as `exists_embedding_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_chainedDiamondsWithTriangleGraph` and lowers it to `chainedDiamondsWithTriangleGraph_admitsPlanarBoundaryAnnulusWitnessAssignment_via_closedWalkCanonicalWitnessChoice`. Thus the positive side is no longer represented by a single one-interior-edge toy instance: canonical witness choice is inhabited on an honest source with two separate interior witness

edges and a connected outer boundary component, and the generic at-most-one theorem itself is non-vacuous on that model. The same model now reaches the one-collar geometry surface as a named concrete object: `chainedDiamondsOneCollarGeometry`. Lean proves `chainedDiamondsOneCollarGeometry_numCollars` and `chainedDiamondsOneCollarGeometry_boundaryData`, so this geometry has exactly one collar and preserves the boundary split extracted from `chainedDiamondsClosedWalkAnnulusBoundarySource`. The source-level existence theorem `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_chainedDiamondsWithTriangleGraph` and graph-level admissibility theorem `chainedDiamondsWithTriangleGraph_admitsPlanarBoundaryAnnulusCollarGeometry_via_closedWalkCanonicalWitnessChoice` show that the larger concrete model inhabits the same one-collar route surface that the theorem-4.9 synthesis layer consumes. The same model now also reaches the downstream certificate surfaces. The vacuous positive-collar placement theorem `chainedDiamondsOneCollarWitnessOnCurrentBoundary` upgrades the one-collar geometry to `chainedDiamondsOneCollarPreviousBoundaryWitnessGeometry`; Lean then constructs `chainedDiamondsOneCollarCollarLayerFacePeelWitnessData`, `chainedDiamondsOneCollarHeightOrderedFacePeelWitnessData`, `chainedDiamondsOneCollarWellFoundedFa` and the attached certificate `chainedDiamondsOneCollarAttachedRootedFacePeelCertificate`. The graph admissibility endpoints through these surfaces, together with `closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_attachedRootedFacePeelCertificate_chainedDiamondsWithTriangle`, show that the larger honest source is not merely an annulus-geometry example: it already inhabits the certificate branch consumed by the current synthesis layer. Finally, `chainedDiamondsWithTriangleEmbedding_theorem49BoundaryRootSynthesis_via_oneCollarGeometry` and `exists_theorem49BoundaryRootSynthesis_chainedDiamondsWithTriangleGraph_via_oneCollarGeometry` package the concrete theorem-4.9 boundary-root synthesis endpoint for any Tait coloring of this finite graph. The coloring quantifier is now marked honest: `not_exists_taitEdgeColoring_chainedDiamondsWithTriangleGraph` proves that this larger graph has no Tait edge coloring, since four distinct edges incident with vertex 3 would need four pairwise distinct nonzero colors while only three are available. Thus the chained model remains valuable as a larger positive honest-source, one-collar, and certificate construction, but its theorem-4.9 synthesis endpoint is formally vacuous as a Tait instance.

This criterion is now calibrated negatively as well. The generic theorem `not_nonempty_planarBoundaryCanonicalWitnessChoice_of_exists_two_distinct_interior_edges_on_faceBoundary` in `Theorem49PlanarBoundaryAnnulusWitness.lean` shows that if one ambient face carries two distinct interior edges, the canonical face-wise witness choice itself cannot exist, since such a choice would upgrade to the already forbidden canonical witness assignment. The abstract boundary-data countermodel is recorded by `not_forall_planarBoundaryAnnulusBoundaryData_implies_nonempty_canonicalWitnessChoice` in `Theorem49PlanarBoundaryAnnulusWitnessRegression.lean`. The honest source countermodel is recorded by `not_forall_exists_planarBoundaryCanonicalWitnessChoice_of_closedWalkAnnulusBoundarySource_sharedInteriorPair` in `Theorem49PlanarBoundaryAnnulusHonestWitnessRegression.lean`. Thus the explicit canonical-witness-choice repair is a genuine extra geometric hypothesis, not a consequence of annulus boundary data or of the honest closed-walk annulus source.

4 Threshold assessment

The current obstruction suite should not be overstated. It gives high confidence, checked in Lean, against three specific shortcuts:

- weak collar geometry alone does not imply parent-witness orientation;
- the honest closed-walk two-component boundary source is incompatible with the current all-roots-on-the-outer-boundary adjacency-distance peel data when both use the same boundary split;
- edge-disjoint peeled annulus data does not imply endpoint no-touch for the raw interior-edge endpoint carrier;
- IDBRAD data covering an interior edge forces a terminal peeled face of boundary cardinality at most one, so source geometries whose peeled faces are all multi-edge faces cannot instantiate the current package directly;
- selected-boundary purification can be vacuous in endpoint-touch models, so non-vacuous use of the purified Theorem 4.9 endpoint needs an explicit nonemptiness or no-touch hypothesis.

Those are serious route blockers for the manuscript as currently written. However, they do not yet meet the $> 90\%$ threshold for declaring the whole Goertzel proof route robustly broken. The reason is also formal: the Lean tree still contains positive replacement surfaces, but their scope is now narrower. The root-distance annulus construction proves parent-witness orientation from `InteriorDualBoundaryRootAdjDistancePeelData`, and the current synthesis module now factors the purified Theorem 4.9 endpoint through generic `PlanarBoundaryInteriorDualBoundaryRootAdjDistancePeelData`, re-exposing it on both `PlanarBoundaryOuterBoundaryRootAdjDistancePeelData` and the two-sided annulus-root surface. The new closed-walk obstruction shows that the old outer-root package cannot be instantiated from the honest closed-walk two-component boundary split while retaining the all-outer-root condition. The new two-sided `PlanarBoundaryAnnulusRootAdjDistancePeelData` surface is the current ground-up replacement for that exact failure mode: it retains the existing interior-dual proof pipeline but removes the all-outer-root localization that the closed-walk source refutes. These surfaces therefore remain useful theorem interfaces, but a continuation must now source the two-sided root data from closed-walk or cyclic boundary-order geometry rather than treating the old outer-root package as the direct annulus endpoint.

So the present verdict is deliberately conservative. The accumulated crux work is enough to say that the weak v23 annulus interfaces cannot be accepted as proof; it is not enough to say that every strengthened annulus/no-touch or selected-boundary purification strategy is blocked. Further crux polishing at the same endpoint is now lower value than ground-up work on the missing geometric source: closed-walk or cyclic boundary-order data, parent-in-peeled annulus geometry, and non-vacuous selected-boundary interior carriers.

5 Status summary

Claim surface	Status	Evidence in hand
Older per-run purification claim $\gamma \cdot 1_R$	Refuted	<code>GoertzelLemma43Obstruction.lean</code> and the archived old branch record
Exact v23 Lemma 4.3 run-invariance statement	Not refuted here	manuscript audit only; no direct Lean counterexample presently attached
Exact v23 Step 2 claim $c \cdot 1_{D \cap \partial f}$ Manuscript spanning-forest leaf reduction	Proved Refuted as stated; repaired in live Lean route	<code>GoertzelLemma44.lean</code> <code>Theorem49ForestRegression.lean</code> , <code>Theorem49InteriorDualForest.</code> <code>lean</code> , <code>PlanarBoundaryAnnulusConstruction.</code> <code>lean</code>
Manuscript adjacency-only well-founded peeling reduction	Refuted as stated; repaired in live Lean route	<code>Theorem49ForestRegression.lean</code> , <code>Theorem49InteriorDualForest.</code> <code>lean</code>
Live boundary-root adjacency-distance / annulus-construction route	Formalized theorem surface	<code>Theorem49InteriorDualForest.</code> <code>lean</code> , <code>Theorem49PlanarBoundaryOuterBoundaryRootInterior.</code> <code>lean</code> , <code>PlanarBoundaryAnnulusConstruction.</code> <code>lean</code>
Weak collar geometry implies parent-witness orientation	Refuted by lowered countermodel	<code>Theorem49PlanarBoundaryAnnulusWitnessRegression.</code> <code>lean</code> , <code>PlanarBoundaryAnnulusConstruction.</code> <code>lean</code>
Honest source plus arbitrary weak collar implies repaired previous-boundary witness placement	Refuted on a connected source embedding	<code>Theorem49HonestSourceForcesPreviousBoundaryWitness.</code> <code>lean</code>
Closed-walk boundary source plus all-outer-root peel data on the same split	Refuted by structural contradiction	<code>Theorem49ClosedWalkOuterRootObstruction.</code> <code>lean</code>
Global > 90% proof-route crux block	Not reached; shift to ground-up geome- try/synthesis obligations	obstruction files plus positive replacement surfaces in <code>PlanarBoundaryAnnulusConstruction.</code> <code>lean</code> and <code>Theorem49Synthesis.lean</code>
Current embedding-side boundary-order dependency Interior-vertex side condition for boundary erasure	Still the main live blocker Purification bridge proved; annulus- construction edge-disjoint peeled faces do not imply the exact peeled-face endpoint	<code>fourcolor-boundary-order-notes.</code> <code>tex</code> and regression files <code>Theorem49InteriorVertices.lean</code> , <code>Theorem49InteriorVerticesEndpointObstruction.</code> <code>lean</code>

6 What this means for continuation

The correct present verdict is:

- the current route is still blocked in the later Theorem 4.9 / boundary-order part;
- the obstruction is *not* just lack of labor;
- the older per-run contradiction remains important historical evidence;
- the exact v23 Step 2 purification claim is now formally supported;
- the manuscript's naive leaf-peeling reductions are false as stated, but the live Lean route already replaces them with stronger adjacency-distance, root-distance, height, and well-founded parent interfaces;
- honest closed-walk source data plus arbitrary same-embedding weak collar geometry does not by itself repair the previous-boundary witness placement; the collar must be tied to the source split, strengthened, or accompanied by the explicit current-boundary witness condition;
- the all-outer-root version of the adjacency-distance route cannot be instantiated from the honest closed-walk two-component boundary split;
- any sound v23.5 theorem targeting the downstream construction face-layer shell must already discharge the local boundary-free-selector / no-forcing obligation there; that burden is not postponed to later decomposition or projection wrappers;
- the remaining blocker is to derive those stronger interfaces from honest annulus / boundary-order geometry.
- the current evidence does not justify declaring the full route $> 90\%$ blocked; it justifies shifting effort from more endpoint counterexamples to ground-up formalization of the stronger annulus geometry.

This also means that v23 should not be described as the first version to have faced a machine-level challenge. Rather, it is the version that followed an earlier contradiction and was supposed to repair that weakness. The present formalization now shows that this specific local repair succeeds, while the later route is still incomplete.

This does *not* by itself prove that no repair of the overall Goertzel route can exist. But it does show that any viable continuation must revise the current theorem surfaces and interfaces. In particular, a repair would have to do some combination of the following:

1. continue from the now-checked local purification claim into the later Theorem 4.9 route without reintroducing stronger unsupported reductions;
2. derive the already-formalized boundary-root adjacency-distance package from the intended annulus geometry, rather than falling back to the refuted leaf heuristic;
3. derive the already-formalized root-distance / height / well-founded peeling data from that geometry, rather than using adjacency-only parent arguments;
4. derive ordered boundary-walk information from stronger embedding-side data, or else postulate and track that stronger data explicitly;
5. identify the intended annulus interior vertex set with the checked selected-boundary purification used to discharge the boundary-erasure side condition.

7 Relevant files

File	Role
.../GoertzelLemma43Obstruction.lean	Older per-run purification obstruction
.../GoertzelLemma44.lean	Exact v23 Step 2 purification theorem
.../Theorem49ForestRegression.lean	Current later-route forest / peeling regressions
.../Theorem49*.lean	Theorem 4.9 theorem surfaces
.../Theorem49InteriorVertices.lean	Selected-boundary interior-vertex purification and discharge
.../fourcolor-boundary-order-notes.tex	Boundary-order gap note
public/literature/math/fourcolor/README.md	Canonical Coq/Rocq 4CT proof line
tmp/ai-agents-merge-check/lean-projects/fourcolor/README.md	Earlier Lean route branch and older Lemma 4.3 contradiction record
tmp/ai-agents-merge-check/isabelle/4CP/README.md	Archival Isabelle route branch

8 Conclusion

The Lean development is currently more valuable as a route-auditing tool than as a completed proof artifact. It has now done three distinct jobs. First, it preserves the historical negative result against the older per-run purification claim and locates that claim precisely. Second, it proves that the revised v23 local purification step is sound on its own theorem surface. Third, it refutes the manuscript’s naive forest-leaf and adjacency-only peeling reductions while also recording a sound replacement for them inside the live Lean route. Fourth, it isolates the interior-vertex side condition for boundary erasure and provides a checked purification bridge instead of silently applying quotient endpoints to arbitrary vertex sets. The main remaining live blocker is now the missing ordered boundary / annulus geometry needed to instantiate that stronger replacement and identify the intended interior vertices.

That is useful progress. It means future work can proceed with a cleaner map:

- do not conflate the older $\gamma \cdot 1_R$ contradiction with the exact v23 Step 2 claim;
- do treat the exact v23 purification statement as locally checked;
- do not build further on the current Theorem 4.9 / boundary-order route without instantiating the already formalized stronger adjacency-distance and root-distance interfaces from honest geometry;
- do not apply boundary-erasure endpoints to an unpurified vertex set unless the selected-boundary avoidance condition has been discharged.

In that sense the present note is not a declaration of final impossibility. It is a sharper ledger of what is already known to fail, what remains unverified, and where honest continuation should focus next.